# MATHEMATICS

# LIMIT

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**LIMIT**

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**EVALUATION OF LIMITS USING STANDARD OPERATION**

1. $\underset{x\to 3}{\text{Limit}} \dfrac{(x^3+27)\ \ln\ (x-2)}{(x^2-9)} =$

(A) – 8 (B) 8 (C) 9 (D) – 9

**Ans.** C

**Sol.** $\underset{x\to 3}{\ell im} \dfrac{(x^3+27)\ \ell n\ (x-2)}{(x^2-9)} = \underset{x\to 3}{\ell im} \dfrac{(x+3)(x^2-3x+9)\,\ell n[1+(x-3)]}{(x-3)(x+3)} = \underset{x\to 3}{\ell im}\ (x^2-3x+9).\dfrac{\ell n[1+(x-3)]}{(x-3)}$

$= (9-9+9)\ (1)\ = 9$

2. $\lim\limits_{x\to 0} \dfrac{(4^x-1)^3}{\sin\left(\dfrac{x}{p}\right)\ \ell n\left(1+\dfrac{x^2}{3}\right)} =$

(A) 9 p (log 4) (B) 3 p $(\log 4)^3$ (C) 12 p $(\log 4)^3$ (D) 27 p $(\log 4)^2$

**Ans.** B

**Sol.** $\underset{x\to 0}{\ell im} \dfrac{(4^x-1)^3}{\sin\left(\dfrac{x}{p}\right)\ \ell n\left(1+\dfrac{x^2}{3}\right)} = \underset{x\to 0}{\ell im} \dfrac{x^3\left(\dfrac{4^x-1}{x}\right)^3}{\left(\dfrac{x}{p}\right)\left[\dfrac{\sin\left(\dfrac{x}{p}\right)}{\dfrac{x}{p}}\right].\ell n\left(1+\dfrac{x^2}{3}\right)} = 3p.\ \underset{x\to 0}{\ell im} \dfrac{\left(\dfrac{4^x-1}{x}\right)^3}{\left[\dfrac{\sin\left(\dfrac{x}{p}\right)}{\dfrac{x}{p}}\right]}.\ \dfrac{\left(\dfrac{x^2}{3}\right)}{\ell n\left(1+\dfrac{x^2}{3}\right)} = 3\ p\,(\ln 4)^3$

3. $\underset{x\to\infty}{\text{Limit}} \dfrac{x^3\ .\ \sin\frac{1}{x}+x+1}{x^2+x+1} =$

(A) 0 (B) 1/2 (C) 1 (D) none

**Ans.** C

**Sol.** $\underset{x\to\infty}{\ell im} \dfrac{x^3\ \sin\dfrac{1}{x}+x+1}{x^2+x+1} = \underset{x\to\infty}{\ell im} \dfrac{\dfrac{\sin\frac{1}{x}}{\frac{1}{x}}+\dfrac{1}{x}+\dfrac{1}{x^2}}{1+\dfrac{1}{x}+\dfrac{1}{x^2}} = \dfrac{1+0+0}{1+0+0} = 1$

4. $\lim\limits_{n\to\infty} \dfrac{-3n+(-1)^n}{4n-(-1)^n}$ is :

(A) $-\frac{3}{4}$

(B) $-\frac{3}{4}$ if n is even ; $\frac{3}{4}$ if n is odd

(C) not exist if n is even ; $-\frac{3}{4}$ if n is odd

(D) + 1 if n is even ; does not exist if n is odd

$\lim\limits_{n\to\infty} \frac{-3n+(-1)^n}{4n-(-1)^n} =$

(A) $-\frac{3}{4}$

(B) $-\frac{3}{4}$ यदि n सम है; $\frac{3}{4}$ यदि n विषम है

(C) यदि n सम है तो विद्यमान नहीं है; $-\frac{3}{4}$ यदि n विषम है

(D) + 1 यदि n सम है; यदि n विषम है तो विद्यमान नहीं है

**Ans.** A

**Sol.** $\lim\limits_{n\to\infty} \frac{-3n+(-1)^n}{4n-(-1)^n} = \lim\limits_{n\to\infty} \frac{-3+(-1)^n\frac{1}{n}}{4-(-1)^n.\frac{1}{n}} = \frac{-3+0}{4-0} = -\frac{3}{4}$

5. $\lim\limits_{x\to 0^+} \left(1+\tan^2\sqrt{x}\right)^{5/x} =$

(A) $e^5$ (B) $e^2$ (C) e (D) none

$\lim\limits_{x\to 0^+} \left(1+\tan^2\sqrt{x}\right)^{5/x} =$

(A) $e^5$ (B) $e^2$ (C) e (D) इनमें से कोई नहीं

**Ans.** A

**Sol.** $\lim\limits_{x\to 0^+} \left(1+\tan^2\sqrt{x}\right)^{\frac{5}{x}} = \lim\limits_{h\to 0} \left(1+\tan^2\sqrt{0+h}\right)^{\frac{5}{0+h}}$

$= \lim\limits_{h\to 0} \left(1+\tan^2\sqrt{h}\right)^{\frac{5}{h}}$ ($1^\infty$ form रूप)

$= e^{\lim\limits_{h\to 0}\frac{5\tan^2\sqrt{h}}{h}} = e^{\lim\limits_{h\to 0} 5\left(\frac{\tan\sqrt{h}}{\sqrt{h}}\right)^2} = e^5$

6. The limiting value of $(\cos x)^{1/\sin x}$ as $x \to 0$ is:

(A) 1 (B) e (C) 0 (D) None of these

यदि $x \to 0$ हो, तो $(\cos x)^{1/\sin x}$ का सीमान्त मान है :

(A) 1 (B) e (C) 0 (D) इनमें से कोई नहीं

**Ans.** A

**Sol.** $\lim\limits_{x\to 0}(\cos x)^{\frac{1}{\sin x}}$ $(1^{\infty}$ form :i)

$= e^{\lim\limits_{x\to 0}\left(\frac{\cos x-1}{\sin x}\right)} = e^{\lim\limits_{x\to 0}\frac{-2\sin^2\frac{x}{2}}{2\sin\frac{x}{2}\cos\frac{x}{2}}} = e^{\lim\limits_{x\to 0}-\tan\left(\frac{x}{2}\right)} = e^0 = 1$

7. $\underset{n\to\infty}{\text{Limit}}\ \frac{5^{n+1}+3^n-2^{2n}}{5^n+2^n+3^{2n+3}} =$

(A) 5 (B) 3 (C) 1 (D) zero

$\underset{n\to\infty}{\text{Limit}}\ \frac{5^{n+1}+3^n-2^{2n}}{5^n+2^n+3^{2n+3}} =$

(A) 5 (B) 3 (C) 1 (D) शून्य

**Ans.** D

**Sol.** $\lim\limits_{n\to\infty}\frac{5^{n+1}+3^n-2^{2n}}{5^n+2^n+3^{2n+3}} = \lim\limits_{n\to\infty}\frac{5.5^n+3^n-4^n}{5^n+2^n+27.9^n} = \lim\limits_{n\to\infty}\frac{5.\left(\frac{5}{9}\right)^n+\left(\frac{3}{9}\right)^n-\left(\frac{4}{9}\right)^n}{\left(\frac{5}{9}\right)^n+\left(\frac{2}{9}\right)^n+27} = \frac{0+0-0}{0+0+27}=0$

8. $\underset{x\to\pi/2}{\text{Limit}}\ \frac{\left(1-\tan\frac{x}{2}\right)(1-\sin x)}{\left(1+\tan\frac{x}{2}\right)(\pi-2x)^3}$ is :

(A) 1/16 (B) – 1/16 (C) 1/32 (D) – 1/32

**Ans.** C

**Sol.** $\lim\limits_{x\to\frac{\pi}{2}}\frac{\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)(1-\sin x)}{(\pi-2x)^3}$

Let $x = \frac{\pi}{2}+y : y\to 0 \Rightarrow \lim\limits_{y\to 0}\frac{\tan\left(-\frac{y}{2}\right)(1-\cos y)}{(-2y)^3}$

$= \lim\limits_{y\to 0}\frac{-\tan\frac{y}{2}.2\sin^2\frac{y}{2}}{(-8)y^3} = \lim\limits_{y\to 0}\frac{1}{32}\frac{\tan\frac{y}{2}}{\left(\frac{y}{2}\right)}.\left[\frac{\sin\frac{y}{2}}{\frac{y}{2}}\right]^2 = \frac{1}{32}$

9. If $f(x)=\sum_{\lambda=1}^{n}\left(x-\frac{1}{\lambda}\right)\left(x-\frac{1}{\lambda+1}\right)$ then $\lim_{n\to\infty} f(0)$ is :

(A) 1 (B) –1 (C) 2 (D) None

यदि $f(x)=\sum_{\lambda=1}^{n}\left(x-\frac{1}{\lambda}\right)\left(x-\frac{1}{\lambda+1}\right)$ हो, तो $\lim_{n\to\infty} f(0)$ का मान है :

(A) 1 (B) –1 (C) 2 (D) इनमें से कोई नहीं

**Ans.** A

**Sol.** We have,

$$\lim_{x\to\infty} f(x)=\lim_{x\to\infty}\left(\frac{x^2+5x+3}{x^2+x+2}\right)^x$$

$$\lim_{x\to\infty} f(x)=\lim_{x\to\infty}\left(\frac{x^2+5x+3}{x^2+x+2}\right)^x$$

10. The value of $\lim_{x\to0}\frac{(1-\cos 2x)\sin 5x}{x^2\sin 3x}$ is :

$\lim_{x\to0}\frac{(1-\cos 2x)\sin 5x}{x^2\sin 3x}$ का मान है :

(A) 10/3 (B) 3/10 (C) 6/5 (D) 5/6

**Ans.** A

**Sol.** $\lim_{x\to0}\frac{(1-\cos 2x)\sin 5x}{x^2\sin 3x}\left(\frac{0}{0}\text{ form}\right)$

$$=\lim_{x\to0}\frac{2\sin^2 x}{x^2}\cdot\frac{3x}{3x}\cdot\frac{5x}{5x}\cdot\frac{\sin 5x}{\sin 3x}=\frac{10}{3}$$

11. $\lim_{x\to\infty}\left(\frac{x^2+5x+3}{x^2+x+3}\right)^x=$

(A) $e^4$ (B) $e^2$ (C) $e^3$ (D) e

**Ans.** A

**Sol.** We have,

$$\lim_{x\to\infty} f(x) \quad =\lim_{x\to\infty}\left(\frac{x^2+5x+3}{x^2+x+2}\right)^x$$

$$=\lim_{x\to\infty}\left(1+\frac{4x+1}{x^2+x+2}\right)^x=e^{\lim_{x\to\infty}\frac{x(4x+1)}{x^2+x+2}}=e^4$$

12. If $\lim_{x\to 0} \frac{\log(3+x)-\log(3-x)}{x} = k$, the value of k is :

यदि $\lim_{x\to 0} \frac{\log(3+x)-\log(3-x)}{x} = k$, तो k का मान है :

(A) $-\frac{2}{3}$ (B) 0 (C) $-\frac{1}{3}$ (D) $\frac{2}{3}$

**Ans.** D

**Sol.** $\lim_{x\to 0} \frac{\log(3+x)-\log(3-x)}{x} = k$

By L-Hospital's rule,

$$\lim_{x\to 0} \frac{\frac{1}{3+x}+\frac{1}{3-x}}{1} = k \Rightarrow \frac{2}{3} = k$$

13. Let f(a) = g(a) = k and their $n^{th}$ derivatives $f^n(a)$, $g^n(a)$ exist and are not equal for some n. Further if $\lim_{x\to a} \frac{f(a)\,g(x)-f(a)-g(a)\,f(x)+g(a)}{g(x)-f(x)} = 4$ then the value of k is :

माना f(a) = g(a) = k तथा उनके n वें अवकलज $f^n(a)$, $g^n(a)$ विद्यमान हो और वे n के किन्हीं मानों के लिऐ बराबर नहीं हों, यदि $\lim_{x\to a} \frac{f(a)\,g(x)-f(a)-g(a)\,f(x)+g(a)}{g(x)-f(x)} = 4$ हो, तो k का मान है :

(A) 0 (B) 4 (C) 2 (D) 1

**Ans.** B

**Sol.** *We have,* $\lim_{x\to a} \frac{l(a)g(x)-l(a)-g(a)l(x)+g(a)}{g(x)-l(x)} = 4$

$$\Rightarrow \lim_{x\to a} \frac{l(a)g'(x)-0-g(a)l'(x)+0}{g'(x)-l'(x)} = 4$$

$$\Rightarrow \frac{l(a)g'(a)-g(a)l'(a)}{g'(a)-l'(a)} = 4$$

$$\Rightarrow \frac{k\{g'(a)-l'(a)\}}{\{g'(a)-l'(a)\}} = 4 \qquad [\because f(a) = g(a) = k]$$

$\Rightarrow k = 4$

14. Let $\alpha$ & $\beta$ be the distinct roots of $ax^2+bx+c=0$, then $\lim_{x\to\alpha} \frac{1-\cos(ax^2+bx+c)}{(x-\alpha)^2}$ is equal to :

यदि $\alpha$ और $\beta$ समीकरण $ax^2 + bx + c = 0$ के दो असमान मूल हैं, तो $\lim_{x\to\alpha} \frac{1-\cos(ax^2+bx+c)}{(x-\alpha)^2}$ का मान है :

(A) $\frac{a^2}{2}(\alpha-\beta)^2$ (B) 0 (C) $\frac{-a^2}{2}(\alpha-\beta)^2$ (D) $\frac{1}{2}(\alpha-\beta)^2$

**Ans.** A

**Sol.** $\lim\limits_{x\to\alpha}\frac{1-\cos(ax^2+bx+c)}{(x-\alpha)^2}$

$= \lim\limits_{x\to\alpha}\left(\frac{1-\cos a(x-\alpha)(x-\beta)}{(x-\alpha)^2(x-\beta)^2a^2}\right)(x-\beta)^2\ a^2$

$= \frac{1}{2}\ a^2(\alpha-\beta)^2$

15. $\lim\limits_{x\to0^+}\frac{\ln(-\cos 3x+\cos x)}{\ln\sin x}$ is equal to :

$\lim\limits_{x\to0^+}\frac{\ln(-\cos 3x+\cos x)}{\ln\sin x}$ का मान होगा :

(A) 0 (B) 1

(C) 2 (D) D.N.E.

**Ans.** C

**Sol.** $\lim\limits_{x\to0^+}\frac{\ln(-4\cos^3x+3\cos x+\cos x)}{\ln\sin x}$

$\lim\limits_{x\to0^+}\frac{\ln\left(4\cos x\sin^2 x\right)}{\ln\sin x}$

$\lim\limits_{x\to0^+}\frac{\ln\left(4\cos x\right)+\ln\sin^2 x}{\ln\sin x}$

$= \lim\limits_{x\to0^+}\frac{2\ln\sin x}{\ln\sin x}+\frac{\ln\left(4\cos x\right)}{\ln\sin x}=2+0=2$

**BASED ON SPECIAL SIGNS LIKE GIF, MOD, SGN ETC**

16. The value of $\lim\limits_{x\to0}(\sin^{-1}[\sin x]+\cos^{-1}[\cos x]-2\tan^{-1}[\tan x])$ is equal to : [Note : [k] denotes largest integer function less than or equal to k.]

(A) $\pi$ (B) $\pi/2$ (C) $3\pi/2$ (D) Non-existent

$\lim\limits_{x\to0}(\sin^{-1}[\sin x]+\cos^{-1}[\cos x]-2\tan^{-1}[\tan x])$ का मान होगा : (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

(A) $\pi$ (B) $\pi/2$ (C) $3\pi/2$ (D) विद्यमान नहीं होगा

**Ans.** B

**Sol.** $\lim\limits_{x\to 0} \sin^{-1}[\sin x]+\cos^{-1}[\cos x]-2\tan^{-1}[\tan x]$

LHL

$=\sin^{-1}(-1)+\cos^{-1}(0)-2\tan^{-1}(-1)$

$=-\frac{\pi}{2}+\frac{\pi}{2}+2\cdot\frac{\pi}{4}=\frac{\pi}{2}$

$\text{RHL}=\sin^{-1}(0)+\cos^{-1}(0)-2\tan^{-1}(0)=\frac{\pi}{2}$

17. Given $l_1=\lim\limits_{x\to\frac{\pi}{4}}\cos^{-1}\left[\sec\left(x-\frac{\pi}{4}\right)\right];l_2=\lim\limits_{x\to\frac{\pi}{4}}\sin^{-1}\left[\operatorname{cosec}\left(x+\frac{\pi}{4}\right)\right];$

$l_3=\lim\limits_{x\to\frac{\pi}{4}}\tan^{-1}\left[\cot\left(x+\frac{\pi}{4}\right)\right];l_4=\lim\limits_{x\to\frac{\pi}{4}}\cot^{-1}\left[\tan\left(x-\frac{\pi}{4}\right)\right]$

where [x] denotes greatest integer function then which of the following limits exist?

(A) $l_1$ and $l_2$ only (B) $l_1$ and $l_3$ only (C) $l_1$ and $l_4$ only (D) All of them

दिया गया है $l_1=\lim\limits_{x\to\frac{\pi}{4}}\cos^{-1}\left[\sec\left(x-\frac{\pi}{4}\right)\right];l_2=\lim\limits_{x\to\frac{\pi}{4}}\sin^{-1}\left[\operatorname{cosec}\left(x+\frac{\pi}{4}\right)\right];$

$l_3=\lim\limits_{x\to\frac{\pi}{4}}\tan^{-1}\left[\cot\left(x+\frac{\pi}{4}\right)\right];l_4=\lim\limits_{x\to\frac{\pi}{4}}\cot^{-1}\left[\tan\left(x-\frac{\pi}{4}\right)\right]$

(जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है) तो निम्न में से कौनसी सीमा विद्यमान होगी

(A) केवल $l_1$ तथा $l_2$ (B) केवल $l_1$ तथा $l_3$ (C) केवल $l_1$ तथा $l_4$ (D) सभी

**Ans.** A

**Sol.** $l_1=\lim\limits_{x\to\frac{\pi}{4}}\cos^{-1}\left[\sec\left(x-\frac{\pi}{4}\right)\right]$

$\text{LHL}=\lim\limits_{h\to 0}\cos^{-1}[\sec(-h)]=0$

$\text{RHL}=\lim\limits_{h\to 0}\cos^{-1}[\text{sech}]=0$

$l_2=\lim\limits_{x\to\frac{\pi}{4}}\sin^{-1}\left[\operatorname{cosec}\left(x+\frac{\pi}{4}\right)\right]$

$\text{LHL}=\lim\limits_{h\to 0}\sin^{-1}\left[\operatorname{cosec}\left(\frac{\pi}{2}-h\right)\right]=\frac{\pi}{2}$

$$RHL = \lim_{h\to 0} \sin^{-1}\left[\cos ec\left(\frac{\pi}{2}+h\right)\right]$$

$$= \lim_{h\to 0} \sin^{-1}\left[\sec(h)\right] = \frac{\pi}{2}$$

$$l_3 = \lim_{x\to \frac{\pi}{4}} \tan^{-1}\left[\cot\left(x+\frac{\pi}{4}\right)\right]$$

$$LHL = \lim_{h\to 0} \tan^{-1}\left[\cot\left(\frac{\pi}{2}-h\right)\right] = \lim_{h\to 0} \tan^{-1}\left[\tanh\right] = 0$$

$$RHL = \lim_{h\to 0} \tan^{-1}\left[\cot\left(\frac{\pi}{2}+h\right)\right] = \lim_{h\to 0} \tan^{-1}\left[-\tanh\right] = -\frac{\pi}{4}$$

$\Rightarrow l_3$ does not exist

$$l_4 = \lim_{x\to \frac{\pi}{4}} \cot^{-1}\left[\tan\left(x-\frac{\pi}{4}\right)\right]$$

$$LHL = \lim_{h\to 0} \cot^{-1}\left[-\tanh\right] = -\frac{\pi}{4}$$

$$RHL = \lim_{h\to 0} \cot^{-1}\left[\tanh\right] = \cot^{-1}(0) = \frac{\pi}{2}$$

$\therefore l_4$ does not exist.

18. $\lim\limits_{x\to \frac{\pi}{2}} \dfrac{\sin x}{\cos^{-1}\left[\frac{1}{4}(3\sin x - \sin 3x)\right]}$ where [ ] denotes greatest integer function, is :

(A) $2/\pi$ (B) 1 (C) $4/\pi$ (D) Does not exist

$\lim\limits_{x\to \frac{\pi}{2}} \dfrac{\sin x}{\cos^{-1}\left[\frac{1}{4}(3\sin x - \sin 3x)\right]}$ का मान होगा (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है) :

(A) $2/\pi$ (B) 1 (C) $4/\pi$ (D) विद्यमान नहीं है

**Ans.** A

**Sol.** $\lim\limits_{x\to \frac{\pi}{2}} \dfrac{\sin x}{\cos^{-1}\left[\frac{1}{4}(3\sin x - \sin 3x)\right]}$

$$= \lim_{x\to\frac{\pi}{2}} \frac{\sin x}{\cos^{-1}\left[\sin^3 x\right]} = \frac{1}{\cos^{-1}(0)} = \frac{2}{\pi}$$

19. The value of $\lim_{x\to 0} \frac{\left(\tan\left(\{x\}-1\right)\right)\sin\{x\}}{\{x\}\left(\{x\}-1\right)}$ where $\{x\}$ denotes the fractional part function :

(A) is 1 (B) is tan 1 (C) is sin 1 (D) is non existent

$\lim_{x\to 0} \frac{\left(\tan\left(\{x\}-1\right)\right)\sin\{x\}}{\{x\}\left(\{x\}-1\right)}$ का मान होगा (जहाँ $\{.\}$ भिन्नात्मक भाग दर्शाता है)

(A) is 1 (B) is tan 1 (C) is sin 1 (D) विद्यमान नहीं है

**Ans.** D

**Sol.** $\underset{x\to 0}{\text{Limit}} \frac{\left(\tan\left(\{x\}-1\right)\right)\sin\{x\}}{\{x\}\left(\{x\}-1\right)} = \underset{x\to 0^+}{\text{Limit}} f(x) = \underset{h\to 0^+}{\text{Limit}} \frac{\tan(h-1).\sinh}{h(h-1)} = \frac{\tan(-1)}{-1} = \tan 1$

$\underset{x\to 0^-}{\text{Limit}} \frac{\tan((1-h)-1)\sin(1-h)}{(1-h)(1-h-1)} = \frac{\sin 1}{1} = \sin 1$

Hence $\text{limit}\, x\to 0\ f(x)$ does not exist ]

20. $\underset{x\to 1}{\text{Limit}}\ (1-x+[x-1]+[1-x]) =$ where $[x]$ denotes greatest integer function :

(A) 0 (B) 1 (C) $-1$ (D) does not exist

$\underset{x\to 1}{\text{Limit}}\ (1-x+[x-1]+[1-x])$, जहाँ $[x]$ महत्तम पूर्णांक फलन है, का मान है :

(A) 0 (B) 1 (C) $-1$ (D) विद्यमान नहीं है।

**Ans.** C

**Sol.** $\lim_{x\to 1}\left(1-x+\left[x-1\right]+\left[1-x\right]\right) = -1$

as

$$\text{LHL} = \lim_{h\to 0}\left(1-(1-h)+\left[-h\right]+\left[1-1+h\right]\right) = -1$$

$$\text{RHL} = \lim_{h\to 0}\left(1-(1+h)+\left[1+h-1\right]+\left[1-(1+h)\right]\right) = -1$$

21. The value of $\lim_{x\to\pi/4} (1+[x])^{1/\ln(\tan x)}$ (where [ . ] denotes the greatest integer function) is equal to :

$\lim_{x\to\pi/4} (1+[x])^{1/\ln(\tan x)}$, जहाँ [ . ] महत्तम पूर्णांक फलन है, का मान है :

(A) 0 (B) 1 (C) e (D) $e^{-1}$

**Ans.** B

**Sol.** $\lim\limits_{x\to\frac{\pi}{4}}(1+[x])^{\frac{1}{\ell n(\tan x)}} = \lim\limits_{x\to\frac{\pi}{4}}(\text{exact}\,1)^{\frac{1}{\ell n(\tan x)}} = 1$

22. The limit $\lim\limits_{\theta\to 0}\left(\left[\frac{n\sin\theta}{\theta}\right]+\left[\frac{n\tan\theta}{\theta}\right]\right)$, where [x] is the greatest integer function and $n \in N$, is :

(A) 2n (B) 2n + 1 (C) 2n – 1 (D) does not exist

$\lim\limits_{\theta\to 0}\left(\left[\frac{n\sin\theta}{\theta}\right]+\left[\frac{n\tan\theta}{\theta}\right]\right)$, जहाँ [x] महत्तम पूर्णांक फलन है और $n \in I$, है :

(A) 2n (B) 2n + 1 (C) 2n – 1 (D) विद्यमान नहीं है

**Ans.** C

**Sol.** $\sin\theta < \theta < \tan\theta \quad , \quad \theta \in \left(0, \frac{\pi}{2}\right)$

$$\frac{\sin\theta}{\theta} < 1 < \frac{\tan\theta}{\theta}$$

$$\frac{n\sin\theta}{\theta} < n < \frac{n\tan\theta}{\theta}$$

$$\lim_{\theta\to 0}\left(\left[\frac{n\sin\theta}{\theta}\right]+\left[\frac{n\tan\theta}{\theta}\right]\right); \quad n \in N$$

$$\text{L.H.L.} = \lim_{\theta\to 0^-}\left(\left[\frac{n\sin\theta}{\theta}\right]+\left[\frac{n\tan\theta}{\theta}\right]\right) = n - 1 + n = 2n - 1$$

$$\text{R.H.L.} = \lim_{\theta\to 0^+}\left(\left[\frac{n\sin\theta}{\theta}\right]+\left[\frac{n\tan\theta}{\theta}\right]\right) = n - 1 + n = 2n - 1$$

$\therefore$ L.H.L. = R.H.L = 2n – 1

**Hindi.** $\sin\theta < \theta < \tan\theta \quad , \quad \theta \in \left(0, \frac{\pi}{2}\right)$

$$\frac{\sin\theta}{\theta} < 1 < \frac{\tan\theta}{\theta}$$

$$\frac{n\sin\theta}{\theta} < n < \frac{n\tan\theta}{\theta}$$

$$\lim_{\theta\to 0}\left(\left[\frac{n\sin\theta}{\theta}\right]+\left[\frac{n\tan\theta}{\theta}\right]\right); \quad n \in N$$

बाँयीं सीमा = $\lim\limits_{\theta \to 0^-} \left( \left[ \frac{n\sin\theta}{\theta} \right] + \left[ \frac{n\tan\theta}{\theta} \right] \right) = n - 1 + n = 2n - 1$

दाँयीं सीमा = $\lim\limits_{\theta \to 0^+} \left( \left[ \frac{n\sin\theta}{\theta} \right] + \left[ \frac{n\tan\theta}{\theta} \right] \right) = n - 1 + n = 2n - 1$

बाँयीं सीमा = दाँयीं सीमा = $2n - 1$

**SANDWICH THEOREM**

23. The value of $\lim\limits_{x \to 0} \left( x^{16} + 4x^{12} - 3x^4 \right) \left[ \frac{1}{x^4} \right]$, (where [.] denotes greatest integer function), is :

(A) 0 (B) –3 (C) 4 (D) Non-existent

$\lim\limits_{x \to 0} \left( x^{16} + 4x^{12} - 3x^4 \right) \left[ \frac{1}{x^4} \right]$ का मान होगा, (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है) :

(A) 0 (B) –3 (C) 4 (D) विद्यमान नहीं होगी

**Ans.** B

**Sol.** $\frac{1}{x^4} - 1 < \left[ \frac{1}{x^4} \right] \le \frac{1}{x^4}$

$$\Rightarrow \left( x^{16} + 4x^{12} - 3x^4 \right) \frac{1 - x^4}{x^4} < \left( x^{16} + 4x^{12} - 3x^4 \right) \left[ \frac{1}{x^4} \right] \le \frac{x^{16} + 4x^{12} - 3x^4}{x^4}$$

$$\Rightarrow \lim_{x \to 0} \left( x^{16} + 4x^{12} - 3x^4 \right) \left( \frac{1 - x^4}{x^4} \right) < \lim_{x \to 0} \left( x^{16} + 4x^{12} - 3x^4 \right) \left[ \frac{1}{x^4} \right] \le \lim_{x \to 0} \frac{x^{16} + 4x^{12} - 3x^4}{x^4}$$

$$\Rightarrow -3 < \lim_{x \to 0} \left( x^{16} + 4x^{12} - 3x^4 \right) \left( \frac{1}{x^4} \right) \le -3$$

$\therefore$ By sandwich theorem

$$\lim_{x \to 0} \left( x^{16} + 4x^{12} - 3x^4 \right) \left( \frac{1}{x^4} \right) = -3$$

**Aliter :** $\lim\limits_{x \to 0} \left( x^{16} + 4x^{12} - 3x^4 \right) \left( \frac{1}{x^4} - \left\{ \frac{1}{x^4} \right\} \right) = -3$

$\because \ 0 \le \left\{ \frac{1}{x^4} \right\} < 1$

therefore limit = – 3

24. If [x] denotes the greatest integer $\le x$, then $\underset{n\to\infty}{\text{Limit}}\frac{1}{n^4}\left(\left([1^3\ x]+[2^3\ x]+...+[n^3\ x]\right)\right)$ equals :

यदि [x], x से छोटे एवं बराबर महत्तम पूर्णांक को प्रदर्शित करता है, तो $\underset{n\to\infty}{\text{Limit}}\frac{1}{n^4}\left(\left([1^3\ x]+[2^3\ x]+...+[n^3\ x]\right)\right)$ का मान है :

(A) x/2 (B) x/3 (C) x/6 (D) x/4

**Ans.** D

**Sol.** $\underset{n\to\infty}{\ell im}\frac{1}{n^4}\left([1^3\ x]+[2^3\ x]+...+[n^3\ x]\right)$

$1^3 x-1<[1^3 x]\le 1^3 x$

$2^3 x-1<[2^3 x]\le 2^3 x$

. . .

. . .

. . .

$n^3 x-1<[n^3 x]\le n^3 x$

Adding all these inequilities इन सभी असमिकाओं को जोड़ने पर

$(1^3+2^3+3^3 .....+n^3)x-n<[1^3x]+[2^3x]+.....[n^3x]\le(1^3+2^3+.....n^3)x$

$$\frac{n^2(n+1)^2}{4}\frac{x-n}{n^4}<\frac{[1^3x]+[2^3x]+....+[n^3x]}{n^4}\le\frac{\left(\frac{n(n+1)}{2}\right)^2 x}{n^4}$$

$$\underset{n\to\infty}{\ell im}\left(1+\frac{1}{n}\right)^2\frac{x}{4}-\frac{1}{n^3}<\underset{n\to\infty}{\ell im}\frac{[1^3x]+[2^3x]+.....+[n^3x]}{n^4}\le\underset{n\to\infty}{\ell im}\left(1+\frac{1}{n}\right)^2\frac{x}{4}$$

$$\frac{x}{4}<\underset{x\to\infty}{\ell im}\frac{[1^3x]+[2^3x]+.....[n^3x]}{n^4}\le\frac{x}{4}$$

$$\therefore\quad\underset{x\to\infty}{\ell im}\frac{[1^3x]+[2^3x]+.....+[n^3x]}{n^4}=\frac{x}{4}$$

25. $\lim\limits_{x\to\infty}\frac{\log x^n-[x]}{[x]}$, $n\in N$, (where [x] denotes greatest integer less than or equal to x)

(A) Has value – 1 (B) Has value 0 (C) Has value 1 (D) Does not exist

$\lim\limits_{x\to\infty}\frac{\log x^n-[x]}{[x]}$, $n\in N$(जहाँ [x] महत्तम पूर्णांक x से कम या बराबर को प्रदर्शित करता है)

(A) का मान – 1 है (B) का मान 0 है (C) का मान 1 है (D) विद्यमान नहीं है

**Ans.** A

**Sol.** $\lim\limits_{x\to\infty}\dfrac{\log x^n-[x]}{[x]}=\lim\limits_{x\to\infty}\dfrac{\log x^n}{[x]}-\lim\limits_{x\to\infty}\dfrac{[x]}{[x]}=0-1=-1$

26. Evaluate $\underset{n\to\infty}{\text{Lim}}\left(\dfrac{1}{\sqrt{n^2}}+\dfrac{1}{\sqrt{n^2+1}}+\dfrac{1}{\sqrt{n^2+2}}+.....+\dfrac{1}{\sqrt{n^2+2n}}\right)$

$\underset{n\to\infty}{\text{Lim}}\left(\dfrac{1}{\sqrt{n^2}}+\dfrac{1}{\sqrt{n^2+1}}+\dfrac{1}{\sqrt{n^2+2}}+.....+\dfrac{1}{\sqrt{n^2+2n}}\right)$ का मान है –

(A) 1 (B) 1/2 (C) 0 (D) 2

**Ans.** D

**Sol.** Let $f(x)=\left(\dfrac{1}{\sqrt{n^2}}+\dfrac{1}{\sqrt{n^2}}+\dfrac{1}{\sqrt{n^2}}+...+\dfrac{1}{\sqrt{n^2}}\right)=\dfrac{2n+1}{\sqrt{n^2}}$

... (2n+1) terms

$\lim\limits_{n\to\infty}f(x)=2$

$g(x)=\left(\dfrac{1}{\sqrt{n^2+2n}}+\dfrac{1}{\sqrt{n^2+2n}}+....+\dfrac{1}{\sqrt{n^2+2n}}\right)$

...(2n+1) terms

$\lim\limits_{n\to\infty}f(x)=2$

$\therefore\ \underset{n\to\infty}{\text{Lim}}\left(\dfrac{1}{\sqrt{n^2}}+\dfrac{1}{\sqrt{n^2+1}}+\dfrac{1}{\sqrt{n^2+2}}+....+\dfrac{1}{\sqrt{n^2+2n}}\right)=2$

27. $\lim\limits_{x\to\infty}\dfrac{3}{x}\left[\dfrac{x}{4}\right]=\dfrac{p}{q}$ (where [.] denotes greatest integer function), then p + q (where p, q are relative prime) is :

$\lim\limits_{x\to\infty}\dfrac{3}{x}\left[\dfrac{x}{4}\right]=\dfrac{p}{q}$ है (जहां [.] महत्तम पूर्णांक फलन को दर्शाता है) तो p + q (जहां p, q परस्पर अभाज्य संख्यायें है) का मान होगा :

(A) 2 (B) 7 (C) 5 (D) 6

**Ans.** B

**Sol.** $\lim\limits_{x\to\infty}\dfrac{3}{x}\left[\dfrac{x}{4}\right]=\dfrac{p}{q}$

$\because\dfrac{x}{4}-1\le\left[\dfrac{x}{4}\right]<\dfrac{x}{4}$

$\Rightarrow\dfrac{3}{x}\dfrac{(x-4)}{4}\le\left[\dfrac{x}{4}\right]\dfrac{3}{x}<\dfrac{3}{4}$

$$\Rightarrow \lim_{x\to\infty} \frac{3}{4}\left(1-\frac{4}{x}\right) \le \lim_{x\to\infty}\left[\frac{x}{4}\right]\cdot\frac{3}{x} < \frac{3}{4}$$

$$\Rightarrow \frac{p}{q} = \frac{3}{4}$$

$$\Rightarrow p + q = 7$$

28. Let $f: R \to R$ be a positive increasing function with $\lim_{x\to\infty} \frac{f(3x)}{f(x)} = 1$. Then $\lim_{x\to\infty} \frac{f(2x)}{f(x)} =$

माना $f: R \to R$ एक धन वर्धमान फलन है जिसमें $\lim_{x\to\infty} \frac{f(3x)}{f(x)} = 1$ है। तो $\lim_{x\to\infty} \frac{f(2x)}{f(x)} =$

(A) 1 (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) 3

**Ans.** A

**Sol.** $x < 2x < 3x$

$f(x) < f(2x) < f(3x)$

$$\frac{f(x)}{f(x)} < \frac{f(2x)}{f(x)} < \frac{f(3x)}{f(x)}$$

$$\lim_{x\to\infty} \frac{f(x)}{f(x)} = 1 \text{ and } \lim_{x\to\infty} \frac{f(3x)}{f(x)} = 1$$

Hence by Sandwich theorem

$$\lim_{x\to\infty} \frac{f(3x)}{f(x)} = 1$$

29. For each $t \in R$, let [t] be the greatest integer less than or equal to t. Then $\lim_{x\to 0+} x\left(\left[\frac{1}{x}\right]+\left[\frac{2}{x}\right]+.....+\left[\frac{15}{x}\right]\right)$

(A) is equal to 120 (B) does not exist (in R)

(C) is equal to 0 (D) is equal to 15

प्रत्येक $t \in R$ के लिए माना [t], t से छोटा महतम पूर्णांक है, तो $\lim_{x\to 0+} x\left(\left[\frac{1}{x}\right]+\left[\frac{2}{x}\right]+.....+\left[\frac{15}{x}\right]\right)$

(A) 120 के बराबर है (B) (R में) इसका अस्तित्व नहीं है

(C) 0 के बराबर है (D) 15 के बराबर है

**Ans.** A

**Sol.** $x - 1 < [x] \le x$

$$\left(\frac{1}{x}-1\right)<\left[\frac{1}{x}\right]\le\frac{1}{x}$$

$$x\left(\frac{1}{x}-1\right)<x\left[\frac{1}{x}\right]\le\frac{x}{x}$$

use sandwich

$$\lim_{x\to0} x\left[\frac{1}{x}\right]=1$$

Simillarly $\lim_{x\to0} x\left[\frac{2}{x}\right]=2$

$\vdots$
$\vdots$
$\vdots$

$$\lim_{x\to0} x\left[\frac{15}{x}\right]=15$$

Ans = 1 + 2 + ..... + 15 = 120

**QUESTIONS BASED ON DERIVATIVES**

30. If $f(1)=1, f'(1)=2$, then $\lim_{x\to1}\frac{\sqrt{f(x)}-1}{\sqrt{x}-1}=$

यदि $f(1)=1, f'(1)=2$, तो $\lim_{x\to1}\frac{\sqrt{f(x)}-1}{\sqrt{x}-1}=$

(A) 2 (B) 1 (C) 3 (D) 4

**Ans.** A

**Sol.** $y=\lim_{x\to1}\frac{\sqrt{l(x)}-1}{\sqrt{x}-1}$

$$\Rightarrow y=\lim_{x\to1}\frac{\left(\sqrt{l(x)}-1\right)}{\left(\sqrt{x}-1\right)}.\frac{\left(\sqrt{l(x)}+1\right)}{\left(\sqrt{x}+1\right)}.\frac{\left(\sqrt{x}+1\right)}{\left(\sqrt{l(x)}+1\right)}$$

$$\Rightarrow y=\lim_{x\to1}\frac{l(x)-1}{x-1}.\frac{\sqrt{x}+1}{\sqrt{l(x)}+1}$$

$$\Rightarrow y=\lim_{x\to1}\frac{l(x)-l(1)}{x-1}.\lim_{x\to1}\frac{\sqrt{x}+1}{\sqrt{l(x)}+1}$$

$$\Rightarrow y = l'(1).\frac{2}{\sqrt{l(1)+1}} \Rightarrow y = 2.\frac{2}{2} = 2$$

**Aliter :** Applying L-Hospital's rule.

31. If function f(x) is differentiable at x = a then $\lim\limits_{x\to a}\frac{x^2f(a)-a^2f(x)}{x-a}$ **:**

यदि फलन f(x) ,x = a पर अवकलनीय है, तो $\lim\limits_{x\to a}\frac{x^2f(a)-a^2f(x)}{x-a}$ बराबर है :

(A) 2a f(a) + $a^2$ f'(a) (B )–$a^2$ f' (a) (C) af(a) – $a^2$ f'(a) (D) 2af(a) – $a^2$ f'(a)

**Ans.** D

**Sol.** $\lim\limits_{x\to a}\frac{x^2l(a)-a^2l(x)}{x-a}$ $\left[\frac{0}{0}\text{ form}\right]$

Use L'Hospital rule

$= \lim\limits_{x\to a}\frac{2x\ l(a)-a^2\,l'(x)}{1}$

$= 2af(a) - a^2f'(a)$

**LIMIT BASED ON EXPANSION**

32. $\lim\limits_{x\to 0}\frac{(\cos x)\ln(1+x)-x\cos x-\ln(1+x)+x}{x^4}$ is equal to :

$\lim\limits_{x\to 0}\frac{(\cos x)\ln(1+x)-x\cos x-\ln(1+x)+x}{x^4}$ का मान होगा :

(A) 0 (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{8}$

**Ans.** C

**Sol.** $\lim\limits_{x\to 0}\frac{\cos x.\ln(1+x)-x\cos x-\ln(1+x)+x}{x^4}=\lim\limits_{x\to 0}\frac{(\ln(1+x)-x)(\cos x-1)}{x^4}=\left(\frac{-1}{2}\right)\left(\frac{-1}{2}\right)=\frac{1}{4}$

33. $\underset{x\to 0}{\text{Limit}}\ \frac{e^{-x^2/2}-\cos x}{x^3\sin x} =$

(A) 1/4 (B) 1/6 (C) 1/12 (D) 1/8

**Ans.** C

**Sol.** $\lim\limits_{x\to 0}\dfrac{e^{-\frac{x^2}{2}}-\cos x}{x^3\sin x}=\lim\limits_{x\to 0}\dfrac{\left[1-\dfrac{x^2}{2}+\dfrac{x^4}{4.2!}-.....\right]-\left[1-\dfrac{x^2}{2!}+\dfrac{x^4}{4!}-.....\right]}{x^3\left[x-\dfrac{x^3}{3!}+.....\right]}$

$=\lim\limits_{x\to 0}\dfrac{x^4\left[\dfrac{1}{8}-\dfrac{1}{4!}\right]-x^6\left[\dfrac{1}{8.3!}-\dfrac{1}{6!}\right]+.....}{x^4\left[1-\dfrac{x^2}{3!}+.....\right]}=\left[\dfrac{1}{8}-\dfrac{1}{24}\right]=\dfrac{1}{12}$

34. If $\lim\limits_{x\to\infty}\left(1+\dfrac{a}{x}+\dfrac{b}{x^2}\right)^{2x}=e^2$, then the values of a and b, are :

यदि $\lim\limits_{x\to\infty}\left(1+\dfrac{a}{x}+\dfrac{b}{x^2}\right)^{2x}=e^2$, तो a व b का मान होगा :

(A) $a\in R, b\in R$ (B) $a=1, b\in R$

(C) $a\in R, b=2$ (D) $a=1$ and $b=2$

**Ans.** B

**Sol.** Since, $\lim\limits_{x\to\infty}\left(1+\dfrac{a}{x}+\dfrac{b}{x^2}\right)^{2x}=e^2$

$\therefore\lim\limits_{x\to\infty}\left[\left(1+\dfrac{ax+b}{x^2}\right)^{\frac{x^2}{ax+b}}\right]^{\frac{2(ax+b)}{x}}=e^2$

$\Rightarrow\lim\limits_{x\to\infty}e^{\frac{2(ax+b)}{x}}=e^2$

$\Rightarrow\lim\limits_{x\to\infty}\dfrac{2(ax+b)}{x}=2\Rightarrow 2a=2\Rightarrow a=1$

Thus $a=1$ and $b\in R$

35. The value of $\underset{x\to 0}{\text{Limit}}\dfrac{\cos(\sin x)-\cos x}{x^4}$ is equal to :

$\underset{x\to 0}{\text{Limit}}\dfrac{\cos(\sin x)-\cos x}{x^4}$ का मान है :

(A) 1/5 (B) 1/6 (C) 1/4 (D) 1/2

**Ans.** B

**Sol.** $\lim\limits_{x\to0}\dfrac{\cos(\sin x)-\cos x}{x^4}=\lim\limits_{x\to0}\dfrac{\cos\left(x-\dfrac{x^3}{3!}+\ldots\right)-\cos x}{x^4}$

$$=\lim_{x\to0}\dfrac{\left[1-\dfrac{\left(x-\dfrac{x^3}{6}+\ldots\right)^2}{2!}+\dfrac{\left(x-\dfrac{x^3}{6}+\ldots\right)^4}{4!}-\ldots\ldots\right]-\left[1-\dfrac{x^2}{2!}+\dfrac{x^4}{4!}-\ldots.\right]}{x^4}$$

$$=\lim_{x\to0}\dfrac{x^4\left(\dfrac{1}{2}.\dfrac{2}{6}+\dfrac{1}{24}\right)-\dfrac{x^4}{24}+\ldots\ldots}{x^4}=\dfrac{1}{6}+\dfrac{1}{24}-\dfrac{1}{24}=\dfrac{1}{6}$$

36. The integer 'n' for which $\underset{x\to0}{\text{Limit}}\ \dfrac{(\cos x-1)(\cos x-e^x)}{x^n}$ is a finite non-zero number is :

पूर्णांक 'n' जिसके लिए $\underset{x\to0}{\text{Limit}}\ \dfrac{(\cos x-1)(\cos x-e^x)}{x^n}$ एक परिमित अशून्य संख्या है, है :

(A) 1 (B) 2 (C) 3 (D) 4

**Ans.** C

**Sol.** $\lim\limits_{x\to0}\dfrac{(\cos x-1)(\cos x-e^x)}{x^n}=1(\text{non zero finite})$

$$\Rightarrow\lim_{x\to0}\dfrac{\left(-\dfrac{x^2}{2}+\dfrac{x^4}{\lfloor4}-\dfrac{x^6}{\lfloor6}+\ldots.\right)\left(1-\dfrac{x^2}{\lfloor2}+\dfrac{x^4}{\lfloor4}+\ldots..\right)-\left(1+\dfrac{x^2}{\lfloor2}+\dfrac{x^3}{\lfloor3}+\ldots.\right)}{x^n}=1$$

$\Rightarrow n=3$

**EVALUATION OF LIMITS USING STANDARD OPERATION**

1. $\lim\limits_{x\to 1} \dfrac{x^2 - x.\ln x + \ln x - 1}{x-1}$

**Ans.** 2

**Sol.**

2. $\lim\limits_{x\to 1} \dfrac{\left[\sum\limits_{K=1}^{100} x^k\right] - 100}{x-1}$

**Ans.** 5050

**Sol.** $\lim\limits_{x\to 1} \dfrac{\left[\sum\limits_{K=1}^{100} x^K - 100\right]}{x-1}$

$$= \lim_{x\to 1}\left[\frac{x + x^2 + x^3 + ..... + x^{100} - 100}{x-1}\right]$$

$$= \lim_{x\to 1}\left[\frac{1 + 2x + 3x^2 + ..... + 100x^{99}}{1}\right]$$

(using L' Hospital)

= 5050

3. $\lim\limits_{x\to \frac{\pi}{4}} \dfrac{1-\tan x}{1-\sqrt{2}\sin x}$

**Ans.** 2

**Sol.** Put $x = \dfrac{\pi}{4} + h$

$\because \quad x \to \dfrac{\pi}{4} \Rightarrow h \to 0$

$$\lim_{h\to 0} \frac{1-\tan\left(\frac{\pi}{4}+h\right)}{1-\sqrt{2}\sin\left(\frac{\pi}{4}+h\right)} = \lim_{h\to 0} \frac{1-\frac{1+\tan h}{1-\tan h}}{1-\sin h-\cos h} = \lim_{h\to 0} \frac{\frac{-2\tan h}{1-\tan h}}{2\sin^2\frac{h}{2} - 2\sin\frac{h}{2}\cos\frac{h}{2}}$$

$$= \lim_{h\to 0} \frac{-2\tan h}{2\sin\frac{h}{2}\left[\sin\frac{h}{2}-\cos\frac{h}{2}\right]} \frac{1}{(1-\tanh)} = \lim_{h\to 0} \frac{\frac{-2\frac{\tanh}{h}}{\sin\frac{h}{2}}}{\frac{h}{2}\left[\sin\frac{h}{2}-\cos\frac{h}{2}\right]} \frac{1}{(1-\tanh)} = \frac{-2}{-1} = 2.$$

4. Find the sum of an infinite geometric series whose first term is the limit of the function $f(x) = \frac{\tan x - \sin x}{\sin^3 x}$ as $x \to 0$ and whose common ratio is the limit of the function $g(x) = \frac{1-\sqrt{x}}{(\cos^{-1} x)^2}$ as $x \to 1$.

ऐसे अनन्त पदों वाले गुणोत्तर श्रेणी का योग ज्ञात करों जिसका प्रथम पद फलन $f(x) = \frac{\tan x - \sin x}{\sin^3 x}$ की $x \to 0$ पर सीमा है तथा सर्वानुपात फलन $g(x) = \frac{1-\sqrt{x}}{(\cos^{-1} x)^2}$ की $x \to 1$ पर सीमा है।.

**Ans.** $a = \frac{1}{2}; r = \frac{1}{4}; S = \frac{2}{3}$

**Sol.** $a = \lim_{x\to 0} \frac{\tan x - \sin x}{\sin^3 x} = \lim_{x\to 0} \frac{1-\cos x}{\sin^2 x \cdot \cos x}$

$$= \lim_{x\to 0} \frac{2\sin^2 \frac{x}{2}}{4\sin^2 \frac{x}{2}\cos^2 \frac{x}{2}\cos x} = \frac{1}{2}$$

$$r = \lim_{x\to 1} \frac{1-\sqrt{x}}{\left(\cos^{-1} x\right)^2} \left(\frac{0}{0} \text{ form}\right)$$

$$= \lim_{x\to 1} \frac{1-x}{\left(1+\sqrt{x}\right)\left(\cos^{-1} x\right)^2} = \frac{1}{2} \lim_{\theta\to 0} \frac{2\sin^2 \frac{\theta}{2}}{\theta^2}$$

$$= \frac{1}{4}$$

$$\Rightarrow S_\infty = \frac{a}{1-r} = \frac{\frac{1}{2}}{1-\frac{1}{4}} = \frac{1\cdot 4}{2\cdot 3} = \frac{2}{3}$$

5. $\lim_{x\to\infty} (x - l\,n \cosh x)$ where $\cosh t = \frac{e^t + e^{-t}}{2}$.

$\lim_{x\to\infty} (x - l\,n \cosh x)$ जहाँ पर $\cosh t = \frac{e^t + e^{-t}}{2}$.

**Ans.** $l\,n\,2$

**Sol.** $\lim_{x\to\infty} \left(x - \ln \cosh x\right)$ $(\infty - \infty$ form)

$$= \lim_{x\to\infty}\left( x - \ln\left(\frac{e^x + e^{-x}}{2}\right)\right)$$

$$= \lim_{x\to\infty}\left( x - \ln\left(\frac{e^{2x}+1}{2e^x}\right)\right)$$

$$= \lim_{x\to\infty}\left(x - \ln\left(e^{2x}+1\right) + \ln 2 + \ln\left(e^x\right)\right)$$

$$= \lim_{x\to\infty}\left(2x - \ln\left(e^{2x}+1\right) + \ln 2\right)$$

$$= \lim_{x\to\infty}\left[\ln\left(\frac{e^{2x}}{e^{2x}+1}\right) + \ln 2\right]$$

$$= \ln 2$$

6. $\lim\limits_{x\to\frac{3\pi}{4}} \dfrac{1+\sqrt[3]{\tan x}}{1-2\cos^2 x}$

**Ans.** $-\dfrac{1}{3}$

**Sol.** $\lim\limits_{x\to\frac{3\pi}{4}} \dfrac{1+(\tan x)^{\frac{1}{3}}}{1-2\cos^2 x} = \lim\limits_{x\to\frac{3\pi}{4}} \dfrac{1+(\tan x)^{\frac{1}{3}}}{-\cos 2x}$

$$= \lim_{h\to 0} \frac{1+\left[\tan\left(\frac{3\pi}{4}+h\right)\right]^{\frac{1}{3}}}{-\cos\left(\frac{3\pi}{2}+2h\right)} = \lim_{h\to 0} \frac{1-\left(\tan\left(\frac{\pi}{4}-h\right)\right)^{\frac{1}{3}}}{-2h}$$

$$= \lim_{h\to 0} \frac{1^{\frac{1}{3}} - \left(\tan\left(\frac{\pi}{4}-h\right)\right)^{\frac{1}{3}}}{-2h} = -\frac{1}{2}\lim_{h\to 0} \frac{1-\tan\left(\frac{\pi}{4}-h\right)}{3\cdot h} = -\frac{1}{3}$$

$\left\{\because a^3+b^3 = (a+b)\left(a^2-ab+b^2\right)\right\}$

7. $\lim\limits_{x\to 0} \dfrac{8}{x^8}\left[1-\cos\dfrac{x^2}{2}-\cos\dfrac{x^2}{4}+\cos\dfrac{x^2}{2}\cos\dfrac{x^2}{4}\right]$

**Ans.** $\dfrac{1}{32}$

**Sol.** $\lim\limits_{x\to 0}\dfrac{8}{x^8}\left[1-\cos\dfrac{x^2}{2}-\cos\dfrac{x^2}{4}+\cos\dfrac{x^2}{2}\cos\dfrac{x^2}{4}\right]$

$=\lim\limits_{x\to 0}\dfrac{8}{x^8}\left(1-\cos\dfrac{x^2}{2}\right)\left(-\cos\dfrac{x^2}{4}+1\right)$

$=\lim\limits_{x\to 0}\dfrac{8}{x^8}2\sin^2\dfrac{x^2}{4}\cdot 2\sin^2\dfrac{x^2}{8}$

$=32\lim\limits_{x\to 0}\left(\dfrac{\sin\dfrac{x^2}{4}}{\dfrac{x^2}{4}}\right)^2\left(\dfrac{\sin\dfrac{x^2}{8}}{\dfrac{x^2}{8}}\right)^2\cdot\dfrac{1}{16\cdot 64}$

$=\dfrac{1}{32}$

8. If $l=\underset{n\to\infty}{\text{Lim}}\sum\limits_{r=2}^{n}\left((r+1)\sin\dfrac{\pi}{r+1}-r\sin\dfrac{\pi}{r}\right)$ then find $\{l\}$. (where $\{\}$ denotes the fractional part function)

यदि $l=\underset{n\to\infty}{\text{Lim}}\sum\limits_{r=2}^{n}\left((r+1)\sin\dfrac{\pi}{r+1}-r\sin\dfrac{\pi}{r}\right)$ तो $\{l\}$ मान ज्ञात करो। (जहाँ पर $\{\}$ भिन्नांश फलन है)

**Ans.** $\pi-3$

**Sol.** $\ell=\lim\limits_{n\to\infty}\sum\limits_{r=2}^{n}\left((r+1)\sin\dfrac{\pi}{r+1}-r\sin\dfrac{\pi}{r}\right)$

$\ell=\lim\limits_{n\to\infty}\sum\limits_{r=2}^{n}\left[r\left(\sin\dfrac{\pi}{r+1}-\sin\dfrac{\pi}{r}\right)+\sin\left(\dfrac{\pi}{r+1}\right)\right]$

$=\lim\limits_{n\to\infty}\left(\begin{array}{c}3\sin\dfrac{\pi}{3}-2\sin\dfrac{\pi}{2}+4\sin\dfrac{\pi}{4}-3\sin\dfrac{\pi}{3}+.....+n\sin\dfrac{\pi}{n}-(n-1)\sin\dfrac{\pi}{n-1}\\ +(n+1)\sin\dfrac{\pi}{n+1}-n\sin\dfrac{\pi}{n}\end{array}\right)$

$=\lim\limits_{n\to\infty}\left((n+1)\sin\dfrac{\pi}{n+1}-2\right)$

$=\lim\limits_{n\to\infty}\left[\pi\cdot\dfrac{\sin\left(\dfrac{\pi}{n+1}\right)}{\dfrac{\pi}{n+1}}-2\right]=\pi-2$

$$\Rightarrow \{\ell\} = \pi - 3$$

9. $\lim\limits_{x\to 1} \dfrac{(\ell n(1+x) - \ell n\, 2)(3.4^{x-1} - 3x)}{[(7+x)^{\frac{1}{3}} - (1+3x)^{\frac{1}{2}}].\sin(x-1)}$

**Ans.** $-\dfrac{9}{4}\ln\dfrac{4}{e}$

**Sol.** $\lim\limits_{x\to 1} \dfrac{\ell n\left(\dfrac{1+x}{2}\right)\times 3.(4^{x-1} - x)}{\left[(7+x)^{\frac{1}{3}} - (1+3x)^{\frac{1}{2}}\right]\sin(x-1)}$

$$= \lim_{h\to 0} \frac{\ell n\left(\dfrac{2+h}{2}\right)\times 3\left(4^h - (1+h)\right)}{\left[(8+h)^{\frac{1}{3}} - (4+3h)^{1/2}\right].\sin h}$$

$$= \lim_{h\to 0} \frac{\dfrac{\ell n\left(1+\dfrac{h}{2}\right)}{2(h/2)} \cdot 3 \cdot \left(4^h - (1+h)\right)}{2\left\{\left[1+\dfrac{h}{8}\times\dfrac{1}{3}+\ldots\right] - \left[1+\dfrac{3h}{4}\times\dfrac{1}{2}+\ldots\right]\right\}\dfrac{\sin h}{h}}$$

$$= \lim_{h\to 0} \frac{\dfrac{3}{4}\left[\dfrac{4^h - 1}{h} - 1\right]}{\dfrac{1}{h}\left[\dfrac{h}{24} - \dfrac{3h}{8}\right]} = \frac{-9}{4}\ell n\frac{4}{e}$$

10. $\lim\limits_{x\to 0} \dfrac{27^x - 9^x - 3^x + 1}{\sqrt{2} - \sqrt{1+\cos x}}$

**Ans.** $8\sqrt{2}(\ln 3)^2$

**Sol.** $\lim\limits_{x\to 0} \dfrac{3^{3x} - 3^{2x} - 3^x + 1}{\sqrt{2} - \sqrt{1+\cos x}} = \lim\limits_{x\to 0} \dfrac{\left(3^x - 1\right)^2\left(3^x + 1\right)}{\sqrt{2}\left(1 - \cos\dfrac{x}{2}\right)}$

$$= \sqrt{2}\lim_{x\to 0} \frac{\left(3^x - 1\right)^2}{2\sin^2\dfrac{x}{4}} \cdot \frac{\dfrac{x^2}{16}}{\dfrac{x^2}{16}}$$

$$= 8\sqrt{2}\lim_{x\to 0}\frac{\left(3^x-1\right)^2}{x^2} = 8\sqrt{2}\lim_{x\to 0}\frac{\left(e^{x\ln 3}-1\right)^2}{x^2}$$

$$= 8\sqrt{2}\left(\ln 3\right)^2$$

11. Let $P_n = a^{P_{n-1}} - 1$, $\forall\, n = 2, 3,.....$and Let $P_1 = a^x - 1$ where $a \in R^+$ then evaluate $\underset{x\to 0}{\text{Lim}}\ \frac{P_n}{x}$.

माना कि $P_n = a^{P_{n-1}} - 1$, $\forall\, n = 2, 3,.....$और $P_1 = a^x - 1$ जहाँ पर $a \in R^+$ तो $\underset{x\to 0}{\text{Lim}}\ \frac{P_n}{x}$ का मान ज्ञात करें।

**Ans.** $(\ell n\, a)^n$

**Sol.** $P_n = a^{P_{n-1}} - 1 \qquad \forall n = 2,3,.....$

$$P_1 = a^x - 1; a \in R^+ = e^{x\ln a} - 1$$

$$P_2 = a^{a^x-1} - 1 = e^{\left(a^x-1\right)\ln a} - 1$$

$$P_3 = a^{a^{a^x-1}-1} - 1 = a^{P_2} - 1$$

$$\Rightarrow \lim_{x\to 0}\frac{P_n}{x} = \lim_{x\to 0}\frac{a^{P_n}-1}{P_{n-1}}\cdot\frac{P_{n-1}}{x}$$

$$\Rightarrow \lim_{x\to 0}\frac{a^{P_{n-1}}-1}{P_{n-1}}\cdot\frac{a^{P_{n-2}}-1}{P_{n-2}}\cdot\frac{a^{P_{n-3}}-1}{P_{n-3}}\cdot.....\cdot\frac{a^{P_2}-1}{P_2}\cdot\frac{a^{P_1}-1}{P_1}\cdot\frac{a^x-1}{x}$$

$$= (\ln(a))^n$$

12. $$\underset{y\to 0}{\text{Lim}}\left[\underset{x\to\infty}{\text{Limit}}\frac{\exp\left(x\ln(1+\frac{ay}{x})\right) - \exp\left(x1n(1+\frac{by}{x})\right)}{y}\right]$$

**Ans.** $a - b$

**Sol.** $$\underset{y\to 0}{\ell\text{im}}\left(\underset{x\to\infty}{\ell\text{im}}\frac{\exp\left(x\,\ell n\left(1+\frac{ay}{x}\right)\right) - \exp\left(x\,\ell n\left(1+\frac{by}{x}\right)\right)}{y}\right) = \underset{y\to 0}{\ell\text{im}}\left(\underset{x\to\infty}{\ell\text{im}}\frac{\left(1+\frac{ay}{x}\right)^x - \left(1+\frac{by}{x}\right)^x}{y}\right)$$

by expansion प्रसार करने पर

$$= \underset{y\to 0}{\ell\text{im}}\left(\underset{x\to\infty}{\ell\text{im}}\frac{\left(1+ay+\frac{x(x-1)}{2!}\cdot\frac{a^2y^2}{x^2}+.....\right) - \left(1+by+\frac{x(x-1)}{2!}\cdot\frac{b^2y^2}{x^2}+....\right)}{y}\right)$$

$$= \lim_{y\to 0}\left[\frac{y(a-b)+\frac{y^2}{2}(a^2-b^2)+.....}{y}\right] = a-b$$

13. Let $\{a_n\}$, $\{b_n\}$, $\{c_n\}$ be sequences such that
**(i)** $a_n + b_n + c_n = 2n + 1$ ; **(ii)** $a_nb_n + b_nc_n + c_na_n = 2n - 1$ ; **(iii)** $a_nb_nc_n = -1$ ; **(iv)** $a_n < b_n < c_n$
Then find the value of $\lim\limits_{n\to\infty}(na_n)$.

यदि $\{a_n\}$, $\{b_n\}$, $\{c_n\}$ इस प्रकार की श्रेढ़ी है कि
**(i)** $a_n + b_n + c_n = 2n + 1$ ; **(ii)** $a_nb_n + b_nc_n + c_na_n = 2n - 1$ ; **(iii)** $a_nb_nc_n = -1$ ; **(iv)** $a_n < b_n < c_n$
तो $\lim\limits_{n\to\infty}(na_n)$ का मान ज्ञात करें।

**Ans.** $-1/2$

**Sol.** $x^3 - (2n+1)x^2 + (2n-1)x + 1 = 0$

$x = 1$ or $x = n - \sqrt{n^2+1}$ or $x = n + \sqrt{n^2+1}$

$\therefore a_n = n - \sqrt{n^2+1}$

$$\Rightarrow a_n = \frac{-1}{n+\sqrt{n^2+1}}$$

$$\Rightarrow \lim_{n\to\infty} na_n = \lim_{n\to\infty}\frac{-1}{1+\sqrt{1+\frac{1}{n^2}}} = -\frac{1}{2}$$

14. $$\lim_{x\to 0}\left(\frac{x-1+\cos x}{x}\right)^{\frac{1}{x}}$$

**Ans.** $e^{-1/2}$

**Sol.** $$\lim_{x\to 0}\left(\frac{x-1+\cos x}{x}\right)^{\frac{1}{x}} \qquad (1^\infty \text{ form})$$

$$= \exp\left[\lim_{x\to 0}\left(\frac{-1+\cos x}{x^2}\right)\right] = \exp\left[\lim_{x\to 0}\frac{-2\sin^2\frac{x}{2}}{\frac{x^2}{4}\cdot 4}\right] = e^{-\frac{1}{2}}$$

15. If $n \in N$ and $a_n = 2^2 + 4^2 + 6^2 + ..... + (2n)^2$ and $b_n = 1^2 + 3^2 + 5^2 + ..... + (2n-1)^2$. Find the value $\lim\limits_{n\to\infty}\frac{\sqrt{a_n}-\sqrt{b_n}}{\sqrt{n}}$.

यदि $n \in N$ तथा $a_n = 2^2 + 4^2 + 6^2 + ..... + (2n)^2$ और $b_n = 1^2 + 3^2 + 5^2 + ..... + (2n-1)^2$ तो $\lim\limits_{n\to\infty} \dfrac{\sqrt{a_n} - \sqrt{b_n}}{\sqrt{n}}$ का मान ज्ञात करो।

**Ans.** $\sqrt{3}/2$

**Sol.** $a_n = 2^2 + 4^2 + 6^2 + .... + (2n)^2 = 2^2\left(1^2 + 2^2 + .... + n^2\right)$

$$= 4\frac{n(n+1)(2n+1)}{6} = \frac{2}{3}n(n+1)(2n+1)$$

$$b_n = 1^2 + 3^2 + 5^2 + .... + (2n-1)^2$$

$$= \frac{2n(2n+1)(4n+1)}{6} - \frac{4n(n+1)(2n+1)}{6}$$

$$\Rightarrow \lim_{n\to\infty} \frac{\sqrt{a_n} - \sqrt{b_n}}{\sqrt{n}} = \lim_{n\to\infty} \frac{(a_n - b_n)}{\sqrt{n}\left(\sqrt{a_n} + \sqrt{b_n}\right)}$$

$$= \lim_{n\to\infty} -2\frac{\dfrac{4n(2n+1)(4n+1)}{6} + \dfrac{8}{6}n(n+1)(2n+1)}{\sqrt{n}\left(\sqrt{\dfrac{4}{6}n(n+1)(2n+1)} + \sqrt{\dfrac{2n(2n+1)(4n+1)}{6} - \dfrac{4}{6}n(n+1)(2n+1)}\right)}$$

$$= -\lim_{n\to\infty} \frac{n^3\left[\dfrac{5}{6}\left(2+\dfrac{1}{n}\right)\left(4+\dfrac{1}{n}\right) - \dfrac{8}{6}\left(1+\dfrac{1}{n}\right)\left(2+\dfrac{1}{n}\right)\right]}{n^2\left(\sqrt{\dfrac{4}{6}\left(1+\dfrac{1}{n}\right)\left(2+\dfrac{1}{n}\right)} + \sqrt{\dfrac{2\left(2+\dfrac{1}{n}\left(4+\dfrac{1}{n}\right)\right)}{6} - \dfrac{4}{6}\left(1+\dfrac{1}{n}\right)\left(2+\dfrac{1}{n}\right)}\right)}$$

$$= -\lim_{n\to\infty} \frac{n\left[\dfrac{2}{6}\left(2+\dfrac{1}{n}\right)\left(4+\dfrac{1}{n}\right) - \dfrac{8}{6}\left(1+\dfrac{1}{n}\right)\left(2+\dfrac{1}{n}\right)\right]}{\left(\sqrt{\dfrac{8}{6}} + \sqrt{\dfrac{16}{6} - \dfrac{8}{6}}\right)}$$

$$= -\frac{1}{2}\sqrt{\frac{6}{8}} \lim_{y\to 0} \frac{\dfrac{1}{3}(2+y)(4+y) - \dfrac{4}{3}(1+y)(2+y)}{y}$$

$$= \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

16. If $|x| < 1$, then $\lim_{n\to\infty}\left((1+x)\left(1+x^2\right).....\left(1+x^{2^n}\right)\right)$ equal to :

यदि $|x| < 1$ है, तो $\lim_{n\to\infty}\left((1+x)\left(1+x^2\right).....\left(1+x^{2^n}\right)\right)$ मान बराबर होगा :

**Ans.** $\frac{1}{1-x}$

**Sol.** Given $|x| < 1$

$$\lim_{n\to\infty}\left((1+x)\left(1+x^2\right).....\left(1+x^{2n}\right)\right)$$

$$= \lim_{n\to\infty} 1 + x + x^2 + .... + x^{2n+1}$$

$$= \frac{1}{1-x}$$

**BASED ON SPECIAL SIGNS LIKE GIF, MOD, SGN ETC.**

17. Let $f(x) = \begin{cases} -5 & , \quad -\infty < x \le -1 \\ x-4 & , \quad -1 < x \le 6 \\ 2 & , \quad 6 < x < \infty \end{cases}$ then $\lim_{x\to n} \frac{[x]^2 - 13[x] + 42}{(x-7)(x-6)}$ is equal to (where 'n' is the number of non zero integers in the range of f(x), [.] denotes greatest integer function)

माना $f(x) = \begin{cases} -5 & , \quad -\infty < x \le -1 \\ x-4 & , \quad -1 < x \le 6 \\ 2 & , \quad 6 < x < \infty \end{cases}$ तो $\lim_{x\to n} \frac{[x]^2 - 13[x] + 42}{(x-7)(x-6)}$ का मान होगा (जहाँ n, f(x) के परिसर में अशून्य पूर्णांको की संख्या है तथा [.] महत्तम पूर्णांक फलन दर्शाता है)

**Ans.** 0

**Sol.** $\lim_{x\to\pi/4} \frac{\sqrt{1-\sqrt{\sin 2x}}}{\pi - 4x}$

$$\text{LHL} = \lim_{h\to 0} \frac{\sqrt{1-\sqrt{\sin\left(\frac{\pi}{2} - 2h\right)}}}{\pi - \pi + 4h}$$

$$= \lim_{h\to 0}\frac{\sqrt{1-\sqrt{\cos 2h}}}{4h} \times \frac{\sqrt{1+\sqrt{\cos 2h}}}{\sqrt{1+\sqrt{\cos 2h}}}$$

$$= \lim_{h\to 0}\frac{\sqrt{1-\cos 2h}}{4h} \times \frac{1}{\sqrt{1+\sqrt{\cos 2h}}}$$

$$= \frac{\sqrt{2}\,|\sin h|}{4h\times\sqrt{2}} = \frac{1}{4}$$

Similarly RHL $= \frac{-1}{4}$

Hence LHL $\neq$ RHL $\quad\therefore$ limit does not exist.

18. The value of $\lim_{x\to 0} 10\left[\frac{\sin^{-1}x}{x}\right]+5\left[\frac{\sin^{-1}2x}{x}\right]+20\left[\frac{x}{\tan^{-1}x}\right]+10\left[\frac{x}{\tan^{-1}2x}\right]$ (where [.] denotes greatest integer function) is equal to –

$\lim_{x\to 0} 10\left[\frac{\sin^{-1}x}{x}\right]+5\left[\frac{\sin^{-1}2x}{x}\right]+20\left[\frac{x}{\tan^{-1}x}\right]+10\left[\frac{x}{\tan^{-1}2x}\right]$ का मान होगा (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

**Ans.** 40

**Sol.** $\lim_{x\to 0} 10\left[\frac{\sin^{-1}x}{x}\right]+5\left[\frac{\sin^{-1}2x}{x}\right]+20\left[\frac{x}{\tan^{-1}x}\right]+10\left[\frac{x}{\tan^{-1}2x}\right] = 10+10+20+0 = 40$

$\left\{\because \frac{\sin^{-1}x}{x}>1, \frac{x}{\tan^{-1}x}>1\right\}$

19. **(a)** $\lim_{x\to\frac{1}{\sqrt{2}}}\frac{\cos^{-1}\left(2x\sqrt{1-x^2}\right)}{x-\frac{1}{\sqrt{2}}}$; **(b)** $\lim_{x\to\frac{\pi}{4}}\frac{\sqrt{1-\sqrt{\sin 2x}}}{\pi-4x}$; **(c)** $\lim_{x\to -7}\frac{[x]^2+15[x]+56}{\sin(x+7)\sin(x+8)}$ where [ ] denotes the greatest integer function

**(a)** $\lim_{x\to\frac{1}{\sqrt{2}}}\frac{\cos^{-1}\left(2x\sqrt{1-x^2}\right)}{x-\frac{1}{\sqrt{2}}}$; **(b)** $\lim_{x\to\frac{\pi}{4}}\frac{\sqrt{1-\sqrt{\sin 2x}}}{\pi-4x}$; **(c)** $\mathit{Lim}_{x\to -7}\frac{[x]^2+15[x]+56}{\sin(x+7)\sin(x+8)}$ जहाँ [ ] महत्तम पूर्णांक फलन है

**Ans.** **(a)** DNE **(b)** DNE **(c)** 0

**Sol.** $\lim\limits_{x\to\frac{1}{2}}\dfrac{\cos^{-1}\left(2x\sqrt{1-x^2}\right)}{x-\dfrac{1}{\sqrt{2}}}=\lim\limits_{\theta\to\frac{\pi}{4}}\dfrac{\cos^{-1}(\sin 2\theta)}{\sin\theta-\dfrac{1}{\sqrt{2}}}$ if $x=\sin\theta$

$$\text{RHL}=\lim_{h\to0}\frac{\cos^{-1}\left(\sin\left(\frac{\pi}{2}+2h\right)\right)}{\sin\left(\frac{\pi}{4}+h\right)-\frac{1}{\sqrt{2}}}=\lim_{h\to0}\frac{2h\cdot\sqrt{2}}{\cosh+\sinh-1}$$

$$=\lim_{h\to0}\frac{2\sqrt{2}h}{h\cdot\left(\frac{\sinh}{h}\right)-\left(\frac{1-\cosh}{h^2}\right)\cdot h^2}=\lim_{h\to0}\frac{2\sqrt{2}}{\frac{\sinh}{h}-\left(\frac{1-\cosh}{h^2}\right)\cdot h}=2\sqrt{2}$$

$$\text{LHL}=\lim_{h\to0}\frac{\cos^{-1}\left(\sin\left(\frac{\pi}{2}-2h\right)\right)}{\sin\left(\frac{\pi}{4}-h\right)-\frac{1}{\sqrt{2}}}=\lim_{h\to0}\frac{2h\cdot\sqrt{2}}{\cosh-\sinh-1}=-2\sqrt{2}$$

$\Rightarrow$ limit DNE

20. $\lim\limits_{x\to-\infty}\dfrac{(3x^4+2x^2)\sin\frac{1}{x}+|x|^3+5}{|x|^3+|x|^2+|x|+1}$

**Ans.** $-2$

**Sol.** $\lim\limits_{x\to\infty}\dfrac{\left(3x^4+2x^2\right)\sin\dfrac{1}{x}+|x|^3+5}{|x|^3+|x|^2+|x|+1}$

$$=\lim_{x\to\infty}\frac{x^4\left[3+\frac{2}{x^2}\right]\frac{\sin\frac{1}{x}}{\frac{1}{x}}\cdot\frac{1}{x}-x^3+5}{x^3\left(-1-\frac{1}{x}-\frac{1}{x^2}+\frac{1}{x^3}\right)}$$

$$= \lim_{x\to\infty} \frac{\left[3+\frac{2}{x^2}\right]\frac{\sin\frac{1}{x}}{\frac{1}{x}} - 1 + \frac{5}{x^3}}{-1-\frac{1}{x}-\frac{1}{x^2}+\frac{1}{x^3}} = -2$$

21. Let $f(x) = \frac{x}{\sin x},\ x > 0$ and $g(x) = x+3,\ x < 1$

$= 2 - x,\ x \le 0$ $= x^2 - 2x - 2,\ 1 \le x < 2$

$= x - 5,\ x \ge 2$

find LHL and RHL of $g(f(x))$ at $x = 0$ and hence find $\lim_{x\to 0} g(f(x))$.

माना $f(x) = \frac{x}{\sin x},\ x > 0$ और $g(x) = x+3,\ x < 1$

$= 2 - x,\ x \le 0$ $= x^2 - 2x - 2,\ 1 \le x < 2$

$= x - 5,\ x \ge 2$

तो $g(f(x))$ का $x = 0$ पर वामपक्ष सीमा (LHL) तथा दाहिना पक्ष सीमा (RHL) ज्ञात करो। फलस्वरूप $\lim_{x\to 0} g(f(x))$ का मान ज्ञात करो।

**Ans.** $-3, -3, -3$

**Sol.** $f(x) = \begin{cases} \frac{x}{\sin x} & x > 0 \\ 2 - x & x \le 0 \end{cases}$ $g(x) = \begin{cases} x+3 & x < 1 \\ x^2 - 2x - 2 & 1 \le x < 2 \\ x - 5 & x \ge 2 \end{cases}$

$$\text{LHL} = \lim_{h\to 0} f(0-h) = \lim_{h\to 0}(2+h) = 2$$

$$\text{RHL} = \lim_{h\to 0} f(0+h) = \lim_{h\to 0} \frac{h}{\sinh} = 1$$

$$\text{LHL} = \lim_{h\to 0} g(f(-h)) = \lim_{h\to 0} g(2+h) = \lim_{h\to 0}(2+h-5) = -3$$

$$\text{RHL} = \lim_{h\to 0} g(f(0+h)) = \lim_{h\to 0} g\left(\frac{h}{\sinh}\right) = \lim_{h\to 0}\left(\frac{h}{\sinh}\right)^2 - 2\left(\frac{h}{\sinh}\right) - 2 = -3$$

$$\Rightarrow \lim_{x\to 0} g(f(x)) = -3$$

22. **(a)** $\lim_{x\to 0} \tan^{-1}\frac{a}{x^2}$, where $a \in R$ ;

**(b)** Plot the graph of the function $f(x) = \lim_{t\to 0}\left(\frac{2x}{\pi}\tan^{-1}\frac{x}{t^2}\right)$

**(a)** $\lim\limits_{x\to 0} \tan^{-1}\dfrac{a}{x^2}$, जहाँ पर $a \in R$ ;

**(b)** फलन $f(x) = \lim\limits_{t\to 0}\left(\dfrac{2x}{\pi}\tan^{-1}\dfrac{x}{t^2}\right)$ का रेखाचित्र खींचो।

**Ans.** (a) $\pi/2$ if $a > 0$ ; 0 if $a = 0$ and $-\pi/2$ if $a < 0$; (b) $f(x) = |x|$

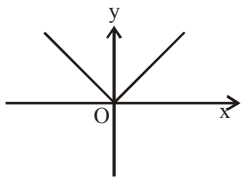
**Sol.** (a) $\lim\limits_{x\to 0}\tan^{-1}\dfrac{a}{x^2}; a \in R$

$$= \begin{cases} \dfrac{\pi}{2} & a > 0 \\ 0 & a = 0 \\ -\dfrac{\pi}{2} & a < 0 \end{cases}$$

(b) $f(x) = \lim\limits_{t\to 0}\left(\dfrac{2x}{\pi}\tan^{-1}\dfrac{x}{t^2}\right)$

$$= \begin{cases} \dfrac{2x}{\pi}\cdot\dfrac{\pi}{2} & x > 0 \\ 0 & x = 0 \\ -\dfrac{2x}{\pi}\cdot\dfrac{\pi}{2} & x < 0 \end{cases}$$

$$= \begin{cases} x & x > 0 \\ 0 & x = 0 = |x| \\ -x & x < 0 \end{cases}$$




23. Find the value of $\lim\limits_{n\to\infty}\left[\dfrac{2}{\pi}(n+1)\cos^{-1}\left(\dfrac{1}{n}\right)-n\right]$.

$\lim\limits_{n\to\infty}\left[\dfrac{2}{\pi}(n+1)\cos^{-1}\left(\dfrac{1}{n}\right)-n\right]$ का मान ज्ञात करो।

**Ans.** $1-\dfrac{2}{\pi}$

**Sol.** $\lim\limits_{n\to\infty}\left[\frac{2}{\pi}(n+1)\cos^{-1}\left(\frac{1}{n}\right)-n\right]$

$=\lim\limits_{n\to\infty}\left[(n+1)-\frac{2}{\pi}(n+1)\sin^{-1}\left(\frac{1}{n}\right)-n\right]$

$=\lim\limits_{n\to\infty}\left[1-\frac{2}{\pi}\left(1+\frac{1}{n}\right)\frac{\sin^{-1}\frac{1}{n}}{\frac{1}{n}}\right]$

$=1-\frac{2}{\pi}$

**SANDWICH THEOREM**

24. Using Sandwich theorem, evaluate

$$\lim_{n\to\infty}\frac{1}{1+n^2}+\frac{2}{2+n^2}+.....+\frac{n}{n+n^2}$$

सैण्डविच प्रमेय (Sandwich theorem) का प्रयोग करते हुये ज्ञात करो।

$$\lim_{n\to\infty}\frac{1}{1+n^2}+\frac{2}{2+n^2}+.....+\frac{n}{n+n^2}$$

**Ans.** 1/2

**Sol.** $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\frac{r}{r+n^2}$

$\because \sum\limits_{r=1}^{n}\frac{r}{n^2+n}\le\sum\limits_{r=1}^{n}\frac{r}{r+n^2}\le\sum\limits_{r=1}^{n}\frac{r}{n^2+1}$

$\Rightarrow \lim\limits_{n\to\infty}\frac{n(n+1)}{2(n^2+n)}\le\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\frac{r}{r+n^2}\le\lim\limits_{n\to\infty}\frac{n(n+1)}{2(n^2+1)}$

$\downarrow \qquad\qquad \downarrow$

$\frac{1}{2} \qquad\qquad \frac{1}{2}$

$\Rightarrow \lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\frac{r}{r+n^2}=\frac{1}{2}$

**QUESTIONS BASED ON DERIVATIVES**

25. $\lim_{h\to 0}\frac{\tan(a+2h)-2\tan(a+h)+\tan a}{h^2}=$

**Ans.** $2\sec^2 a.\tan a$

**Sol.** $\lim_{h\to 0}\frac{\tan(a+2h)-2\tan(a+h)+\tan a}{h^2}$

$=\frac{d^2}{dx^2}(\tan x)\Big|_{x=a}=2\sec^2 a\tan a$

26. $\lim_{h\to 0}\frac{\sin\left(\frac{\pi}{3}+4h\right)-4\sin\left(\frac{\pi}{3}+3h\right)+6\sin\left(\frac{\pi}{3}+2h\right)-4\sin\left(\frac{\pi}{3}+h\right)+\sin\frac{\pi}{3}}{h^4}=$

**Ans.** $\frac{\sqrt{3}}{2}$

**Sol.** $\lim_{h\to 0}\frac{\sin\left(\frac{\pi}{3}+4h\right)-4\sin\left(\frac{\pi}{3}+3h\right)+6\sin\left(\frac{\pi}{3}+2h\right)-4\sin\left(\frac{\pi}{3}+h\right)+\sin\frac{\pi}{3}}{h^4}=\frac{d^4}{dx^4}\sin(x)\Big|_{x=\frac{\pi}{3}}=\frac{\sqrt{3}}{2}$

**LIMITS BASED ON EXPANSION**

27. Let $l_1=\lim_{x\to 0}\frac{\sin(\sin^{-1}x)-\sin x}{x^3}, l_2=\lim_{x\to 0}\frac{\tan(\tan^{-1}x)-\ln(1+x)}{x^2}, l_3=\lim_{x\to 0}\frac{e^x-e^{\ln x}-1}{x^2}$ & $l_4=\lim_{x\to 0}\frac{e^x-e^{-x}-2x}{x^3}$

$l_5=\lim_{x\to 0}\frac{e^x+e^{-x}-x^2-2}{x^4}$ . Find the value of $3(l_1+2l_2+3l_3+l_4\div l_5)$

माना $l_1=\lim_{x\to 0}\frac{\sin(\sin^{-1}x)-\sin x}{x^3}, l_2=\lim_{x\to 0}\frac{\tan(\tan^{-1}x)-\ln(1+x)}{x^2}, l_3=\lim_{x\to 0}\frac{e^x-e^{\ln x}-1}{x^2}$ & $l_4=\lim_{x\to 0}\frac{e^x-e^{-x}-2x}{x^3}$

$l_5=\lim_{x\to 0}\frac{e^x+e^{-x}-x^2-2}{x^4}$ . $3(l_1+2l_2+3l_3+l_4\div l_5)$ का मान ज्ञात करों–

**Ans.** 20

**Sol.** $l_1=\lim_{x\to 0}\frac{\sin(\sin^{-1}x)-\sin x}{x^3}=\lim_{x\to 0}\frac{x-\sin x}{x^3}=\frac{1}{6}$

$l_2=\lim_{x\to 0}\frac{\tan(\tan^{-1}x)-\ln(1+x)}{x^2}=\lim_{x\to 0}\frac{x-\ln(1+x)}{x^2}=\frac{1}{2}$

$$l_3 = \lim_{x\to 0}\frac{e^x - e^{\ln x} - 1}{x^2} = \lim_{x\to 0}\frac{e^x - x - 1}{x^2} = \frac{1}{2}$$

$$l_4 = \lim_{x\to 0}\frac{e^x - e^{-x} - 2x}{x^3} = \lim_{x\to 0}\frac{\left(1 + x + \frac{x^2}{\lfloor 2} + \frac{x^3}{\lfloor 3} + \ldots\right) - \left(1 - x + \frac{x^2}{\lfloor 2} - \frac{x^3}{\lfloor 3} + \ldots.\right) - 2x}{x^3}$$

$$= \frac{2}{\lfloor 3} = \frac{1}{3}$$

$$l_5 = \lim_{x\to 0}\frac{e^x + e^{-x} - x^2 - 2}{x^4} = \frac{2}{\lfloor 4} = \frac{1}{12}$$

$$\therefore 3\left(l_1 + 2l_2 + 3_3 + \frac{l_4}{l_5}\right) = 20$$

28. $\lim\limits_{x\to 1}\left(\frac{p}{1-x^p} - \frac{q}{1-x^q}\right) p, q \in N$

**Ans.** $\frac{p-q}{2}$

**Sol.** $\lim\limits_{x\to 1}\left(\frac{p}{1-x^p} - \frac{1}{1-x^q}\right) = \lim\limits_{x\to 1}\left(\frac{p - px^q - q + qx^p}{\left(1-x^p\right)\left(1-x^q\right)}\right)$

$$= \lim_{x\to 1}\frac{(p-q) + qx^p - px^q}{\left(1-x^p\right)\left(1-x^q\right)}$$

$$= \lim_{h\to 0}\frac{(p-q) + q(1+h)^p - p(1+h)^q}{\left(1-(1+h)^q\right)\left(1-(1+h)^p\right)}$$

$$= \lim_{h\to 0}\frac{(p-q) + q\left(1 + ph + \frac{p(p-1)h^2}{\lfloor 2} + \ldots.\right) - p\left(1 + qh + \frac{q(q-1)h^2}{\lfloor 2} + \ldots..\right)}{\left(1 - \left(1 + qh + \frac{q(q-1)h^2}{\lfloor 2} + \ldots....\right)\right)\left(1 - \left(1 + ph + \frac{p(p-1)h^2}{\lfloor 2} + \ldots....\right)\right)}$$

$$= \frac{\frac{pq}{\lfloor 2} - \frac{pq(q-1)}{\lfloor 2}}{pq} = \frac{p-q}{2}$$

29. $\lim\limits_{x\to\infty} x^2\left(\sqrt{\frac{x+2}{x}}-\sqrt[3]{\frac{x+3}{x}}\right)$

**Ans.** 1/2

**Sol.** $\lim\limits_{x\to\infty} x^2\left(\sqrt{\frac{x+2}{x}}-\left(\frac{x+3}{x}\right)^{\frac{1}{3}}\right)$

$$=\lim_{x\to\infty}\frac{\sqrt{1+\frac{2}{x}}-\left(1+\frac{3}{x}\right)^{\frac{1}{3}}}{\frac{1}{x^2}}$$

$$=\lim_{x\to\infty}\frac{\left(1+\frac{2}{x}\right)^{\frac{1}{2}}-\left(1+\frac{3}{x}\right)^{\frac{1}{3}}}{\frac{1}{x^2}}$$

$$=\frac{1+\frac{1}{2}\left(\frac{2}{x}\right)+\frac{-1}{4}\left(\frac{2}{x}\right)^2\cdot\frac{1}{\lfloor 2}+....-\left(1+\frac{1}{3}\left(\frac{3}{x}\right)+\frac{1}{3}\left(\frac{1}{3}-1\right)\left(\frac{3}{x}\right)^2\cdot\frac{1}{\lfloor 2}+....\right)}{\frac{1}{x^2}}$$

$$=-\frac{1}{2}+1=\frac{1}{2}$$

30. Find a & b if : (i) $\lim\limits_{x\to\infty}\left[\frac{x^2+1}{x+1}-ax-b\right]=0$ (ii) $\lim\limits_{x\to-\infty}\left[\sqrt{x^2-x+1}-ax-b\right]=0$

a व b ज्ञात करिये जहाँ पर : (i) $\lim\limits_{x\to\infty}\left[\frac{x^2+1}{x+1}-ax-b\right]=0$ (ii) $\lim\limits_{x\to-\infty}\left[\sqrt{x^2-x+1}-ax-b\right]=0$

**Ans.** **(i)** a =1, b= –1 **(ii)** a=–1,b = $\frac{1}{2}$

**Sol.** We put x = – t

$$\lim_{t\to\infty}\left(\sqrt{t^2+t+1}+at-b\right)=0$$

[For $\infty-\infty$, a must be negative.]

$$\lim_{t\to\infty}\left(\frac{t^2+t+1-(at-b)^2}{\sqrt{t^2+t+1}-(at-b)}\right)=0$$

$$\lim_{t\to 0}\left(\frac{t^2(1-a^2)+t(1+2ab)+1-b^2}{\sqrt{t^2+t+1}-(at-b)}\right)$$

For limit coefficient of $t^2$ in numerator should be zero

$1 - a^2 = 0$ & $1 + 2ab = 0$

$\Rightarrow \quad a^2 = \pm 1 \;\Rightarrow\; a = -1 \;\Rightarrow b = \frac{1}{2}$

(a=1 is rejected)

31. If the $\underset{x\to 0}{\text{Lim}}\ \frac{1}{x^3}\left(\frac{1}{\sqrt{1+x}}-\frac{1+ax}{1+bx}\right)$ exists and has the value equal to *l*, then find the value of $\frac{1}{a}-\frac{2}{l}+\frac{3}{b}$.

यदि $\underset{x\to 0}{\text{Lim}}\ \frac{1}{x^3}\left(\frac{1}{\sqrt{1+x}}-\frac{1+ax}{1+bx}\right)$ विद्यमान हो और इसका मान *l* हो तो $\frac{1}{a}-\frac{2}{l}+\frac{3}{b}$ का मान ज्ञात करो।

**Ans.** 72

**Sol.** $\lim_{x\to 0}\frac{1}{x^3}\left(\frac{1+bx-(1+ax)\sqrt{1+x}}{\sqrt{1+x}(1+bx)}\right)$

$$=\lim_{x\to 0}\frac{1}{x^3}\left(\frac{(1+bx)^2-(1+ax)^2(1+x)}{\sqrt{1+x}(1+bx)\left(1+bx+(1+ax)\sqrt{1+x}\right)}\right)$$

$$=\lim_{x\to 0}\frac{1}{x^3}\frac{1+b^2x^2+2bx-\left(1+a^2x^2+2ax+2ax^2+a^2x^3+x\right)}{\sqrt{1+x}(1+bx)\left(1+bx+(1+ax)\sqrt{1+x}\right)}$$

$$=\lim_{x\to 0}\frac{-a^2x^3+\left(b^2-a^2-2a\right)x^2+(2b-2a-1)x}{2x^3}$$

$\Rightarrow l = -\frac{a^2}{2}$ & $b^2 - a^2 - 2a = 0$ & $2b - 2a - 1 = 0$

$\Rightarrow b^2 = a^2 + \underbrace{2a \quad \&2b}_{\downarrow\ a=\frac{1}{4},b=\frac{3}{4}} = 2a+1$

$$\Rightarrow \frac{1}{a}-\frac{2}{l}+\frac{3}{b}=4+\frac{4}{a^2}+\frac{3}{\frac{3}{4}} = 8 + 64 = 72$$

32. Let $f(x) = ax^3 + bx^2 + cx + d$ and $g(x) = x^2 + x - 2$.

If $\lim\limits_{x\to 1}\dfrac{f(x)}{g(x)}=1$ and $\lim\limits_{x\to -2}\dfrac{f(x)}{g(x)}=4$, then find the value of $\dfrac{c^2+d^2}{a^2+b^2}$.

माना $f(x)=ax^3+bx^2+cx+d$ तथा $g(x)=x^2+x-2$ है।

यदि $\lim\limits_{x\to 1}\dfrac{f(x)}{g(x)}=1$ तथा $\lim\limits_{x\to -2}\dfrac{f(x)}{g(x)}=4$ है, तो $\dfrac{c^2+d^2}{a^2+b^2}$ का मान ज्ञात कीजिये।

**Ans.** 16

**Sol.** $g(x)=x^2+x-2=(x+2)(x-1)$

Now $\lim\limits_{x\to 1}\dfrac{f(x)}{(x+2)(x-1)}$ and $\lim\limits_{x\to -2}\dfrac{f(x)}{(x+2)(x-1)}$ both exist

hence $f(x)=(x-1)(x+2)(k_1x+k_2)$

$\therefore \quad \lim\limits_{x\to 1}(k_1x+k_2)=1 \quad\Rightarrow\quad k_1+k_2=1 \quad ....(1)$

and $\quad \lim\limits_{x\to -2}(k_1x+k_2)=4 \quad\Rightarrow\quad k_2-2k_1=4 \quad ....(2)$

On solving (1) and (2), we get

$k_1=-1$ and $k_2=2$

Hence cubic $f(x)=(x-1)(x+2)(2-x)=-x^3+x^2+4x-4$

$\therefore \quad a=-1,\ b=1,\ c=4,\ d=-4$

Hence $\dfrac{c^2+d^2}{a^2+b^2}=\dfrac{(4)^2+(-4)^2}{(-1)+(1)^2}=16$

33. Find the values of constants a, b & c so that $\lim\limits_{x\to 0}\dfrac{axe^x-b\log(1+x)+cxe^{-x}}{x^2\sin x}=2.$

अचर a, b एवं c के मान ज्ञात कीजिए ताकि $\lim\limits_{x\to 0}\dfrac{axe^x-b\log(1+x)+cxe^{-x}}{x^2\sin x}=2$ हो.

**Ans.** $a=3,\ b=12,\ c=9$

**Sol.** $\lim\limits_{x\to 0}\dfrac{ax\left(1+x+\frac{x^2}{2!}+.....\right)-b\left(x-\frac{x^2}{2}+\frac{x^3}{3}-.....\right)+cx\left(1-x+\frac{x^2}{2!}-\frac{x^3}{3!}+.....\right)}{x^2\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}.....\right)}=2$

$\Rightarrow \quad \lim\limits_{x\to 0}\dfrac{(a-b+c)+x\left(a+\frac{b}{2}-c\right)+x^2\left(\frac{a}{2}-\frac{b}{3}+\frac{c}{2}\right)+x^3 \text{ and term containing higher power}}{x^2-\frac{x^4}{3!}+\frac{x^6}{5!}-.....}=2$

then $a - b + c = 0$ .....(1)

$2a + b - 2c = 0$ .....(2)

$3a - 2b + 3c = 12$ .....(3)

On solving (1), (2), (3)

we get $a = 3,\ b = 12,\ c = 9$

34. If $\lim\limits_{x\to\infty} \frac{a(2x^3 - x^2) + b(x^3 + 5x^2 - 1) - c(3x^3 + x^2)}{a(5x^4 - x) - bx^4 + c(4x^4 + 1) + 2x^2 + 5x} = 1$, then the value of $(a + b + c)$ can be expressed in the lowest form as $\frac{p}{q}$. Find the value of $(p + q)$.

यदि $\lim\limits_{x\to\infty} \frac{a(2x^3 - x^2) + b(x^3 + 5x^2 - 1) - c(3x^3 + x^2)}{a(5x^4 - x) - bx^4 + c(4x^4 + 1) + 2x^2 + 5x} = 1$, तो $(a + b + c)$ का मान $\frac{p}{q}$ के सरलतम रूप में व्यक्त किया जा सकता है तो $(p + q)$ का मान ज्ञात करों।

**Ans.** 167

**Sol.** $\lim\limits_{x\to\infty} \frac{(2a + b - 3c)x^3 + x^2(5b - c - a) - b}{(5a - b + 4c)x^4 + 2x^2 + (5 - a)x + c} = 1$

$\Rightarrow \left.\begin{matrix} 2a + b - 3c = 0 \\ 5a - b + 4c = 0 \\ 5b - c - a = 2 \end{matrix}\right] \rightarrow \left.\begin{matrix} 7a + c = 0 \\ 4a + 3c + 4b = 2 \end{matrix}\right] \rightarrow 4(a + b) - 21a = 2$

$4b - 17a = 2$

$8a + 4b - 12c = 0 \Rightarrow 8a + 4b + 84a = 0$

$\Rightarrow 4b = -92a$

$\Rightarrow b = -23a$

$\Rightarrow a = \frac{-2}{109}$ $\Rightarrow a + (-23a) + -7a = -29a$

$= \frac{58}{109}$

$\Rightarrow p + q = 167$

35. $f(x)$ is the function such that $\lim\limits_{x\to0} \frac{f(x)}{x} = 1$. If $\lim\limits_{x\to0} \frac{x(1 + a\cos x) - b\sin x}{(f(x))^3} = 1$, then find the value of $a$ and $b$.

$f(x)$ एक ऐसा फलन है कि $\lim\limits_{x\to0} \frac{f(x)}{x} = 1$ यदि $\lim\limits_{x\to0} \frac{x(1 + a\cos x) - b\sin x}{(f(x))^3} = 1$, तो $a$ व $b$ का मान ज्ञात करो।

**Ans.** $a = -5/2,\ b = -3/2$

**Sol.** $\lim\limits_{x\to 0}\dfrac{f(x)}{x}=1$

$$\lim_{x\to 0}\frac{x(1+a\cos x)-b\sin x}{(f(x))^3}=1$$

$$\Rightarrow \lim_{x\to 0}\frac{x+ax\left(1-\frac{x^2}{\lfloor 2}+\frac{x^4}{\lfloor 4}+.....\right)-b\left(x-\frac{x^3}{\lfloor 3}+.....\right)}{x^3.\frac{(f(x))^3}{x^3}}=1$$

$$\Rightarrow \lim_{x\to 0}\frac{x(1+a-b)+\left(\frac{b}{6}-\frac{a}{2}\right)x^3+.....}{x^3}=1$$

$$\Rightarrow a-b+1=0 \qquad \& \qquad \frac{b}{6}-\frac{a}{2}=1$$

$$\Rightarrow \underbrace{a-b+1=0 \quad \& \quad b-3a=6}_{\downarrow}$$

$$\Rightarrow \qquad a-(6+3a)+1=0$$

$$\Rightarrow -2a-5=0$$

$$a=-\frac{5}{2}$$

$$b=-\frac{15}{2}+6=\frac{12-15}{2}=-\frac{3}{2}$$

36. If $\lim\limits_{x\to 1}\left(1+ax+bx^2\right)^{\frac{c}{x-1}}=e^3$, then the value of $\left|\dfrac{5b}{a}\right|+4bc$ is equal to.

यदि $\lim\limits_{x\to 1}\left(1+ax+bx^2\right)^{\frac{c}{x-1}}=e^3$, तो $\left|\dfrac{5b}{a}\right|+4bc$ का मान बराबर होगा।

**Ans.** 17

**Sol.** $\lim\limits_{x\to 1}\left(1+ax+bx^2\right)\dfrac{c}{x-1}=e^3$

$\Rightarrow 1 + a + b = 1$

$\Rightarrow \exp\left(\lim_{x\to 1} \frac{acx + bcx^2}{x-1}\right) = e^3$

$\Rightarrow (a + b)\, c = 0$

$\Rightarrow \lim_{x\to 1} c\left(\frac{ax + bx^2}{x-1}\right) = 3$

$\Rightarrow c \lim_{x\to 1} \frac{a(x-1+1) + b(x-1)(x+1) + b}{x-1} = 3$

$\Rightarrow c \lim_{x\to 1}\left(a + b(x+1) + \frac{a+b}{x-1}\right) = 3$

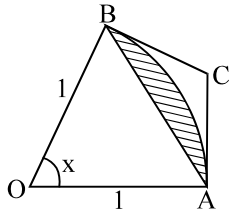
$\Rightarrow (a + 2b)c = 3$

$\Rightarrow bc = 3$

$\Rightarrow \left|\frac{5b}{a}\right| + 4bc = 17$

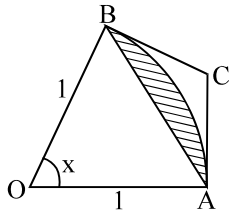
**GEOMETRICAL LIMITS**

37. A circular arc of radius 1 subtends an angle of x radians, $0 < x < \frac{\pi}{2}$ as shown in the figure. The point C is the intersection of the two tangent lines at A & B. Let T(x) be the area of triangle ABC & let S(x) be the area of the shaded region. Compute:




(a) T(x) (b) S(x) (c) the limit of $\frac{T(x)}{S(x)}$ as $x \to 0$.

त्रिज्या 1 का एक वृत्ताकार चाप चित्रानुसार x रेडियन का कोण बनाता है, जहाँ $0 < x < \frac{\pi}{2}$. बिन्दु A व B पर खीची गयी स्पर्श रेखा बिन्दु C पर प्रतिच्छेद करती है। यदि T(x) त्रिभुज ABC का क्षेत्रफल तथा S(x) रेखांकित क्षेत्र का क्षेत्रफल है तो निम्नलिखित के मान की गणना करो।




(a) T(x) (b) S(x) (c) $\frac{T(x)}{S(x)}$ की $x \to 0$ पर सीमा

**Ans.** $T(x) = \frac{1}{2}\tan^2\frac{x}{2}.\sin x$ or $\tan\frac{x}{2} - \frac{\sin x}{2}$, $S(x) = \frac{1}{2}x - \frac{1}{2}\sin x$, limit $= \frac{3}{2}$

**Sol.** $T(x) = \frac{1}{2}BC \cdot AC \sin(\pi - x)$

$= \frac{1}{2}\left(\tan^2\frac{x}{2}\right)\cdot\sin x = \tan\frac{x}{2} - \frac{\sin x}{2}$

s(x) = area of sector OAB – area of $\Delta OAB$

$= \frac{1}{2}(1)^2 \cdot x - \frac{1}{2}(1)^2 \sin x$

$\lim\limits_{x\to 0} \frac{\tan\frac{x}{2} - \frac{\sin x}{2}}{\frac{x - \sin}{2}} = \frac{3}{2}$

## COMPREHENSION TYPE QUESTIONS

**Paragraph for Question Nos. 38 to 40**

Consider two functions $f(x) = \lim\limits_{n\to\infty}\left(\cos\frac{x}{\sqrt{n}}\right)^n$ and $g(x) = -x^{4b}$ where $b = \lim\limits_{x\to\infty}\left(\sqrt{x^2+x+1} - \sqrt{x^2+1}\right)$. Then :

माना दो फलन $f(x) = \lim\limits_{n\to\infty}\left(\cos\frac{x}{\sqrt{n}}\right)^n$ तथा $g(x) = -x^{4b}$ है, जहाँ $b = \lim\limits_{x\to\infty}\left(\sqrt{x^2+x+1} - \sqrt{x^2+1}\right)$. तो :

38. f(x) is :

f(x) है :

(A) $e^{-x^2}$ (B) $e^{\frac{-x^2}{2}}$ (C) $e^{x^2}$ (D) $e^{\frac{x^2}{2}}$

**Ans.** B

39. g(x) is

g(x) है

(A) $-x^2$ (B) $x^2$ (C) $x^4$ (D) $-x^4$

**Ans.** A

40. Number of solutions of $f(x) + g(x) = 0$ is :

$f(x) + g(x) = 0$ के हलों की संख्या है –

(A) 2 (B) 4 (C) 0 (D) 1

**Ans.** A

**Sol.** **(38 - 40)**

$$f(x) = \lim_{n\to\infty} \left(\cos\frac{x}{\sqrt{n}}\right)^n = e^{\lim_{n\to\infty}\left(\cos\frac{x}{\sqrt{n}}-1\right).n}$$

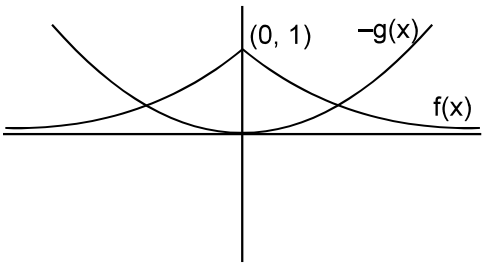
Substituting, $n = \frac{1}{t^2}$

$$f(x) = e^{\lim_{t\to0}\left(\frac{\cos tx-1}{t^2}\right)} = e^{\lim_{t\to0}-x^2\left(\frac{1-\cos tx}{t^2x^2}\right)} = e^{-\frac{1}{2}x^2}$$

$g(x) = -x^{4b}$

$$b = \lim_{x\to\infty}\left(\sqrt{x^2+x+1}-\sqrt{x^2+1}\right) = \lim_{x\to\infty}\left(\frac{x^2+x+1-x^2-1}{\sqrt{x^2+x+1}+\sqrt{x^2+1}}\right) = \frac{1}{2}$$

$\therefore \quad g(x) = -x^{4.\frac{1}{2}} = -x^2$




By observation, graphs of $f(x)$ and $-g(x)$ intersect each other at two points

$\therefore$ Number of solutions is 2.

**Hindi. (38-40)**

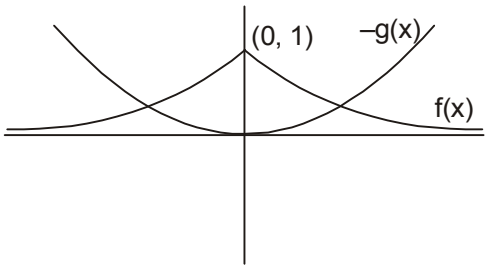
$$f(x) = \lim_{n\to\infty} \left(\cos\frac{x}{\sqrt{n}}\right)^n = e^{\lim_{n\to\infty}\left(\cos\frac{x}{\sqrt{n}}-1\right).n}$$ $n = \frac{1}{t^2}$ प्रतिस्थापित करने पर

$$f(x) = e^{\lim_{t\to0}\left(\frac{\cos tx-1}{t^2}\right)} = e^{\lim_{t\to0}-x^2\left(\frac{1-\cos tx}{t^2x^2}\right)} = e^{-\frac{1}{2}x^2}$$

$g(x) = -x^{4b}$

$$b = \lim_{x\to\infty}\left(\sqrt{x^2+x+1}-\sqrt{x^2+1}\right) = \lim_{x\to\infty}\left(\frac{x^2+x+1-x^2-1}{\sqrt{x^2+x+1}+\sqrt{x^2+1}}\right) = \frac{1}{2}$$

$\therefore \quad g(x) = -x^{4\cdot\frac{1}{2}} = -x^2$




देखने पर पता चलता है कि आरेख f(x) तथा – g(x) एक दूसरे को दो बिन्दुओं पर प्रतिच्छेद करतें हैं।

$\therefore$ हलों की संख्या 2 है।

**Paragraph for Question Nos. 41 to 43**

Consider $f(x) = \frac{\sin x + ae^x + be^{-x} + c\ \ell n(1+x)}{x^3}$, where a, b, c are real numbers.

माना $ff(x) = \frac{\sin x + ae^x + be^{-x} + c\ \ell n(1+x)}{x^3}$, जहाँ a, b, c वास्तविक संख्याऐं है।

41. If $\lim_{X\to0^+} f(x)$ is finite, then the value of $a + b + c$ is

यदि $\lim_{X\to0^+} f(x)$ परिमित है तो $a + b + c$ का मान होगा –

(A) 0 (B) 1 (C) 2 (D) – 2

**Ans.** A

42. If $\lim_{X\to0^+} f(x) = \ell$ (finite), then the value of $\ell$ is

यदि $\lim_{X\to0^+} f(x) = \ell$ (परिमित), तो $\ell$ का मान होगा –

(A) – 2 (B) $-\frac{1}{2}$ (C) – 1 (D) $-\frac{1}{3}$

**Ans.** D

43. Using the values of a, b, c as found above, the value of $\lim_{x\to0^+} x\,f(x)$ is :

उपरोक्त ज्ञात किये गये a, b, c के मानों का उपयोग करते हुये $\lim_{x\to0^+} x\,f(x)$ का मान होगा –

(A) 0 (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) 2

**Ans.** A

**Sol.** **(41-43)**

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \frac{\sin x + ae^x + be^{-x} + c\ \ell n(1+x)}{x^3}$$

$$= \lim_{x\to 0^+} \frac{\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} + \ldots\right) + a\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots\ldots\right) + b\left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \ldots.\right) + c\left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots.\right)}{x^3}$$

$$= \lim_{x\to 0^+} \frac{(a+b) + x(1+a-b) + x^2\left(\frac{a}{2!} + \frac{b}{2!} - \frac{c}{2}\right) + x^3\left(-\frac{1}{3!} + \frac{a}{3!} - \frac{b}{3!} + \frac{c}{3}\right) + x^4\left(\ \ \right) + \ldots.}{x^3}$$

$\because$ limit is finite $\Rightarrow$ $\left.\begin{matrix} a+b=0 \\ 1+a-b=0 \\ a+b-c=0 \end{matrix}\right\}$ $\Rightarrow$ $a = -\frac{1}{2}$, $b = \frac{1}{2}$ and $c = 0$

Also $\lim_{x\to 0^+} f(x) = -\frac{1}{3!} + \frac{a}{3!} - \frac{b}{3!} + \frac{c}{3} = -\frac{1}{6}(1 - a + b - 2c) = -\frac{1}{3}$

**4.** $a + b + c = -\frac{1}{2} + \frac{1}{2} + 0 = 0$

**5.** $\lim_{x\to 0^+} f(x) = -\frac{1}{3}$

**6.** $\lim_{x\to 0^+} x\, f(x) = \lim_{x\to 0^+} \frac{\sin x + ae^x + be^{-x} + c\ell n(1+x)}{x^2} = \frac{a+b-c}{2} = 0$

**Paragraph for Question Nos. 44 to 45**

Let $f(x) = \begin{cases} x+3 & ; -2 < x < 0 \\ 4 & ; x = 0 \\ 2x+5 & ; 0 < x < 1 \end{cases}$, then :

माना $f(x) = \begin{cases} x+3 & ; -2 < x < 0 \\ 4 & ; x = 0 \\ 2x+5 & ; 0 < x < 1 \end{cases}$, है तो

44. $\lim_{x\to 0^-} f([x - \tan x])$ is : (where [.] denotes greatest integer function)

(A) 2 (B) 4 (C) 5 (D) None of these

$\lim_{x\to 0^-} f([x-\tan x])$ का मान होगा : (जहां [.] महत्तम पूर्णांक फलन को दर्शाता है)

(A) 2 (B) 4 (C) 5 (D) इनमें से कोई नहीं

**Ans.** B

45. $\lim_{x\to 0^+} f\left(\left\{\frac{x}{\tan x}\right\}\right)$ is : (where {.}denotes fractional part of function)

(A) 4 (B) 5 (C) 7 (D) None of these

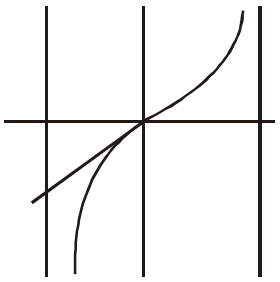
$\lim_{x\to 0^+} f\left(\left\{\frac{x}{\tan x}\right\}\right)$ का मान होगा : (जहां {.} भिन्नात्मक भाग फलन को दर्शाता है)

(A) 4 (B) 5 (C) 7 (D) कोई नहीं

**Ans.** C

**Sol.** **(44 – 45)**

$$f(x)=\begin{cases} x+3 & -2<x<0 \\ 4 & x=0 \\ 2x+5 & 0<x<1 \end{cases}$$



$\lim_{x\to 0^-} f([x-\tan x])$

$\because x \geq \tan x$

$\Rightarrow x-\tan x \geq 0$

$$x-\tan x=\begin{cases} <0 & \text{if} \quad x>0 \\ 0 & x=0 \\ >0 & \text{if} \quad x<0 \end{cases}$$

$$x-\tan x=\begin{cases} -1 & \text{if} \quad x>0 \\ 0 & x=0 \\ 0 & x<0 \end{cases}$$

$\Rightarrow \lim_{x\to 0^-} f(0)=4$

$\lim_{x\to 0^+} f\left(\left\{\frac{x}{\tan x}\right\}\right)=\lim_{x\to 0^-} 2\left\{\frac{x}{\tan x}\right\}+5$

$=\lim_{x\to 0^+} \frac{2x}{\tan x}+5=7$

## ONE OR MORE THAN ONE CORRECT TYPE

46. If $\underset{x\to 0}{\text{Limit}}\ \frac{\sin 2x + a\sin x}{x^3} = p$ (finite), then :

यदि $\underset{x\to 0}{\text{Limit}}\ \frac{\sin 2x + a\sin x}{x^3} = P$ (परिमित) हो, तो :

(A) $a = -2$ (B) $a = -1$ (C) $p = -2$ (D) $p = -1$

**Ans.** AD

**Sol.** $\underset{x\to 0}{\ell im}\ \frac{2\sin x\cos x + a\sin x}{x^3} = p$ (finite परिमित)

$\Rightarrow \quad \underset{x\to 0}{\ell im}\ \frac{\sin x}{x}\cdot\left(\frac{2\cos x + a}{x^2}\right) = p \quad \Rightarrow \quad \underset{x\to 0}{\ell im}\ \frac{\sin x}{x}\cdot\underset{x\to 0}{\ell im}\ \frac{2\cos x + a}{x^2} = p$

$\Rightarrow \quad \underset{x\to 0}{\ell im}\ \frac{2\cos x + a}{x^2} = p$

For $\left(\frac{0}{0}\right)$ form $\left(\frac{0}{0}\right)$ रूप के लिए

$2 + a = 0$

$a = -2$

$\Rightarrow \quad p = \underset{x\to 0}{\ell im}\ \frac{2(\cos x - 1)}{x^2} \quad \Rightarrow \quad p = \underset{x\to 0}{\ell im}\ \frac{-2\left(2\sin^2\frac{x}{2}\right)}{x^2} \quad \Rightarrow \quad p = -1$

47. Let $f(x) = \frac{|x + \pi|}{\sin x}$, then

(A) $f(-\pi^+) = -1$ (B) $f(-\pi^-) = 1$

(C) $\underset{x\to -\pi}{\text{Limit}}\ f(x)$ does not exists (D) $\underset{x\to \pi}{\text{Limit}}\ f(x)$ does not exists

माना $f(x) = \frac{|x + \pi|}{\sin x}$ हो, तो–

(A) $f(-\pi^+) = -1$ (B) $f(-\pi^-) = 1$

(C) $\underset{x\to -\pi}{\text{Limit}}\ f(x)$ विद्यमान नहीं है। (D) $\underset{x\to \pi}{\text{Limit}}\ f(x)$ विद्यमान नहीं है।

**Ans.** ABCD

**Sol.** $f(x) = \frac{|x + \pi|}{\sin x}$

(A) $f(-\pi^+) = \underset{h\to 0}{\ell im}\ \frac{|-\pi + h + \pi|}{\sin(-\pi + h)} = \underset{h\to 0}{\ell im}\ \frac{|h|}{-\sin h} = -1$

(B) $f(-\pi^-) = \underset{h\to 0}{\ell im}\ \frac{|-\pi - h + \pi|}{\sin(-\pi - h)} = \underset{h\to 0}{\ell im}\ \frac{|h|}{\sin h} = 1$

(C) $f(-\pi^+) \neq f(-\pi^-)$ so $\lim\limits_{x \to -\pi} f(x)$ does not exist

(D) for $\lim\limits_{x\to\pi} f(x)$

$$LHL = \lim_{x\to\pi^-} \frac{|x+\pi|}{\sin x} = \lim_{h\to 0} \frac{2\pi - h}{\sinh} = \frac{2\pi}{0} = \infty$$

$$RHL = \lim_{x\to\pi^+} \frac{|x+\pi|}{\sin x} = \lim_{h\to 0} \frac{2\pi + h}{-\sinh} = -\frac{2\pi}{0} = -\infty$$

$LHL \neq RHL$ so $\lim\limits_{x\to\pi} f(x)$ does not exist.

**Hindi.** $f(x) = \dfrac{|x+\pi|}{\sin x}$

(A) $f(-\pi^+) = \lim\limits_{h\to 0} \dfrac{|-\pi + h + \pi|}{\sin(-\pi + h)} = \lim\limits_{h\to 0} \dfrac{|h|}{-\sin h} = -1$

(B) $f(-\pi^-) = \lim\limits_{h\to 0} \dfrac{|-\pi - h + \pi|}{\sin(-\pi - h)} = \lim\limits_{h\to 0} \dfrac{|h|}{\sin h} = 1$

(C) $f(-\pi^+) \neq f(-\pi^-)$

अतः $\lim\limits_{x \to -\pi} f(x)$ विद्यमान नहीं है।

(D) $\lim\limits_{x \to \pi} f(x)$ के लिये

$$LHL = \lim_{x\to\pi^-} \frac{|x+\pi|}{\sin x} = \lim_{h\to 0} \frac{2\pi - h}{\sinh} = \frac{2\pi}{0} = \infty$$

$$RHL = \lim_{x\to\pi^+} \frac{|x+\pi|}{\sin x} = \lim_{h\to 0} \frac{2\pi + h}{-\sinh} = -\frac{2\pi}{0} = -\infty$$

$LHL \neq RHL$

अतः $\lim\limits_{x\to\pi} f(x)$ विद्यमान नहीं है।

48. $\lim\limits_{x\to\infty} \dfrac{(ax+1)^n}{x^n + A} =$

(A) $a^n$ if $n \in N$

(B) $\infty$ if $n \in Z^-$ & $a = A = 0$

(C) $\dfrac{1}{1+A}$ if $n = 0$

(D) $a^n$ if $n \in Z^-$, $A = 0$ & $a \neq 0$

$\lim\limits_{x\to\infty} \dfrac{(ax+1)^n}{x^n + A} =$

(A) $a^n$ यदि $n \in N$

(B) $\infty$ यदि $n \in Z^-$ तथा $a = A = 0$

(C) $\dfrac{1}{1+A}$ यदि $n = 0$

(D) $a^n$ यदि $n \in Z^-$, $A = 0$ तथा $a \neq 0$

**Ans.** ABCD

**Sol.** $\lim\limits_{x\to\infty}\dfrac{(ax+1)^n}{x^n+A}$

(A) If $n \in N$ $\Rightarrow$ $\lim\limits_{x\to\infty}\dfrac{\left(a+\dfrac{1}{x}\right)^n}{1+\dfrac{A}{x^n}} = \dfrac{(a+0)^n}{1+0} = a^n$

(B) If $n \in Z^-$ & $a = A = 0$ then $\lim\limits_{x\to\infty}\dfrac{(ax+1)^n}{x^n+A} = \lim\limits_{x\to\infty}\dfrac{1}{x^n} = \infty$ $n \in Z^-$

(C) If $n = 0$ then $\lim\limits_{x\to\infty}\dfrac{(ax+1)^n}{x^n+A} = \lim\limits_{x\to\infty}\dfrac{1}{1+A} = \dfrac{1}{1+A}$

(D) If $n \in Z^-, A = 0$ & $a \neq 0$ then $\lim\limits_{x\to\infty}\dfrac{(ax+1)^n}{x^n+A} = \lim\limits_{x\to\infty}\dfrac{(ax+1)^n}{x^n} = \lim\limits_{x\to\infty}\left(a+\dfrac{1}{x}\right)^n$

$= (a+0)^n = a^n$

**Hindi.** $\lim\limits_{x\to\infty}\dfrac{(ax+1)^n}{x^n+A}$

(A) यदि $n \in N$ $\Rightarrow$ $\lim\limits_{x\to\infty}\dfrac{\left(a+\dfrac{1}{x}\right)^n}{1+\dfrac{A}{x^n}} = \dfrac{(a+0)^n}{1+0} = a^n$

(B) यदि $n \in Z^-$ & $a = A = 0$

तब $\lim\limits_{x\to\infty}\dfrac{(ax+1)^n}{x^n+A} = \lim\limits_{x\to\infty}\dfrac{1}{x^n} = \infty$ $n \in Z^-$

(C) यदि $n = 0$

तब $\lim\limits_{x\to\infty}\dfrac{(ax+1)^n}{x^n+A} = \lim\limits_{x\to\infty}\dfrac{1}{1+A} = \dfrac{1}{1+A}$

(D) यदि $n \in Z^-, A = 0$ & $a \neq 0$

तब $\lim\limits_{x\to\infty}\dfrac{(ax+1)^n}{x^n+A} = \lim\limits_{x\to\infty}\dfrac{(ax+1)^n}{x^n} = \lim\limits_{x\to\infty}\left(a+\dfrac{1}{x}\right)^n = (a+0)^n = a^n$

49. Let $f(x) = \dfrac{\sqrt{x^2+2}}{3x-6}$ =

माना $f(x) = \dfrac{\sqrt{x^2+2}}{3x-6}$ हो, तो–

(A) $\lim\limits_{x\to-\infty} f(x) = -\frac{1}{3}$ (B) $\lim\limits_{x\to\infty} f(x) = \frac{1}{3}$

(C) $\lim\limits_{x\to-\infty} f(x) = \frac{1}{3}$ (D) $\lim\limits_{x\to\infty} f(x) = -\frac{1}{3}$

**Ans.** AB

**Sol.** $f(x) = \dfrac{\sqrt{x^2+2}}{3x-2} = \dfrac{|x|\sqrt{1+\frac{2}{x}}}{x\left(3-\frac{2}{x}\right)}$

$\Rightarrow f(x) = \dfrac{|x|}{x}\cdot\dfrac{\sqrt{1+\frac{2}{x}}}{\left(3-\frac{2}{x}\right)}$

$\Rightarrow \lim\limits_{x\to\infty} f(x) = \frac{1}{3}$ & $\lim\limits_{x\to\infty} f(x) = \frac{-1}{3}$

50. If $\ell = \underset{x\to0}{\text{Lim}} \dfrac{x(1+a\cos x)-b\sin x}{x^3} = \underset{x\to0}{\text{Lim}} \dfrac{1+a\cos x}{x^2} - \lim\limits_{x\to0} \dfrac{b\sin x}{x^3}$, where $\ell \in R$, then

(A) (a, b) = (–1, 0) (B) a & b are any real numbers

(C) $\ell = 0$ (D) $\ell = \frac{1}{2}$

यदि $\ell = \underset{x\to0}{\text{Lim}} \dfrac{x(1+a\cos x)-b\sin x}{x^3} = \underset{x\to0}{\text{Lim}} \dfrac{1+a\cos x}{x^2} - \underset{x\to0}{\text{Lim}} \dfrac{b\sin x}{x^3}$, जहाँ $\ell \in R$ हो, तो –

(A) (a, b) = (–1, 0) (B) a एवं b कोई भी वास्तविक संख्याएँ हैं।

(C) $\ell = 0$ (D) $\ell = \frac{1}{2}$

**Ans.** AD

**Sol.** $\ell = \lim\limits_{x\to0} \dfrac{x(1+a\cos x)-b\sin x}{x^3}$

$= \lim\limits_{x\to0} \dfrac{x\left(1+a\left(1-\frac{x^2}{\underline{|2}}+\frac{x^4}{\underline{|4}}+.....\right)\right)-b\left(x-\frac{x^3}{\underline{|3}}+\frac{x^5}{\underline{|5}}+.....\right)}{x^3}$

$= \lim\limits_{x\to0} \dfrac{x(1+a-b)+x^3\left(\frac{b}{\underline{|3}}-\frac{a}{\underline{|2}}\right)+....}{x^3}$

If $\ell$ is finite then $\ell = \frac{b}{\lfloor 3} - \frac{a}{\lfloor 2}$ & $a - b + 1 = 0$

If $\ell$ is not finite then $a - b + 1 \neq 0$

Now, $\ell = \lim_{x\to 0} \frac{1 + a\cos x}{x^2} - \lim_{x\to 0} \frac{b\sin x}{x^3} \Rightarrow \ell$ should be finite

$$= \lim_{x\to 0} \frac{1 + a\left(1 - \frac{x^2}{\lfloor 2} + \frac{x^4}{\lfloor 4} + .....\right)}{} - \lim_{x\to 0} \frac{b\left(x - \frac{x^3}{\lfloor 3} + .....\right)}{x^3}$$

$\downarrow$ $\downarrow$

should be finite $\Rightarrow 1 + a = 0$ should be finite $\Rightarrow b = 0$

$\Rightarrow a = -1$ $\Rightarrow \ell = \frac{1}{2}$

51. $\lim_{x\to c} f(x)$ does not exist when

$\lim_{x\to c} f(x)$ विद्यमान नहीं है जब

(A) $f(x) = [[x]] - [2x - 1], c = 3$ (B) $f(x) = [x] - x, c = 1$

(C) $f(x) = \{x\}^2 - \{-x\}^2, c = 0$ (D) $f(x) = \frac{\tan(\text{sgn}\, x)}{\text{sgn}\, x}, c = 0$

where [x] denotes step up function & {x} fractional part function.

जहां [x] महत्तम पूर्णांक फलन को तथा {x} भिन्नात्मक भाग फलन को दर्शाता है

**Ans.** BC

**Sol.** (A) $\lim_{x\to 3}\big[[x] - [2x - 1]\big]$ exists

as

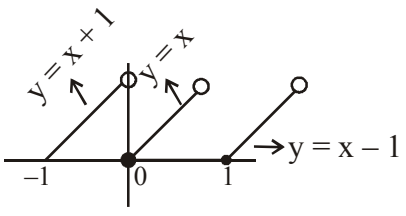
$\text{LHL} = \lim_{h\to 0}\big[[3 - h]\big] - [6 - 2h - 1] = 2 - 4 = -2$

$\text{RHL} = \lim_{h\to 0}\big[[3 + h]\big] - [6 + 2h - 1] = 3 - 5 = -2$

(B) $f(x) = [x] - x$

$\lim_{x\to 1} f(x) = -\lim_{x\to 1}\{x\}$

does not exist




(C) $\lim_{x\to 0}\{x\}^2 - \{-x\}^2$ does not exist as

$$LHL = \lim_{h\to 0}\{-h\}^2 - \{h\}^2 = \lim_{h\to 0}(-h+1)^2 - h^2$$

$= 1$

$$RHL = \lim_{h\to 0}\{h\}^2 - \{-h\}^2 = \lim_{h\to 0}\left(h^2 - (-h+1)^2\right)$$

(D) $\lim\limits_{x\to 0}\dfrac{\tan(\operatorname{sgn} x)}{\operatorname{sgn} x} = \tan 1$ exists

52. Which of the following limits vanish ?

निम्न में से किन सीमाओं के मान शून्य है –

(A) $\underset{x\to\infty}{\text{Limit}}\ x^{\frac{1}{4}}\sin\dfrac{1}{\sqrt{x}}$

(B) $\underset{x\to\pi/2}{\text{Limit}}\ (1-\sin x)\,.\tan x$

(C) $\underset{x\to\infty}{\text{Limit}}\ \dfrac{2x^2+3}{x^2+x-5}\,.\operatorname{sgn}(x)$

(D) $\underset{x\to 3^+}{\text{Limit}}\ \dfrac{[x]^2-9}{x^2-9}$

where [ ] denotes greatest integer function

जहां [ ] महत्तम पूर्णांक फलन है

**Ans.** ABD

**Sol.** $\lim\limits_{x\to 0}\dfrac{(1-\sin x)\sin x}{\cos x}\cdot\dfrac{1+\sin x}{1+\sin x}$

$\lim\limits_{x\to\pi/2}\dfrac{\sin x\cos x}{1+\sin x} = 0]$

53. If $\lim\limits_{x\to 0}(p\tan qx^2 - 3\cos^2 x + 4)^{1/(3x^2)} = e^{5/3}$; $p, q \in R$ then :

यदि $\lim\limits_{x\to 0}(p\tan qx^2 - 3\cos^2 x + 4)^{1/(3x^2)} = e^{5/3}$; $p, q \in R$ है तो –

(A) $p = \sqrt{2}, q = \dfrac{1}{2\sqrt{2}}$

(B) $p = \dfrac{1}{\sqrt{2}}, q = 2\sqrt{2}$

(C) $p = 1, q = 2$

(D) $p = 2, q = 4$

**Ans.** BC

**Sol.** $\lim\limits_{x\to 0}\left(p\tan qx^2 - 3\cos^2 x + 4\right)^{\frac{1}{3x^2}}$ $(1^\infty$ form)

$$= \exp\left(\lim_{x\to 0}\frac{\left(3\sin^2 x + p\tan qx^2\right)}{3x^2}\right)$$

$$= \exp\left(\lim_{x\to 0}\left(\frac{\sin^2 x}{x^2} + \frac{p}{3}\cdot q\cdot\frac{\tan qx^2}{qx^2}\right)\right)$$

$= e^{\frac{3+pq}{3}} = e^{\frac{5}{3}}$

$\Rightarrow pq = 2$

54. Let $f : R \to [-1, 1]$ be defined as $f(x) = \cos(\sin x)$, then which of the following is/are correct :

(A) f is periodic with fundamental period $2\pi$

(B) Range of f = [cos 1, 1]

(C) $\lim\limits_{x\to\frac{\pi}{2}}\left(f\left(\frac{\pi}{2}-x\right)+f\left(\frac{\pi}{2}+x\right)\right)=2$

(D) f is neither even nor odd function

माना $f : R \to [-1, 1]$ में फलन $f(x) = \cos(\sin x)$ परिभाषित है तो निम्न में से कौनसा/कौनसे सत्य होगें –

(A) f एक आवर्ती फलन है जिसका मूलभूत आवर्तकाल $2\pi$ है

(B) f का परिसर = [cos 1, 1]

(C) $\lim\limits_{x\to\frac{\pi}{2}}\left(f\left(\frac{\pi}{2}-x\right)+f\left(\frac{\pi}{2}+x\right)\right)=2$

(D) f ना तो सम और ना ही विषम फलन है

**Ans.** BC

**Sol.** $f : R \to [-1, 1]$

$f(x) = \cos(\sin x)$

(A) $f(x + \pi) = \cos(-\sin x) = f(x)$

(B) $\because -1 \le \sin x \le 1$

$\Rightarrow f(x) \in [\cos 1, 1]$

(C) $\lim\limits_{x\to\frac{\pi}{2}} 2\cos(\cos x) = 2$

(D) $f(-x) = f(x)$

55. If $\lim\limits_{x\to 0}\frac{(\sin(\sin x)-\sin x)}{ax^3+bx^5+c}=-\frac{1}{12}$, then :

यदि $\lim\limits_{x\to 0}\frac{(\sin(\sin x)-\sin x)}{ax^3+bx^5+c}=-\frac{1}{12}$ है तो –

(A) a = 2 (B) a = –2 (C) c = 0 (D) $b \in R$

**Ans.** ACD

**Sol.** $\lim\limits_{x\to 0}\frac{\sin(\sin x)-\sin x}{ax^3+bx^5+c}=-\frac{1}{12}$

$\Rightarrow c = 0$

$$\Rightarrow \lim_{x\to 0} \frac{\sin(\sin x) - \sin x}{x^3\left(a + bx^2\right)} = -\frac{1}{12}$$

$$\Rightarrow \lim_{x\to 0} \frac{\sin x - \frac{\sin^3 x}{\lfloor 3} + ..... - \sin x}{x^3\left(a + bx^2\right)} = -\frac{1}{12}$$

$$\Rightarrow \frac{-1}{6a} = \frac{-1}{12}$$

$\Rightarrow a = 2$ & $b \in R$

**MATCH THE COLUMN**

56. $\lim_{x\to 0} f(x)$, where $f(x)$ is as in column-I is :

| | **Column – I** | **Column – II** |
|---|---|---|
| (A) | $f(x) = \frac{\tan[e^2]x^2 - \tan[-e^2]x^2}{\sin^2 x}$ where [x] is the greatest integer function. | (P) $\sqrt{2}/8$ |
| (B) | $f(x) = \frac{[5/2 + \tan x + \tan^2 x] - [5/2]}{\tan x}$ where [x] is the greatest integer function. | (Q) 15 |
| (C) | $f(x)\ \frac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x + x^2}$ | (R) 0 |
| (D) | $f(x) = \frac{\sqrt{2} - \sqrt{1+\cos x}}{\sin^2 x}$ | (S) 1/2 |

$\lim_{x\to 0} f(x)$, जहाँ $f(x)$ स्तम्भ - **I** में है :

| | **स्तम्भ-I** | **स्तम्भ-II** |
|---|---|---|
| (A) | $f(x) = \frac{\tan[e^2]x^2 - \tan[-e^2]x^2}{\sin^2 x}$ जहाँ [ . ] महत्तम पूर्णांक फलन को दर्शाता है। | (P) $\sqrt{2}/8$ |
| (B) | $f(x) = \frac{[5/2 + \tan x + \tan^2 x] - [5/2]}{\tan x}$ जहाँ [ . ] महत्तम पूर्णांक फलन को दर्शाता है। | (Q) 15 |

(C) $f(x)\ \dfrac{\sqrt[3]{1+x^2}-\sqrt[4]{1-2x}}{x+x^2}$ (R) 0

(D) $f(x)=\dfrac{\sqrt{2}-\sqrt{1+\cos x}}{\sin^2 x}$ (S) 1/2

**Ans.** (A) Q; (B) R; (C) S; (D) P

**Sol.** (A) $\lim\limits_{x\to 0}\dfrac{\tan\left[e^2\right]x^2-\tan\left[-e^2\right]x^2}{\sin^2 x}$

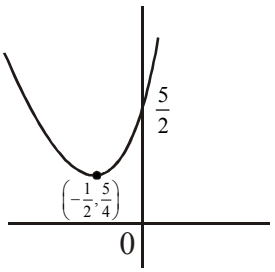
$=\lim\limits_{x\to 0}\dfrac{\tan 7x^2+\tan 8x^2}{\sin^2 x}$

$=\lim\limits_{x\to 0}\dfrac{\tan 7x^2}{x^2}+\dfrac{\tan 8x^2}{x^2}=15$

(B) $\lim\limits_{x\to 0}\dfrac{\left[\dfrac{5}{2}+\tan x+\tan^2 x\right]-\left[\dfrac{5}{2}\right]}{\tan x}=0$

$\because \tan^2 x+\tan x+\dfrac{5}{2}\in\left[\dfrac{9}{4},\infty\right)$

$\Rightarrow\left[\dfrac{5}{2}+\tan x+\tan^2 x\right]\in[2,\infty)$




(C) $\lim\limits_{x\to 0}\dfrac{\left(1+x^2\right)^{\frac{1}{3}}-(1-2x)^{\frac{1}{4}}}{x+x^2}$

$=\lim\limits_{x\to 0}\dfrac{1+\dfrac{1}{3}\left(x^2\right)+\dfrac{1}{3}\left(\dfrac{1}{3}-1\right)\dfrac{x^4}{\underline{|2}}+.....-\left(1-\dfrac{1}{4}(2x)+\dfrac{1}{4}\left(\dfrac{1}{4}-1\right)\dfrac{(2x)^2}{\underline{|2}}+.....\right)}{x(1+x)}$

$=+\dfrac{1}{2}$

(D) $\lim\limits_{x\to 0}\dfrac{\sqrt{2}-\sqrt{1+\cos x}}{\sin^2 x}=\lim\limits_{x\to 0}\dfrac{\sqrt{2}\left(1-\cos\dfrac{x}{2}\right)}{\sin^2 x}$

$$= \lim_{x\to 0} \frac{\sqrt{2}\cdot 2\sin^2\frac{x}{4}}{\sin^2 x}$$

$$= 2\sqrt{2}\lim_{x\to 0} \frac{\sin^2\frac{x}{4}}{\left(\frac{x}{4}\right)^2}\cdot\frac{x^2}{16\sin^2 x}$$

$$= \frac{\sqrt{2}}{8}$$

57. Match the following column :

| **Column-I** | **Column-II** |
|---|---|
| (A) $\lim_{x\to\infty}\left(\sqrt{x+\sqrt{x}}-\sqrt{x-\sqrt{x}}\right)$ equals | (P) – 2 |
| (B) The value of the limit, $\lim_{x\to 0}\frac{\sin 2x-2\tan x}{ln(1+x^3)}$ is | (Q) – 1 |
| (C) $\lim_{x\to 0^+}\left(ln\sin^3 x - ln(x^4+ex^3)\right)$ equals | (R) 0 |
| (D) Let $\tan(2\pi\lvert\sin\theta\rvert) = \cot(2\pi\lvert\cos\theta\rvert)$, where $\theta\in R$ and $f(x) = (\lvert\sin\theta\rvert+\lvert\cos\theta\rvert)^x$. The value of $\lim_{x\to\infty}\left[\frac{2}{f(x)}\right]$ equals (Here [ ] represents greatest integer function) | (S) 1 |

निम्न कॉलम को सुमेलित कीजिए :

| **Column-I** | **Column-II** |
|---|---|
| (A) $\lim_{x\to\infty}\left(\sqrt{x+\sqrt{x}}-\sqrt{x-\sqrt{x}}\right)$ का मान होगा | (P) – 2 |
| (B) सीमा $\lim_{x\to 0}\frac{\sin 2x-2\tan x}{ln(1+x^3)}$ का मान होगा | (Q) – 1 |
| (C) $\lim_{x\to 0^+}\left(ln\sin^3 x - ln(x^4+ex^3)\right)$ का मान होगा | (R) 0 |
| (D) माना कि $\tan(2\pi\lvert\sin\theta\rvert) = \cot(2\pi\lvert\cos\theta\rvert)$, जहाँ $\theta\in R$ तथा $f(x) = (\lvert\sin\theta\rvert+\lvert\cos\theta\rvert)^x$ तो $\lim_{x\to\infty}\left[\frac{2}{f(x)}\right]$ का मान होगा (जहाँ [ ] महत्तम पूर्णांक फलन है) | (S) 1 |

**Ans.** (A) S; (B) P ; (C) Q; (D) R

**Sol.** **(A)** $\lim\limits_{x\to\infty} \dfrac{(x+\sqrt{x})-(x-\sqrt{x})}{\sqrt{x+\sqrt{x}}+\sqrt{x-\sqrt{x}}} = \lim\limits_{x\to\infty} \dfrac{2\sqrt{x}}{\sqrt{x+\sqrt{x}}+\sqrt{x-\sqrt{x}}} = 1$ Ans.

(C) $\lim\limits_{x\to 0^+} \left(\ln \sin^3 x - \ln(x^4+ex^3)\right) = \lim\limits_{x\to 0} \ln\left(\dfrac{\sin^3 x}{x^3(x+e)}\right) = \ln\left(\dfrac{1}{e}\right) = -1$

(D) $\tan(2\pi|\sin\theta|) = \tan\left(\dfrac{\pi}{2} - |\cos\theta| 2\pi\right)$

$2\pi|\sin\theta| = n\pi + \dfrac{\pi}{2} - |\cos\theta| 2\pi$

$2\pi(|\sin\theta| + |\cos\theta|) = n\pi + \dfrac{\pi}{2}$

$|\sin\theta| + |\cos\theta| = \dfrac{n}{2} + \dfrac{1}{4}$ ....(1)

since $1 \le |\sin\theta| + |\cos\theta| \le \sqrt{2}$; $1 \le \dfrac{n}{2} + \dfrac{1}{4} \le \sqrt{2}$

$4 \le 2n + 1 \le 4\sqrt{2}$

$\dfrac{3}{2} \le n \le \dfrac{4\sqrt{2}-1}{2}$; Thus n = 2 is only possible value.

putting in (i) $|\sin\theta| + |\cos\theta| = \dfrac{5}{4}$

58. Match the following column :

निम्न कॉलम को सुमेलित कीजिए :

| **Column-I** | **Column-II** |
|---|---|
| (A) $\lim\limits_{x\to 1} \dfrac{\ln x}{x^4-1}$ | (P) 2 |
| (B) $\lim\limits_{x\to 0} \dfrac{3e^x - x^3 - 3x - 3}{\tan^2 x}$ | (Q) $\dfrac{2}{3}$ |
| (C) $\lim\limits_{x\to\infty} \dfrac{\pi - 2\tan^{-1} x}{\ln\left(1+\dfrac{1}{x}\right)}$ | (R) $\dfrac{3}{2}$ |
| (D) $\lim\limits_{x\to 0} \dfrac{2\sin x - \sin 2x}{x(\cos x - \cos 2x)}$ | (S) $\dfrac{1}{4}$ |
| (E) $\lim\limits_{x\to 0} \dfrac{e^x - e^{-x} - 2x}{x - \sin x}$ | |

**Ans.** (A) S; (B) R; (C) P; (D) Q ; (E) P

**Sol.** (A) $\lim\limits_{x\to1}\dfrac{\ln x}{x^4-1}=\lim\limits_{x\to1}\dfrac{\ln\left(1+\left(x-1\right)\right)}{\left(x-1\right)\left(x^3+x^2+x+1\right)}$

$=\dfrac{1}{4}$

(B) $\lim\limits_{x\to0}\dfrac{3e^x-x^3-3x-3}{\tan^2 x}=\lim\limits_{x\to0}\dfrac{3e^x-x^3-3x-3}{x^2}$

$=\lim\limits_{x\to0}\dfrac{3\left(1+x+\dfrac{x^2}{\underline{|2}}+\dfrac{x^3}{\underline{|3}}+.....\right)-x^3-3x-3}{x^2}$

$=\dfrac{3}{2}$

(C) $\lim\limits_{x\to\infty}\dfrac{\pi-2\tan^{-1}x}{\ln\left(1+\dfrac{1}{x}\right)}=\lim\limits_{x\to\infty}\dfrac{\pi-2\tan^{-1}x}{\dfrac{1}{x}}$

$=\lim\limits_{y\to0}\dfrac{\pi-2\tan^{-1}\left(\dfrac{1}{y}\right)}{y}$

(D) $\lim\limits_{x\to0}\dfrac{2\sin x\left(1-\cos x\right)}{x2\sin\left(\dfrac{3x}{2}\right)\sin\dfrac{x}{2}}$

$=2\lim\limits_{x\to0}\dfrac{2\sin^2\dfrac{x}{2}}{2\sin\dfrac{x}{2}\cdot\sin\dfrac{3x}{2}}=\dfrac{2}{3}\lim\limits_{x\to0}\dfrac{\sin\dfrac{x}{2}}{\dfrac{x}{2}}\cdot\dfrac{\dfrac{3x}{2}}{\sin\dfrac{3x}{2}}$

$=\dfrac{2}{3}$

(E) $\lim\limits_{x\to0}\dfrac{e^{2x}-1-2xe^x}{e^x\left(x-\sin x\right)}$

$$= \lim_{x\to 0} \frac{\left(1+2x+\frac{(2x)^2}{\underline{|2}}+....\right)-1-2x\left(1+x\frac{x^2}{\underline{|2}}+....\right)}{\left(x-\left(x-\frac{x^3}{\underline{|3}}+....\right)\right)}$$

$$= \lim_{x\to 0} \frac{x^3\left(\frac{8}{6}-1\right)}{x^3\left(\frac{1}{\underline{|3}}+....\right)} = \frac{2}{6}\times 6 = 2$$

1. The value of $\lim\limits_{x\to 0}\left(\dfrac{x}{\sqrt[8]{1-\sin x}-\sqrt[8]{1+\sin x}}\right)$ is equal to : **[JEE(Main)July2021]**

$\lim\limits_{x\to 0}\left(\dfrac{x}{\sqrt[8]{1-\sin x}-\sqrt[8]{1+\sin x}}\right)$ का मान बराबर है : **[JEE(Main)July2021]**

(A) 4 (B) – 4 (C) – 1 (D) 0

Ans. B

**Sol.** $\lim\limits_{n\to o}\left(\dfrac{x}{\sqrt[8]{1-\sin x}-\sqrt[8]{1+\sin x}}\right)$

$$\lim_{n\to o}\left(\frac{x}{(1-\sin x)^{\frac{1}{8}}-(1+\sin x)^{\frac{1}{8}}}\right)$$

$$=\lim_{n\to 0}\left(\frac{x}{\left(1-\frac{1}{8}\sin x+\frac{\frac{1}{8}\left(\frac{1}{8}-1\right)}{2!}\sin^2 x-\frac{\frac{1}{8}\left(\frac{1}{8}-1\right)\left(\frac{1}{8}-2\right)}{3!}\sin^3 x.....\right)-\left(1+\frac{1}{8}\sin x+\frac{\frac{1}{8}\left(\frac{1}{8}-1\right)}{2!}\sin^2 x+\frac{\frac{1}{8}\left(\frac{1}{8}-1\right)\left(\frac{1}{8}-2\right)}{3!}\sin^3 x+.....\right)}\right)$$

$$=\lim_{n\to o}\frac{x}{\frac{-2}{8}\sin x-\frac{2\cdot\frac{1}{8}\left(\frac{1}{8}-1\right)\left(\frac{1}{8}-2\right)}{3!}\sin^3 x.....}$$

$$\lim_{n\to o}\frac{x}{\frac{-2}{8}\sin x\left[1+\frac{\left(\frac{1}{8}-1\right)\left(\frac{1}{8}-2\right)}{3!}\sin^2 x+.....\right]}$$

$$=\frac{1}{\frac{-2}{8}\cdot[1+0]}=-4$$

2. If the value of $\lim_{x\to 0}(2-\cos x\sqrt{\cos 2x})^{\left(\frac{x+2}{x^2}\right)}$ is equal $e^a$, then a is equal to _____.

**[JEE(Main)July2021]**

यदि $\lim_{x\to 0}(2-\cos x\sqrt{\cos 2x})^{\left(\frac{x+2}{x^2}\right)}$ का मान $e^a$ है, तो a बराबर है _____। **[JEE(Main)July2021]**

**Ans. 3**

**Sol.** $1^\infty$ Form

$$\lim_{x\to 0}(2-\cos x\sqrt{\cos 2x})^{\frac{x+2}{x^2}}=e^{\lim_{x\to 0}\left(\frac{x+2}{x^2}\right)(1-\cos x\sqrt{\cos 2x})}$$

$$\text{So } a=\lim_{x\to 0}\frac{(x+2)}{x^2}(1-\cos x\sqrt{\cos 2x})$$

$$=\lim_{x\to 0}\frac{2}{x^2}\frac{(1-\cos^2 x\cos 2x)}{(1+\cos x\sqrt{\cos 2x})}$$

$$=\lim_{x\to 0}\frac{2(1-\cos^2 x(2\cos^2 x-1))}{2x^2}$$

$$=\lim_{x\to 0}\frac{1+\cos^2 x-2\cos^4 x}{x^2}$$

$$=\lim_{x\to 0}\frac{(1-\cos^2 x)(1+2\cos^2 x)}{x^2}$$

$$=\lim_{x\to 0}\frac{3\sin^2 x}{x^2}=3$$

3. If $\lim\limits_{x\to 0}\dfrac{axe^x-\beta\log_e(1+x)+\gamma x^2e^{-x}}{x\sin^2 x}=10$, $\alpha, \beta, \gamma \in R$ then the value of $\alpha+\beta+\gamma$ is _____.

**[JEE(Main)July2021]**

यदि $\lim\limits_{x\to 0}\dfrac{axe^x-\beta\log_e(1+x)+\gamma x^2e^{-x}}{x\sin^2 x}=10$, $\alpha, \beta, \gamma \in R$ है, तो $\alpha+\beta+\gamma$ का मान है _____ ।

**[JEE(Main)July2021]**

**Ans. 3**

**Sol.** $\lim\limits_{n\to 0}\dfrac{\alpha x\left(1+x+\dfrac{x^2}{2!}+.....\right)-\beta\left(x+\dfrac{x^2}{2}+\dfrac{x^3}{3}+.....\right)-\gamma x^2\left(1+x+\dfrac{x^2}{2!}-.....\right)}{x^3}$

$\alpha-\beta=0 \qquad \Rightarrow \alpha=\beta$

$\alpha+\dfrac{\beta}{2}+\gamma=0 \qquad \Rightarrow \gamma=\dfrac{-3\beta}{2}$

$\dfrac{\alpha}{2}-\dfrac{\beta}{3}-\gamma=10$

$\beta-6,\ \alpha=6,\ \gamma=-9$

$\alpha+\beta+\gamma=3\ \alpha$

4. $\lim\limits_{x\to 2}\left(\sum\limits_{n=1}^{9}\dfrac{x}{n(n+1)x^2+2(2n+1)x+4}\right)$ is equal to : **[JEE(Main)Aug.2021]**

$\lim\limits_{x\to 2}\left(\sum\limits_{n=1}^{9}\dfrac{x}{n(n+1)x^2+2(2n+1)x+4}\right)$ बराबर है : **[JEE(Main)Aug.2021]**

(A) $\dfrac{9}{44}$ (B) $\dfrac{5}{24}$ (C) $\dfrac{1}{5}$ (D) $\dfrac{7}{36}$

**Ans. A**

**Sol.** $\lim\limits_{x\to 2}\sum\limits_{x=1}^{9}\left(\dfrac{x}{x(x+1)x^2+2(2x+1)x+2}\right)$

$\lim\limits_{x\to 2}\sum\limits_{x=1}^{9}\left(\dfrac{x}{(hx+2)x^2+(x+1)x+2}\right)$

$$\lim_{x\to 2}\sum_{x=1}^{9}\left(\frac{1}{hx+2}-\frac{1}{(x+1)x+2}\right)$$

$$\Rightarrow \lim_{x\to 2}\left(\frac{1}{x+2}-\frac{1}{10x+2}\right)=\frac{9}{44}$$

5. If $\alpha, \beta$ are the distinct roots of $x^2 + bx + c = 0$, then $\lim\limits_{x\to\beta}\dfrac{e^{2(x^2+bx+c)}-1-2(x^2+bx+c)}{(x-\beta)^2}$ is equal to :

**[JEE(Main)Aug.2021]**

यदि $\alpha, \beta$ समीकरण $x^2 + bx + c = 0$ के दो भिन्न मूल हैं, तो $\lim\limits_{x\to\beta}\dfrac{e^{2(x^2+bx+c)}-1-2(x^2+bx+c)}{(x-\beta)^2}$ बराबर है :

**[JEE(Main)Aug.2021]**

(A) $2(b^2 + 4c)$ (B) $b^2 - 4c$ (C) $b^2 + 4c$ (D) $2(b^2 - 4c)$

**Ans. D**

**Sol.** $$\lim_{x\to\beta}\frac{\left\{1+2(x^2+bx+c)+\frac{\{2(x^2+bx+c)\}^2}{2!}+.....\right\}-1-2(x^2+bx+c)}{(x-\beta)^2}$$

$$\lim_{x\to\beta}\frac{2\left(x^2+bx+c\right)^2}{(x-\beta)^2}=\lim_{x\to\beta}\frac{2(x-\alpha)^2(x-\beta)^2}{(x-\beta)^2}$$

$$\Rightarrow 2(\beta-\alpha)^2=2\left((\alpha+\beta)^2-4\alpha\beta\right)$$

$$=2\left(b^2-4c\right)$$

6. If $\lim\limits_{x\to\infty}\left(\sqrt{x^2-x+1}-ax\right)=b$, then the ordered pair (a, b) is: **[JEE(Main)Aug.2021]**

यदि $\lim\limits_{x\to\infty}\left(\sqrt{x^2-x+1}-ax\right)=b$ है, तो क्रमित युग्म (a, b) है : **[JEE(Main)Aug.2021]**

(A) $\left(-1,\frac{1}{2}\right)$ (B) $\left(-1,-\frac{1}{2}\right)$ (C) $\left(1,\frac{1}{2}\right)$ (D) $\left(1,-\frac{1}{2}\right)$

**Ans. D**

**Sol.** $$\lim_{x\to\infty}\sqrt{x^2-x+1}-a\,x=b$$

$$\lim_{x\to\infty}\frac{x^2-x+1-a^2x^2}{\sqrt{x^2-x+1}+ax}=b$$

$$\lim_{x\to\infty}\frac{x^2(1-a^2)+1-x}{\sqrt{x^2-x+1}+ax}=b$$

If limit exits $a^2=1$

Then $a=\pm1$

$$\lim_{x\to\infty}\frac{1-x}{\sqrt{x^2-x+1}+ax}=b$$

$$\lim_{x\to\infty}\frac{\frac{1}{x}-1}{\sqrt{1-\frac{1}{x}+\frac{1}{x^2}}+a}=b$$

$$\frac{-1}{1+a}=b \qquad (\text{here } a\neq-1)$$

So possible values of b

$$-\frac{1}{2}=b$$

$$\left(1,-\frac{1}{2}\right)$$

7. $\lim_{x\to0}\frac{\sin^2(\pi\cos^4x)}{x^4}$ is equal to : **[JEE(Main)Aug.2021]**

$\lim_{x\to0}\frac{\sin^2(\pi\cos^4x)}{x^4}$ बराबर है : **[JEE(Main)Aug.2021]**

(A) $2\pi^2$ (B) $4\pi^2$ (C) $\pi^2$ (D) $4\pi$

**Ans. B**

**Sol.** $\lim_{x\to0}\frac{\sin^2(\pi-\pi\cos^4x)}{x^4}\left(\frac{0}{0}\right)$

$$\Rightarrow\lim_{x\to0}\frac{(\pi-\pi\cos^4x)^2}{x^4}$$

$$=\lim_{x\to0}\frac{\pi^2\sin^4x(1+\cos^2x)^2}{x^4}$$

$$=4\pi^2$$

8. If $\alpha = \lim\limits_{x\to\pi/4} \dfrac{\tan^3 x - \tan x}{\cos\left(x+\dfrac{\pi}{4}\right)}$ and $\beta = \lim\limits_{x\to 0}(\cos x)^{\cot x}$ are the roots of the equation, $ax^2 + bx - 4 = 0$, then the ordered pair (a, b) is : **[JEE(Main)Aug.2021]**

यदि समीकरण $ax^2 + bx - 4 = 0$ के मूल $\alpha = \lim\limits_{x\to\pi/4} \dfrac{\tan^3 x - \tan x}{\cos\left(x+\dfrac{\pi}{4}\right)}$ तथा $\beta = \lim\limits_{x\to 0}(\cos x)^{\cot x}$ हैं, तो क्रमित युग्म (a, b) है :

**[JEE(Main)Aug.2021]**

(A) (–1, –3) (B) (1, –3) (C) (1, 3) (D) (–1, 3)

**Ans.** **C**

**Sol.** $\alpha = \lim\limits_{x\to\frac{\pi}{4}} \dfrac{\tan^3 x - \tan x}{\cos\left(x+\dfrac{\pi}{4}\right)}$

$= \lim\limits_{x\to\frac{\pi}{4}} \dfrac{3\tan^2 x \times \sec^2 x - \sec^2 x}{-\sin\left(x+\dfrac{\pi}{4}\right)}$

$= \dfrac{3\times 1\times 2 - 2}{-1} = -4 = \alpha$

$\beta = \lim\limits_{x\to 0}(\cos x)^{\cot x}$

$= e^{\lim\limits_{x\to 0}\cot x(\cos x - 1)}$

$= e^{\lim\limits_{x\to 0} -\left(\frac{1-\cos x}{\tan x}\right)}$

$= e^{\lim\limits_{x\to 0} -x\left(\frac{1-\cos x}{x^2}\right) = 1 = \beta}$

$-\dfrac{b}{a} = -4 + 1$

$-\dfrac{b}{a} = -3 \Rightarrow \quad b = 3a$

$-\dfrac{4}{a} = -4 \times 1$

$a = 1$

$b = 3$

$(a, b) = (1, 3)$

9. $\lim\limits_{x\to\frac{\pi}{2}}\left(\tan^2 x\left(\left(2\sin^2 x+3\sin x+4\right)^{\frac{1}{2}}-\left(\sin^2 x+6\sin x+2\right)^{\frac{1}{2}}\right)\right)$ is equal to : **[JEE(Main)June2022]**

$\lim\limits_{x\to\frac{\pi}{2}}\left(\tan^2 x\left(\left(2\sin^2 x+3\sin x+4\right)^{\frac{1}{2}}-\left(\sin^2 x+6\sin x+2\right)^{\frac{1}{2}}\right)\right)$ बराबर है : **[JEE(Main)June2022]**

(A) $\frac{1}{12}$ (B) $-\frac{1}{18}$ (C) $-\frac{1}{12}$ (D) $\frac{1}{6}$

Ans. A

Sol. $\lim\limits_{x\to\frac{\pi}{2}}\tan^2 x\left(\sqrt{2\sin^2+3\sin x+4}-\sqrt{\sin^2 x+6\sin x+2}\right)$

$\lim\limits_{x\to\frac{\pi}{2}}\tan^2 x\left(\frac{\left(2\sin^2 x+3\sin x+4\right)-\left(\sin^2 x+6\sin xx+2\right)}{\sqrt{2\sin^2 x+3\sin x+14}+\sqrt{\sin^2 x+6\sin x+2}}\right)$

$\lim\limits_{x\to\frac{\pi}{2}}\frac{\tan^2 x.\left(\sin x-1\right)\left(\sin x-2\right)}{\left(3+3\right)}$

$\lim\limits_{x\to\frac{\pi}{2}}\frac{-1}{6}.\tan^2 x\left(\sin x-1\right)$

put $x=\frac{\pi}{2}-h$

$\lim\limits_{h\to 0}\frac{-1}{6}\cot^2 h\left(\cos h-1\right)$

$\lim\limits_{h\to 0}\frac{1}{6}\frac{\left(1-\cos h\right)}{\tan^2 h}$

$\lim\limits_{h\to 0}\frac{1}{6}\frac{\left(1-\cos h\right)}{h^2}=\frac{1}{6}.\frac{1}{2}=\frac{1}{12}$

10. $\lim\limits_{x\to\frac{1}{\sqrt{2}}}\frac{\sin\left(\cos^{-1}x\right)-x}{1-\tan\left(\cos^{-1}x\right)}$ is equal to : **[JEE(Main)June2022]**

$\lim\limits_{x\to\frac{1}{\sqrt{2}}}\frac{\sin\left(\cos^{-1}x\right)-x}{1-\tan\left(\cos^{-1}x\right)}$ बराबर है : **[JEE(Main)June2022]**

(A) $\sqrt{2}$ (B) $-\sqrt{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $-\frac{1}{\sqrt{2}}$

Ans. D

Sol. $\lim\limits_{x\to\frac{1}{\sqrt{2}}}\frac{\sin\left(\cos^{-1}x\right)-x}{1-\tan\left(\cos^{-1}x\right)}$

$$\lim_{x\to\frac{1}{\sqrt{2}}}\frac{\sin\left(\sin^{-1}\sqrt{1-x^2}\right)-x}{1-\tan\left(\tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)\right)}$$

$$\lim_{x\to\frac{1}{\sqrt{2}}}\frac{\sqrt{1-x^2}-x}{1-\left(\frac{\sqrt{1-x^2}}{x}\right)}$$

$$\lim_{x\to\frac{1}{\sqrt{2}}}(-x)=-\frac{1}{\sqrt{2}}$$

11. $\lim_{x\to0}\frac{\cos(\sin x)-\cos x}{x^4}$ is equal to : **[JEE(Main)June2022]**

$\lim_{x\to0}\frac{\cos(\sin x)-\cos x}{x^4}$ बराबर है : **[JEE(Main)June2022]**

(A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{1}{6}$ (D) $\frac{1}{12}$

Ans. C

Sol. $\lim_{x\to0}\frac{\cos(\sin x)-\cos x}{x^4}$

$$\lim_{x\to0}\frac{-2\sin\left(\frac{x+\sin x}{2}\right)\sin\left(\frac{\sin x-x}{2}\right)}{x^4}$$

$$=-2\lim_{x\to0}\frac{\sin\left(\frac{\sin x+x}{2}\right)}{\left(\frac{\sin x+x}{2}\right)}\cdot\frac{\sin\left(\frac{\sin x-x}{2}\right)}{\left(\frac{\sin x+x}{2}\right)}\frac{\left(\sin^2x-x^2\right)}{4x^2}$$

$$=\frac{-2}{4}\lim_{x\to0}\left(\frac{\sin^2x-x^2}{x^4}\right);\left(\frac{0}{0}\text{form}\right)\text{L 'Hospitla rule}$$

$$=\frac{-1}{2}\lim_{x\to0}\left(\frac{\sin2x-2x}{4x^3}\right);\left(\frac{0}{0}\text{form}\right)\text{L 'Hospitla rule}$$

$$=\frac{-1}{2}\lim_{x\to0}\left(\frac{2\cos2x-2}{12x^2}\right)$$

$$= \frac{-1}{24}\lim_{x\to 0}\left(\frac{2-4\sin^2 x-2}{x^2}\right)$$

$$= \frac{4}{24}\lim_{x\to 0}\left(\frac{\sin^2 x}{x^2}\right)$$

$$= \frac{1}{6}$$

12. Let a be an integer such that $\lim_{x\to 7}\frac{18-[1-x]}{[x-3a]}$ exists, where [t] is greatest integer $\le$ t. Then a is equal to :

**[JEE(Main)June2022]**

माना a एक पूर्णांक है जिसके लिए $\lim_{x\to 7}\frac{18-[1-x]}{[x-3a]}$, जहाँ [t] महत्तम पूर्णांक $\le$ t है, का अस्तित्व है, तो a बराबर है :

**[JEE(Main)June2022]**

(A) $-6$ (B) $-2$ (C) 2 (D) 6

Ans. A

Sol. $\lim_{x\to 7}\frac{18-[1-x]}{[x-3a]}$

$\lim_{x\to 7}\frac{18-1-[x]}{[x]-3a}$

$\lim_{x\to 7}\frac{17-[-x]}{[x]-3a}$

$\text{R.H.L.} = \frac{17-(-7)}{6-3a} = \frac{24}{6-3a}$

$\text{L.H.L.} = \frac{17-(-8)}{7-3a} = \frac{25}{7-3a}$

L.H.L. = R.H.L.

$\frac{24}{6-3a} = \frac{25}{7-3a}$

$3a = -18$

$a = -6$

13. The value of $\lim_{n\to\infty} 6\tan\left\{\sum_{r=1}^{n}\tan^{-1}\left(\frac{1}{r^2+3r+3}\right)\right.$ is equal to : **[JEE(Main)June2022]**

$\lim_{n\to\infty} 6\tan\left\{\sum_{r=1}^{n}\tan^{-1}\left(\frac{1}{r^2+3r+3}\right)\right.$ का मान बराबर है : **[JEE(Main)June2022]**

(A) 1 (B) 2 (C) 3 (D) 6

Ans. C

Sol. $6\lim\limits_{n\to\infty}\tan\left\{\sum\limits_{r=1}^{n}\tan^{-1}\left(\frac{1}{r^2+3r+3}\right)\right\}$

$=6\lim\limits_{n\to\infty}\tan\left\{\sum\limits_{r=1}^{n}\tan^{-1}\left[\frac{1}{1+\left(r^2+3r+2\right)}\right]\right\}$

$=6\lim\limits_{n\to\infty}\tan\left[\sum\limits_{r=1}^{n}\tan^{-1}\left[\frac{1}{1+(r+2)(r+1)}\right]\right]$

$=6\lim\limits_{n\to\infty}\tan\left[\sum\limits_{r=1}^{n}\tan^{-1}\left[\frac{(r+2)-(r+1)}{1+(r+2)(r+1)}\right]\right]$

$=6\lim\limits_{n\to\infty}\tan\left[\sum\limits_{r=1}^{n}\left(\tan^{-1}(r+2)-\tan^{-1}(r+1)\right)\right]$

$=6\lim\limits_{n\to\infty}\tan\left[\left(\tan^{-1}(n+2)-\tan^{-1}2\right)\right]$

$=6\tan\left[\lim\limits_{n\to\infty}\tan^{-1}(n+2)-\lim\limits_{n\to\infty}\tan^{-1}(2)\right]$

$=6\tan\left[\tan^{-1}(\infty)-\tan^{-1}2\right]=6\tan\left[\frac{\pi}{2}-\tan^{-1}2\right]$

$=\tan\left(\cot^{-2}2\right)$

$=6\tan\left(\tan^{-1}\frac{1}{2}\right)$

$=6\times\frac{1}{2}=3$

14. The value of $\lim\limits_{x\to1}\frac{\left(x^2-1\right)\sin^2(\pi x)}{x^4-2x^3+2x-1}$ is equal to : **[JEE(Main)June2022]**

$\lim\limits_{x\to1}\frac{\left(x^2-1\right)\sin^2(\pi x)}{x^4-2x^3+2x-1}$ का मान बराबर है : **[JEE(Main)June2022]**

(A) $\frac{\pi^2}{6}$ (B) $\frac{\pi^2}{3}$ (C) $\frac{\pi^2}{2}$ (D) $\pi^2$

Ans. D

Sol. $\lim\limits_{x\to1}\frac{\left(x^2-1\right)\sin^2(\pi x)}{x^4-2x^3+2x-1}$

$$\lim_{x\to 1}\frac{(x-1)(x+1)\sin^2(\pi x)}{(x-1)^3(x+1)}$$

$$\lim_{x\to 1}\frac{\sin^2(\pi x)}{(x-1)^2}$$

Let $x-1\to t$ $\qquad\qquad \lim_{x\to 0}\frac{\sin^2(\pi(\pi x)}{t^2}$

$x-1\to 0$

$$\Rightarrow \lim_{x\to 0}\frac{\sin^2(t)}{n^2t^2}\pi^2=\pi^2$$

15. Let [t] denote the greatest integer $\le t$ and {t} denote the fractional part of t. The integral value of $\alpha$ for which the left hand limit of the function $f(x)=[1+x]+\frac{\alpha^{2[x]+\{x\}}+[x]-1}{2[x]+\{x\}}$ at $x=0$ is equal to $\alpha-\frac{4}{3}$, is _____.

**[JEE(Main)June2022]**

माना [t] महत्तम पूर्णांक $\le$ t है, तथा t का भिन्नात्मक भाग {t} है। $\alpha$ का वह पूर्णांक मान, जिसके लिए फलन $f(x)=[1+x]+\frac{\alpha^{2[x]+\{x\}}+[x]-1}{2[x]+\{x\}}$ की $x=0$ पर बाएँ पक्ष की सीमा $\alpha-\frac{4}{3}$ है, है _____

**[JEE(Main)June2022]**

Ans. 3

Sol. $\lim_{x\to 0}[1+x]+\frac{\alpha^{2[x]+\{x\}}+[x]-1}{2[x]+\{x\}}$

$$\lim_{x\to 0^-}0+\frac{\alpha^{-2+1}-1-1}{-2+1}$$

$$\frac{\alpha^{-1}-2}{-1}=\alpha-\frac{4}{3}$$

$$2-\frac{1}{\alpha}=\alpha-\frac{4}{3}$$

$$\alpha+\frac{1}{\alpha}=\frac{10}{3}$$

$$3\alpha^2-10\alpha+3=0$$

$\alpha=3,\frac{1}{3}$ $\qquad$ Integral value $=3$

16. If $\lim\limits_{x\to 1}\dfrac{\sin\left(3x^2-4x+1\right)-x^2+1}{2x^3-7x^2+ax+b}=-2$, then the value of $(a-b)$ is equal to _____.

**[JEE(Main)June2022]**

यदि $\lim\limits_{x\to 1}\dfrac{\sin\left(3x^2-4x+1\right)-x^2+1}{2x^3-7x^2+ax+b}=-2$ है, तो $(a-b)$ का मान बराबर है _____

**[JEE(Main)June2022]**

Ans. 11

Sol. $\lim\limits_{x\to 1}\dfrac{\sin\left(3x^2-4x+1\right)-x^2+1}{2x^3-7x^2+ax+b}=-2$

this will be $\dfrac{0}{0}$ form for finite answer denominator must be zero.

$\because\ 2-7+a+b=0$

$\therefore\ a+b=5$ _____(1)

As this will be $\dfrac{0}{0}$ form use De LHoptal Rule,

$$\lim_{x\to 1}\frac{\cos\left(3x^2-4x+1\right)(6x-4)-2x}{6x^2-14x+a}$$

Again for finite answer denominator must be zero.

$\therefore\ a=8$

from eq. (1)

$b=-3$

required $= a-b=8-(-3)$

$=8+3=11$

17. If $\lim\limits_{n\to\infty}\left(\sqrt{n^2-n-1}+n\alpha+\beta\right)=0$, then $8(\alpha+\beta)$ is equal to : **[JEE(Main)July2022]**

यदि $\lim\limits_{n\to\infty}\left(\sqrt{n^2-n-1}+n\alpha+\beta\right)=0$ है, तो $8(\alpha+\beta)$ बराबर है : **[JEE(Main)July2022]**

(A) 4 (B) –8 (C) –4 (D) 8

Ans. C

Sol. $\lim\limits_{n\to\infty}\left(\sqrt{n^2-n-1}+n\alpha+\beta\right)=0$

$$=\lim_{n\to\infty}\left[\sqrt{1-\frac{1}{n}-\frac{1}{n^2}}+\alpha+\frac{\beta}{n}\right]=0$$

$\therefore\ \alpha=-1$

Now,

$$\lim_{n\to\infty} n\left[\left\{1-\left(\frac{1}{n}+\frac{1}{n^2}\right)\right\}^{1/2}+\frac{\beta}{n}-1\right]=0$$

$$=\lim_{n\to\infty}\frac{\left(1-\frac{1}{2}\left(\frac{1}{n}+\frac{1}{n^2}\right)+.....\right)+\frac{\beta}{n}-1}{\frac{1}{n}}=0$$

$$\Rightarrow \beta-\frac{1}{2}=0$$

$$\therefore \beta=\frac{1}{2}$$

Now, $8(\alpha+\beta)=8\left(-\frac{1}{2}\right)=-4$

18. $\lim\limits_{x\to\frac{\pi}{4}}\frac{8\sqrt{2}-(\cos x+\sin x)^7}{\sqrt{2}-\sqrt{2}\sin 2x}$ is equal to : **[JEE(Main)July2022]**

$\lim\limits_{x\to\frac{\pi}{4}}\frac{8\sqrt{2}-(\cos x+\sin x)^7}{\sqrt{2}-\sqrt{2}\sin 2x}$ बराबर है : **[JEE(Main)July2022]**

(A) 14 (B) 7 (C) $14\sqrt{2}$ (D) $7\sqrt{2}$

Ans. A

Sol. $\lim\limits_{x\to\frac{\pi}{4}}\frac{8\sqrt{2}-\left(\sqrt{2}\right)^7\left(\cos\left(x-\frac{\pi}{4}\right)\right)^7}{\sqrt{2}(1-\sin 2x)}$

$$x=\frac{\pi}{4}+h$$

$$\lim_{h\to 0}\frac{8\sqrt{2}\left(1-\cos^7 h\right)}{\sqrt{2}(1-\cos 2h)}$$

L–Hospital

$$\lim_{h\to 0}8\left(\frac{-7\cos^6 h(-\sinh)}{\sin 2h.2}\right)$$

$$\lim_{h\to 0}\frac{8\times 7\sinh\cos^6 h}{4\sin\cosh}$$

$$= \lim_{h\to 0} \frac{8\times 7}{4}\cos^5 = 14$$

19. Let $f : R \to R$ be a continuous function such that $f(3x) - f(x) = x$. If $f(8) = 7$, then $f(14)$ is equal to : **[JEE(Main)July2022]**

माना $f : R \to R$ एक संतत फलन है जिसके लिए $f(3x) - f(x) = x$ है। यदि $f(8) = 7$ है, तो $f(14)$ बराबर है : **[JEE(Main)July2022]**

(A) 4 (B) 10 (C) 11 (D) 16

Ans. B

Sol. $f(x) - f(x/3) = x/3$

$f(x/3) - f(x/3^2) = x/3^2$

.... on adding

$$f(x) - \lim_{n\to\infty}\left(\frac{x}{3^n}\right) = x\left(\frac{1}{3} + \frac{1}{3^2}....\infty\right)$$

$$f(x) - f(0) = \frac{x}{2}$$

$f(8) = 7$ ; $f(0) = 3$

$f(x) = x/2 + 3$

$f(14) = 10$

20. Let $\beta = \lim_{x\to 0} \frac{\alpha x - \left(e^{3x} - 1\right)}{\alpha x\left(e^{3x} - 1\right)}$ for some $\alpha \in R$. Then the value of $\alpha + \beta$ is : **[JEE(Main)July2022]**

माना किसी $\alpha \in R$ के लिए $\beta = \lim_{x\to 0} \frac{\alpha x - \left(e^{3x} - 1\right)}{\alpha x\left(e^{3x} - 1\right)}$ है। तो $\alpha + \beta$ का मान है : **[JEE(Main)July2022]**

(A) $\frac{14}{5}$ (B) $\frac{3}{2}$ (C) $\frac{5}{2}$ (D) $\frac{7}{2}$

Ans. C

Sol. $$\beta = \lim_{x\to 0} \frac{\alpha x - \left(e^{3x} - 1\right)}{\alpha x\left(e^{3x} - 1\right)}$$

$$\beta = \lim_{x\to 0} \frac{1 + \alpha x - \left[1 + 3x + \frac{9x^2}{2!} + ....\right]}{(\alpha x)\frac{\left(e^{3x} - 1\right)}{3x}3x}$$

$$\beta = \lim_{x\to 0} \frac{(\alpha x - 3x) - \frac{9x^2}{2!}}{3\alpha x^2}$$

For existence of limit $\alpha - 3 = 0$

$\alpha = 3$

Limit $\beta = \frac{-3}{2\alpha}$

$$\beta = -\frac{1}{2}$$

Now,

$$\alpha + \beta = \frac{5}{2}$$

21. If $\lim_{x\to 0} \frac{\alpha e^x + \beta e^{-x} + \gamma \sin x}{x \sin^2 x} = \frac{2}{3}$, where $\alpha, \gamma \in R$, then which of the following is NOT correct?

**[JEE(Main)July2022]**

यदि $\lim_{x\to 0} \frac{\alpha e^x + \beta e^{-x} + \gamma \sin x}{x \sin^2 x} = \frac{2}{3}$ है, जहाँ $\alpha, \gamma \in R$ हैं, तो निम्न में से कौन सा सही नहीं है?

**[JEE(Main)July2022]**

(A) $\alpha^2 + \beta^2 + \gamma^2 = 6$ (B) $\alpha\beta + \beta\gamma + \gamma\alpha + 1 = 0$

(C) $\alpha\beta^2 + \beta\gamma^2 + \gamma\alpha^2 + 3 = 6$ (D) $\alpha^2 - \beta^2 + \gamma^2 = 4$

Ans. C

Sol. $$\lim_{x\to 0} \frac{\alpha\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}\right) + \beta\left(1 - x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\right) + \gamma\left(x - \frac{x^3}{3!} + \ldots\right)}{x^3}$$

constant terms should be zero

$\Rightarrow \alpha + \beta = 0$

coeff of x should be zero

$\Rightarrow \alpha - \beta + \gamma = 0$

coeff of $x^2$ should be zero

$$\lim_{x\to 0} \frac{x^3\left(\frac{\alpha}{3!} - \frac{\beta}{3!} - \frac{\gamma}{3!}\right) + x^4\left(\frac{\alpha}{3!} - \frac{\beta}{3!} - \frac{\gamma}{3!}\right)}{x^3} = \frac{2}{3}$$

$$\Rightarrow \frac{\alpha}{2} + \frac{\beta}{2} = 0$$

$$\frac{\alpha}{6} - \frac{\beta}{6} - \frac{\gamma}{6} = 2/3$$

$\Rightarrow \alpha = 1, \beta = -1, \gamma = -2$

22. $\lim\limits_{x\to 0}\left(\dfrac{(x+2\cos x)^3+2(x+2\cos x)^2+3\sin(x+2\cos x)}{(x+2)^3+2(x+2)^2+3\sin(x+2)}\right)^{\frac{100}{x}}$ is equal to _____.

**[JEE(Main)July2022]**

$\lim\limits_{x\to 0}\left(\dfrac{(x+2\cos x)^3+2(x+2\cos x)^2+3\sin(x+2\cos x)}{(x+2)^3+2(x+2)^2+3\sin(x+2)}\right)^{\frac{100}{x}}$ बराबर है_____

**[JEE(Main)July2022]**

Ans. 1

Sol. $\lim\limits_{x\to 0}\left(\dfrac{(x+2\cos x)^3+2(x+2\cos x)^2+3\sin(x+2\cos x)}{(x+2)^3+2(x+2)^2+3\sin(x+2)}\right)^{\frac{100}{x}}$

Form $1^\infty$

$= e^{\lim\limits_{x\to 0}\left[\left(\frac{(x+2\cos x)^3+2(x+2\cos x)^2+3\sin(x+2\cos x)}{(x+2)^3+2(x+2)^2+3\sin(x+2)}\right)-1\right]\times\frac{100}{x}}$

$= e^{\lim\limits_{x\to 0}\left[\frac{100}{x}\left(\frac{(x+2\cos x)^3+2(x+2\cos x)^2+3\sin(x+2\cos x)-\left((x+2)^3+2(x+2)^2+3\sin(x+2)\right)}{(x+2)^3+2(x+2)^2+3\sin(x+2)}\right)\right]}$

$= e^{\lim\limits_{x\to 0}\frac{100}{x}\left[\left(\frac{(x+2\cos x)^3+(x+2)^3+2(x+2\cos x)^2-2(x+2)^2+3\sin(x+2\cos x)-3\sin(x+2)}{8+8+3\sin^2}\right)\right]}$

$= e^{\frac{100}{16+3\sin^2}\lim\limits_{x\to 0}\frac{3(x+2\cos x)^2\times(1+2\sin x)-3(x+2)^2-4(x+2\cos x)}{x(1-2\sin x)-4(x+2)+3\cos(x+2\cos x)\times(1-2\sin x)-3\cos(x+2)}}$

$= e^{\frac{100}{16+3\sin 2}\left(\frac{12-3(4)+8\times 1-8+3\cos 2-3\cos 2}{1}\right)}$

Using L'H rule.

$= e^0 = 1$

## Exercise J-A

1. Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation $\left(\sqrt[3]{1+a}-1\right)x^2+\left(\sqrt{1+a}-1\right)x+\left(\sqrt[6]{1+a}-1\right)=0$ Where $a > -1$, then $\lim\limits_{a\to 0^+}\alpha(a)$ and $\lim\limits_{a\to 0^+}\beta(a)$ are : **[JEE-2012]**

माना समीकरण $\left(\sqrt[3]{1+a}-1\right)x^2+\left(\sqrt{1+a}-1\right)x+\left(\sqrt[6]{1+a}-1\right)=0$, जहाँ $a>-1$, के मूल $\alpha(a)$ तथा $\beta(a)$ है, तो $\lim\limits_{a\to 0^+}\alpha(a)$ तथा $\lim\limits_{a\to 0^+}\beta(a)$ का मान होगा : **[JEE-2012]**

(A) $-\frac{5}{2}\&1$ (B) $-\frac{1}{2}\&-1$ (C) $-\frac{7}{2}\&2$ (D) $-\frac{9}{2}\&3$

**Ans.** B

**Sol.** $\left(\left(1+\frac{a}{3}\right)-1\right)x^2+\left(\left(1+\frac{a}{2}\right)-1\right)x+\left(1+\frac{a}{6}-1\right)=0$

$a\left(\frac{x^2}{3}+\frac{x}{2}+\frac{1}{6}\right)=0 \Rightarrow 2x^2+3x+1=0$

$\Rightarrow x=-\frac{1}{2},-1$

$\Rightarrow \lim\limits_{a\to 0^+}\alpha(a)$ and $\lim\limits_{a\to 0^+}\beta(a)$ are $-\frac{1}{2}$ and $-1$

2. If $\lim\limits_{x\to\infty}\left(\frac{x^2+x+1}{x+1}-ax-b\right)=4$ then : **[JEE-2012]**

यदि $\lim\limits_{x\to\infty}\left(\frac{x^2+x+1}{x+1}-ax-b\right)=4$ तो : **[JEE-2012]**

(A) $a=1, b=4$ (B) $a=1, b=-4$ (C) $a=2, b=-3$ (D) $a=2, b=3$

**Ans.** B

**Sol.** $\lim\limits_{x\to\infty}\left(\frac{x^2+x+1}{x+1}-ax-b\right)=4$

$\lim\limits_{x\to\infty}\left(\frac{x^2(1-a)+x(1-a-b)+(1-b)}{x+1}\right)=4$

Limit is finite

It exists when $1-a=0 \quad \Rightarrow a=1$

then $\lim\limits_{x\to\infty}\left(\frac{1-a-b+\frac{1-b}{x}}{1+\frac{1}{x}}\right)=4$

$\therefore \quad 1 - a - b = 4 \quad \Rightarrow \quad b = -4$

3. The largest value of the nonnegative integer a for which $\lim\limits_{x\to 1}\left\{\dfrac{-ax+\sin(x-1)+a}{x+\sin(x-1)-1}\right\}^{\frac{1-x}{1-\sqrt{x}}} = \dfrac{1}{4}$ is :

**[JEE(Adv.)2014]**

a का अधिकतम अऋणात्मक पूर्णांक मान ज्ञात कीजिये जिसके लिये $\lim\limits_{x\to 1}\left\{\dfrac{-ax+\sin(x-1)+a}{x+\sin(x-1)-1}\right\}^{\frac{1-x}{1-\sqrt{x}}} = \dfrac{1}{4}$ होगा :

**[JEE(Adv.)2014]**

**Ans.** Zero

**Sol.** $\lim\limits_{x\to 1}\left\{\dfrac{-ax+\sin(x-1)+a}{x+\sin(x-1)-1}\right\}^{\frac{1-x}{1-\sqrt{x}}} = \dfrac{1}{4}$

$\Rightarrow \left\{\dfrac{-ax+\sin(x-1)+a}{x+\sin(x-1)-1}\right\}^{x+\sqrt{x}} = \dfrac{1}{4}$

Hence $\left\{\dfrac{-ax+\sin(x-1)+a}{(x-1)+\sin(x-1)}\right\}^{1+\sqrt{x}} = \dfrac{1}{4}$

put $x = 1 + h$,

$\lim\limits_{h\to 0}\left\{\dfrac{-ah+\sinh}{h+\sinh}\right\}^{1+\sqrt{1+h}} = \dfrac{1}{4}$

or $\quad \dfrac{-a+1}{2} = \dfrac{1}{2} \quad$ or $-\dfrac{1}{2} \quad \Rightarrow \quad a = 0 \quad$ or 2

But at $a = 2$, $\dfrac{-ah+\sinh}{h+\sinh}$ tends to negative value

So correct Answer is $a = 0$

However $a = 2$ may be accepted if this is not considered

4. Let $f(x) = \sin\left(\dfrac{\pi}{6}\sin\left(\dfrac{\pi}{2}\sin x\right)\right)$ for all $x \in R$ and $g(x) = \dfrac{\pi}{2}\sin x$ for all $x \in R$. Let $(f \circ g)(x)$ denote $f(g(x))$ and $(g \circ f)(x)$ denote $g(f(x))$. Then which of the following is (are) true? **[JEE(Adv.)2015]**

(A) Range of f is $\left[-\dfrac{1}{2}, \dfrac{1}{2}\right]$

(B) Range of $f \circ g$ is $\left[-\dfrac{1}{2}, \dfrac{1}{2}\right]$

(C) $\lim\limits_{x\to 0}\dfrac{f(x)}{g(x)} = \dfrac{\pi}{6}$

(D) There is an $x \in R$ such that $(g \circ f)(x) = 1$

माना कि सभी $x \in R$ के लिए $f(x) = \sin\left(\frac{\pi}{6}\sin\left(\frac{\pi}{2}\sin x\right)\right)$ और सभी $x \in R$ के लिए $g(x) = \frac{\pi}{2}\sin x$ है। माना कि $(f \circ g)(x)$ और $(g \circ f)(x)$ क्रमशः $f(g(x))$ और $g(f(x))$ को दर्शाते हैं, तब निम्नलिखित में से कौन सही है ? **[JEE(Adv.)2015]**

(A) f का परिसर $\left[-\frac{1}{2}, \frac{1}{2}\right]$ है

(B) $f \circ g$ का परिसर $\left[-\frac{1}{2}, \frac{1}{2}\right]$ है

(C) $\lim_{x\to 0}\frac{f(x)}{g(x)} = \frac{\pi}{6}$

(D) R में एक x ऐसा है जिसके लिए $(g \circ f)(x) = 1$

**Ans.** ABC

**Sol.** $f(x) = \sin\left(\frac{\pi}{6}\sin\left(\frac{\pi}{2}\sin x\right)\right)$

Let $\frac{\pi}{2}\sin x = \theta \qquad \because \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\therefore \qquad f(x) = \sin\left(\frac{\pi}{6}\sin\theta\right)$

Let $\frac{\pi}{6}\sin\theta = \phi \therefore \qquad \phi \in \left[-\frac{\pi}{6}, \frac{\pi}{6}\right]$

$\therefore \qquad f(x) = \sin\phi \in \left[-\frac{1}{2}, \frac{1}{2}\right] \qquad \therefore$ **(A)**

Now $fog(x) = \sin\left(\frac{\pi}{6}\sin\left(\frac{\pi}{2}\sin\left(\frac{\pi}{2}\sin x\right)\right)\right)$

Clearly, range of fog is also $\left(-\frac{1}{2}, \frac{1}{2}\right) \qquad \therefore$ **(B)**

Now, $\lim_{x\to 0}\frac{\sin\left(\frac{\pi}{6}\sin\left(\frac{\pi}{2}\sin x\right)\right)}{\frac{\pi}{2}\sin x}$

$= \lim_{x\to 0}\frac{2}{\pi}\frac{\sin\left(\frac{\pi}{6}\sin\left(\frac{\pi}{2}\sin x\right)\right)}{\frac{\pi}{6}\sin\left(\frac{\pi}{2}\sin x\right)} \times \frac{\frac{\pi}{6}\sin\left(\frac{\pi}{2}\sin x\right)}{\frac{\sin x}{x}\times x}$

$= \lim_{x\to 0}\frac{2}{\pi}\times\frac{\pi}{6}\times\frac{\sin\left(\frac{\pi}{2}\sin x\right)}{\frac{\pi}{2}\sin x}\times\frac{\frac{\pi}{2}\sin x}{x}$

$\frac{1}{3}\times\frac{\pi}{2}=\frac{\pi}{6}$ $\quad\therefore\quad$ **(C)**

Now, $gof(x)=1$

$\Rightarrow \quad \frac{\pi}{2}\sin\left(\frac{\pi}{2}\sin\left(\frac{\pi}{2}\sin x\right)\right)=1$

$\Rightarrow \quad \sin\left(\frac{\pi}{6}\sin\left(\frac{\pi}{2}\sin x\right)\right)=\frac{2}{\lambda}\cong\frac{2}{3.14}>\frac{1}{2}$ $\quad\therefore\quad$ **(D)**

5. Let m and n be two positive integers greater than 1. If $\lim\limits_{\alpha\to 0}\left(\frac{e^{\cos(\alpha^n)}-e}{\alpha^m}\right)=-\left(\frac{e}{2}\right)$. Then the value of $\frac{m}{n}$ is :

**[JEE(Adv.)2015]**

माना कि दो धनात्मक पूर्णांक m तथा n एक (1) से बड़े हैं (greater than 1) । यदि $\lim\limits_{\alpha\to 0}\left(\frac{e^{\cos(\alpha^n)}-e}{\alpha^m}\right)=-\left(\frac{e}{2}\right)$ तब m/n का मान है:

**[JEE(Adv.)2015]**

**Ans.** 2

**Sol.** $\because$ $m\geq 2$ and और $n\geq 2$

$$=\lim_{a\to 0}\frac{e(e^{\cos(a^n)-1}-1)}{(\cos(a^n)-1)}\times\left(\frac{\cos(a^n)-1}{(a^n)^2}\right)\frac{a^{2n}}{a^m}$$

$$=e\times\lim_{a\to 0}\left(\frac{e^{\cos(a^n)-1}-1)}{\cos(a^n)-1)}\right)\times\lim_{a\to 0}\left(\frac{\cos(a^n)-1}{a^{2n}}\right)\times\lim_{a\to 0}a^{2n-m}$$

$$=e\times 1\times-\frac{1}{2}\times\lim_{a\to 0}a^{2n-m}$$

Now $\lim\limits_{a\to 0}a^{2n-m}$ must be equal to 1.

i.e. अर्थात, $2n-m=0$

$$\frac{m}{n}=2$$

6. Let a, b $\in$ R be such that $\lim\limits_{x\to 0}\frac{x^2\sin(\beta x)}{\alpha x-\sin x}=1$. Then $6(\alpha+\beta)$ equals : **[JEE(Adv.)2016]**

माना कि a, b $\in$ R इस प्रकार हैं कि $\lim\limits_{x\to 0}\frac{x^2\sin(\beta x)}{\alpha x-\sin x}=1$ है। तब $6(\alpha+\beta)$ का मान है : **[JEE(Adv.)2016]**

**Ans.** 7

**Sol.** $\lim\limits_{x\to 0}\dfrac{x^2\left(\dfrac{\sin(\beta x)}{\beta x}\right)\beta x}{\alpha x-\sin x}=1$

$\Rightarrow \lim\limits_{x\to 0}\dfrac{\beta x^3}{\alpha x-\left(x-\dfrac{x^3}{\underline{|x}}+\dfrac{x^5}{\underline{|5}}+.....\right)}=1 \quad \left(\because \lim\limits_{x\to 0}\dfrac{\sin x}{x}=1\right)$

For finite limit $= \alpha - 1 = 0 \Rightarrow \alpha = 1$

Limit $6\beta = 1, \beta = \dfrac{1}{6}$

$\Rightarrow 6(\alpha+\beta)=6\left(1+\dfrac{1}{6}\right)=7$

7. Let $f(x)=\dfrac{1-x(1+|1-x|)}{|1-x|}\cos\left(\dfrac{1}{1-x}\right)$ for $x\neq 1$. Then : **[JEE(Adv.)2017]**

(A) $\lim_{x\to 1+} f(x)=0$ (B) $\lim_{x\to 1-} f(x)=0$

(C) $\lim_{x\to 1+} f(x)$ does not exist (D) $\lim_{x\to 1-} f(x)$ does not exist

माना कि $x \neq 1$ के लिये, $f(x)=\dfrac{1-x(1+|1-x|)}{|1-x|}\cos\left(\dfrac{1}{1-x}\right)$ । तब : : **[JEE(Adv.)2017]**

(A) $\lim_{x\to 1+} f(x)=0$ (B) $\lim_{x\to 1-} f(x)=0$

(C) $\lim_{x\to 1+} f(x)$ का अस्तित्व नहीं है (D) $\lim_{x\to 1-} f(x)$ का अस्तित्व नहीं है

**Ans.** BC

**Sol.** $f(1^+)=\lim\limits_{h\to 0}\dfrac{1-(1+h)(1+h)}{h}\cos\dfrac{1}{h}$

$=\lim\limits_{h\to 0}\dfrac{1-(1+h)^2}{h}\cos\dfrac{1}{h}$

$=\lim\limits_{h\to 0}\dfrac{-h^2-2h}{h}\cos\dfrac{1}{h}$

$=\lim\limits_{h\to 0}(-h-2)\cos\dfrac{1}{h}$

$\Rightarrow \lim\limits_{h\to 0}(1^+)$ does not exist

$f(1^-)\lim\limits_{h\to 0}\dfrac{1-(1-h)(1+h)}{h}\cos\dfrac{1}{h}$

$$=\lim_{h\to 0}\frac{h^2}{h}\cos\frac{1}{h}$$

$$=\lim_{h\to 0} h\cos\frac{1}{h}=0$$

8. Let e denote the base of the natural logarithm. The value of the real number a for which the right hand limit

$\lim\limits_{x\to 0^+}\dfrac{(1-x)^{\frac{1}{x}}-e^{-1}}{x^a}$ is equal to a nonzero real number, is _____. **[JEE(Adv.)2020]**

माना कि e प्राकृतिक लघुगणक के आधार को दर्शाता है। वास्तविक संख्या a का वो मान जिसके लिए दायें पक्ष की सीमा

$\lim\limits_{x\to 0^+}\dfrac{(1-x)^{\frac{1}{x}}-e^{-1}}{x^a}$ एक शून्येतर वास्तविक संख्या के बराबर है, है _____ **[JEE(Adv.)2020]**

**Ans.** 1

**Sol.** $\lim\limits_{x\to 0^+}\dfrac{(1-x)^{\frac{1}{x}}-e^{-1}}{x^a}\neq 0$

$$\lim_{x\to 0^+}\frac{e^{\frac{1}{x}ln(1-x)}-e^{-1}}{x^a}\neq 0$$

$$\lim_{x\to 0^+}\frac{e^{-1}(e^{1+\frac{1}{x}ln(1-x)}-1)}{x^a}\neq 0$$

$$\lim_{x\to 0^+}\frac{1+\frac{1}{x}ln(1-x)}{ex^a}\neq 0$$

$$\lim_{x\to 0^+}\frac{x+ln(1-x)}{ex^{a+1}}\neq 0$$

$$\lim_{x\to 0^+}\frac{x+\left(-x-\frac{x^2}{2}....\right)}{ex^{a+1}}\neq 0$$

$\Rightarrow a+1=2$

$\Rightarrow a=1$

if $a<1\Rightarrow \lim\limits_{x\to 0^+}\dfrac{x+\left(-x-\frac{x^2}{2}....\right)}{ex^{a+1}}=0$

if $a > 1$ $\lim\limits_{x\to 0^+} \dfrac{x+\left(-x-\dfrac{x^2}{2}....\right)}{ex^{a+1}} = DNE$

9. The value of the limit $\lim\limits_{x\to\frac{\pi}{2}} \dfrac{4\sqrt{2}\left(\sin 3x+\sin x\right)}{\left(2\sin 2x\sin\dfrac{3x}{2}+\cos\dfrac{5x}{2}\right)-\left(\sqrt{2}+\sqrt{2}\cos 2x+\cos\dfrac{3x}{2}\right)}$ : **[JEE(Adv.)2020]**

सीमा $\lim\limits_{x\to\frac{\pi}{2}} \dfrac{4\sqrt{2}\left(\sin 3x+\sin x\right)}{\left(2\sin 2x\sin\dfrac{3x}{2}+\cos\dfrac{5x}{2}\right)-\left(\sqrt{2}+\sqrt{2}\cos 2x+\cos\dfrac{3x}{2}\right)}$ का मान है_____ **[JEE(Adv.)2020]**

**Ans.** 8

**Sol.** $\lim\limits_{x\to\frac{\pi}{2}} \dfrac{4\sqrt{2}\left(\sin 3x+\sin x\right)}{\left(2\sin 2x\sin\dfrac{3x}{2}+\cos\dfrac{5x}{2}\right)-\left(\sqrt{2}+\sqrt{2}\cos 2x+\cos\dfrac{3x}{2}\right)}$

$\lim\limits_{x\to\frac{\pi}{2}} \dfrac{4\sqrt{2}\left(2\sin 2x\cos x\right)}{\left[\cos\dfrac{x}{2}-\cos\dfrac{7x}{2}+\cos\dfrac{5x}{2}\right]-\left[\sqrt{2}\left(2\cos^2 x\right)+\cos\dfrac{3x}{2}\right]}$

$\lim\limits_{x\to\frac{\pi}{2}} \dfrac{4\sqrt{2}\left(2\sin 2x\cos x\right)}{\cos\dfrac{x}{2}-\cos\dfrac{3x}{2}+\cos\dfrac{5x}{2}-\cos\dfrac{7x}{2}-2\sqrt{2}\cos^2 x}$

$\lim\limits_{x\to\frac{\pi}{2}} \dfrac{16\sqrt{2}\sin x\cos^2 x}{2\sin x\sin\dfrac{x}{2}+2\sin 3x\sin\dfrac{x}{2}-2\sqrt{2}\cos^2 x}$

$\lim\limits_{x\to\frac{\pi}{2}} \dfrac{16\sqrt{2}\sin x\cos^2 x}{2\sin\dfrac{x}{2}\left(\sin 3x+\sin x\right)-2\sqrt{2}\cos^2 x}$

$\lim\limits_{x\to\frac{\pi}{2}} \dfrac{16\sqrt{2}\sin x\cos^2 x}{2\sin\dfrac{x}{2}\left(2\sin 2x\cos x\right)-2\sqrt{2}\cos^2 x}$

$\lim\limits_{x\to\frac{\pi}{2}} \dfrac{16\sqrt{2}\sin x\cos^2 x}{8\sin\dfrac{x}{2}\sin x\cos^2 x-2\sqrt{2}\cos^2 x}$

$$\lim_{x\to\frac{\pi}{2}} \frac{16\sqrt{2}}{8\times\frac{1}{\sqrt{2}}-2\sqrt{2}}=8$$

10. For any positive integer n, let $S_n : (0, \infty) \to R$ be defined by $S_n(x)=\sum_{k=1}^{n}\cot^{-1}\left(\frac{1+k(k+1)x^2}{x}\right)$, where for any $x \in R, \cot^{-1}(x) \in (0,\pi)$ and $\tan^{-1}(x) \in \left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ Then which of the following statements is (are) TRUE?

**[JEE(Adv.)2021]**

(A) $S_{10}(x)=\frac{\pi}{2}-\tan^{-1}\left(\frac{1+11x^2}{10x}\right)$, for all $x>0$

(B) $\lim_{n\to\infty}\cot(S_n(x))=x$, for all $x>0$

(C) The equation $S_3(x)=\frac{\pi}{4}$ has a root in $(0, \infty)$

(D) $\tan(S_n(x)) \le \frac{1}{2}$, for all $n \ge 1$ and $x>0$

किसी धनात्मक पूर्णांक n के लिए, माना $S_n :(0, \infty) \to R$, $S_n(x)=\sum_{k=1}^{n}\cot^{-1}\left(\frac{1+k(k+1)x^2}{x}\right)$ द्वारा परिभाषित है जहाँ किसी $x \in R$ के लिए $\cot^{-1}(x) \in (0,\pi)$ तथा $\tan^{-1}(x) \in \left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ तब निम्न में से कौनसा/कौनसे कथन सही हैं/है?

**[JEE(Adv.)2021]**

(A) सभी $x>0$ के लिए, $S_{10}(x)=\frac{\pi}{2}-\tan^{-1}\left(\frac{1+11x^2}{10x}\right)$

(B) सभी $x>0$ के लिए $\lim_{n\to\infty}\cot(S_n(x))=x$

(C) समीकरण $S_3(x)=\frac{\pi}{4}$ का एक मूल $(0, \infty)$ में हैं

(D) $\tan(S_n(x)) \le \frac{1}{2}$, सभी $n \ge 1$ तथा $x>0$ के लिए

**Ans.** AB

**Sol.** $S_n(x) = \sum_{k=1}^{n} \cot^{-1}\left(\frac{1+k(k+1)x^2}{x}\right)$

$$= \sum_{k=1}^{n} \tan^{-1}\left(\frac{(k+1)x - kx}{1+(k+1)xkx}\right)$$

$$= \sum_{k=1}^{n} \tan^{-1}(k+1)x - \tan^{-1} kx$$

$$= (\tan^{-1} 2x - \tan^{-1} x) + (\tan^{-1} 3x - \tan^{-1} 2x) + ..... + (\tan^{-1}(n+1)x - \tan^{-1} nx)$$

$S_n(x) = \tan^{-1}(n+1)x - \tan^{-1} x$

(A) $S_{10}(x) = \tan^{-1} 11x - \tan^{-1} x,\ x > 0$

$$= \tan^{-1}\left(\frac{11x - x}{1+11x.x}\right)$$

$$= \tan^{-1}\left(\frac{10x}{1+11x^2}\right)$$

$$= \cot^{-1}\left(\frac{1+11x^2}{10x}\right)$$

$$= \frac{\pi}{2} - \tan^{-1}\left(\frac{1+11x^2}{10x}\right) \Rightarrow \text{(A) is correct}$$

(B) $\lim_{n\to\infty} \cot(S_n(x)) = \lim_{n\to\infty} \cot(\tan^{-1}(n+1)x - \tan^{-1} x),\ x > 0$

$$= \lim_{n\to\infty} \cot\left(\tan^{-1}\frac{(n+1)x - x}{1+(n+1)x.x}\right)$$

$$= \lim_{n\to\infty} \cot\left(\tan^{-1}\frac{nx}{1+(n+1)x^2}\right)$$

$$= \lim_{n\to\infty} \cot\left[\tan^{-1}\frac{x}{\frac{1}{n}+\left(1+\frac{1}{n}\right)x^2}\right] = \cot\tan^{-1}\left(\frac{x}{x^2}\right)$$

$$= \cot(\cot^{-1}x) = x$$

$\Rightarrow$ (B) is correct

(C) $S_3(x) = \frac{\pi}{4}$ has a root in $(0, \infty)$

$\tan^{-1} 4x - \tan^{-1} x = \frac{\pi}{4}$

$\Rightarrow \frac{3x}{1+4x^2} = 1 \Rightarrow 4x^2 + 1 = 3x \Rightarrow 4x^2 - 3x + 1 = 0$

$D < 0$

$\Rightarrow$ no real

$\Rightarrow$ (C) option is incorrect

(D) $S_n(x) = \tan^{-1}(n+1)x - \tan^{-1} x$

$$= \tan^{-1}\left(\frac{nx + x - x}{1+(n+1)x^2}\right)$$

$$= \tan^{-1}\left(\frac{nx}{1+(n+1)x^2}\right)$$

$$\tan(S_n(x)) = \frac{nx}{1+(n+1)x^2} = \frac{n}{\frac{1}{x}+(n+1)x}$$

$\Rightarrow$ A.M. $\geq$ G.M.

$$\frac{\frac{1}{x}+(n+1)x}{2} \geq \sqrt{\frac{1}{x}.(n+1)x}$$

$$\frac{1}{x}+(n+1)x \geq 2\sqrt{n+1}$$

$$\Rightarrow \tan(S_n(x)) = \frac{nx}{1+(n+1)x^2} \leq \frac{n}{2\sqrt{n+1}}$$

for $n = 2$

$$\frac{n}{2\sqrt{n+1}} = \frac{1}{\sqrt{3}} > \frac{1}{2}$$

D is incorrect

11. If $\beta = \lim\limits_{x\to 0} \dfrac{e^{x^3} - \left(1-x^3\right)^{\frac{1}{3}} + \left(\left(1-x^2\right)^{\frac{1}{2}} - 1\right)\sin x}{x\sin^2 x}$ then the value of $6\beta$ is ____. **[JEE(Adv.)2022]**

यदि $\beta = \lim\limits_{x\to 0} \dfrac{e^{x^3} - \left(1-x^3\right)^{\frac{1}{3}} + \left(\left(1-x^2\right)^{\frac{1}{2}} - 1\right)\sin x}{x\sin^2 x}$, तब $6\beta$ का मान ज्ञात कीजिए। **[JEE(Adv.)2022]**

Ans. 5

Sol. $$\beta=\lim_{x\to0}\frac{e^{x^3}-\left(1-x^3\right)^{1/3}}{\frac{x\sin^2x}{x^2}x^2}+\frac{\left(\left(1-x^2\right)^{1/2}-1\right)\sin x}{x\frac{\sin^2x}{x^2}x^2}$$

use expansion

$$\beta=\lim_{x\to0}\frac{\left(1+x^3\right)-\left(1-\frac{x^3}{3}\right)}{x^3}+\lim_{x\to0}\frac{\left(\left(1-\frac{x^2}{2}\right)-1\right)}{x^2}\frac{\sin x}{x}$$

$$\beta=\lim_{x\to0}\frac{4x^3}{3x^3}+\lim_{x\to0}\frac{-x^2}{2x^2}$$

$$\beta=\frac{4}{3}-\frac{1}{2}=\frac{5}{6}$$

$6\beta = 5$

12. Let $k \in R$. If $\lim\limits_{x\to0+}\left(\sin\left(\sin kx\right)+\cos x+x\right)^{\frac{2}{x}}=e^6$, then the value of k is : **[JEE(Adv.)2024]**

माना कि $k \in R$ है। यदि $\lim\limits_{x\to0+}\left(\sin\left(\sin kx\right)+\cos x+x\right)^{\frac{2}{x}}=e^6$, तब k का मान है : **[JEE(Adv.)2024]**

(A) 1 (B) 2 (C) 3 (D) 4

**Ans.** B

**Sol.** $$\lim_{x\to0^+}\left(\sin\left(\sin kx\right)+\cos x+x\right)^{\frac{2}{x}}=e^6$$

$$e^{\lim\limits_{x\to0^+}2\left\{\frac{\sin(\sin kx)+\cos x+x-1}{x}\right\}}=e^6$$

$$\lim_{x\to0^+}\frac{\sin\left(\sin kx\right)+\cos x+x-1}{x}=3$$

$$\lim_{x\to0^+}\frac{\cos\left(\sin kx\right)\cdot\cos\left(kx\right)\cdot k-\sin x+1}{1}=3$$

$k + 1 = 3 \Rightarrow k = 2$

13. Let S be the set of all $(\alpha, \beta) \in R \times R$ such that $\lim\limits_{x\to\infty}\frac{\sin\left(x^2\right)\left(\log_e x\right)^\alpha\sin\left(\frac{1}{x^2}\right)}{x^{\alpha\beta}\left(\log_e(1+x)\right)^\beta}=0$. Then which of the following

is (are) correct? **[JEE(Adv.)2024]**

माना कि S उन सभी $(\alpha, \beta) \in R \times R$ का समुच्चय है कि $\lim\limits_{x\to\infty} \frac{\sin\left(x^2\right)\left(\log_e x\right)^{\alpha} \sin\left(\frac{1}{x^2}\right)}{x^{\alpha\beta}\left(\log_e(1+x)\right)^{\beta}} = 0$ है। तब निम्नलिखित में से कौनसा(से) सही है (हैं)? **[JEE(Adv.)2024]**

(A) $(-1, 3) \in S$ (B) $(-1, 1) \in S$ (C) $(1, -1) \in S$ (D) $(1, -2) \in S$

**Ans.** BC

**Sol.** $$\lim_{x\to\infty} \frac{\sin\left(x^2\right)\left(\ell n\left(x\right)\right)^{\alpha} \sin\left(\frac{1}{x^2}\right)}{x^{\alpha\beta}\left(\ell n\left(1+x\right)\right)^{\beta}} = 0$$

$$\lim_{x\to\infty} \frac{\sin\left(x^2\right) \ell n\left(x\right)^{\alpha} \left(\frac{1}{x^2}\right)}{x^{\alpha\beta}\left(\ell n\left(1+x\right)\right)^{\beta}} = 0$$

$$\lim_{x\to\infty} \frac{\sin\left(x^2\right)\left(\ell nx\right)^{\alpha}}{x^{\alpha\beta+2}\left(\ell n\left(1+x\right)\right)^{\beta}} = 0$$

$\Rightarrow \alpha\beta + 2 > 0$ or $\alpha\beta + 2 = 0$ & $\beta > \alpha$

$\alpha\beta > -2$ $\alpha\beta = -2$ & $\beta > \alpha$

(A) $\alpha\beta = -3$ Incorrect

(B) $\alpha\beta = -1$ Correct

(C) $\alpha\beta = -1$ Correct

(D) $\alpha\beta = -2$ but $\alpha > \beta$ Incorrect

14. Let $x_0$ be the real number such that $e^{x_0} + x_0 = 0$. For a given real number $\alpha$, define

$$g\left(x\right) = \frac{3xe^{x} + 3x - \alpha e^{x} - \alpha x}{3\left(e^{x} + 1\right)}$$

for all real numbers x.

Then which one of the following statements is TRUE? **[JEE(Adv.)2025]**

(A) For $\alpha = 2$, $\lim\limits_{x \to x_0} \left| \dfrac{g(x) + e^{x_0}}{x - x_0} \right| = 0$

(B) For $\alpha = 2$, $\lim\limits_{x \to x_0} \left| \dfrac{g(x) + e^{x_0}}{x - x_0} \right| = 1$

(C) For $\alpha = 3$, $\lim\limits_{x \to x_0} \left| \dfrac{g(x) + e^{x_0}}{x - x_0} \right| = 0$

(D) For $\alpha = 3$, $\lim\limits_{x \to x_0} \left| \dfrac{g(x) + e^{x_0}}{x - x_0} \right| = \dfrac{2}{3}$

मान लीजिए कि $x_0$ वह वास्तविक संख्या (real number) है कि $e^{x_0} + x_0 = 0$ है। किसी दी गयी एक वास्तविक संख्या $\alpha$, तथा सभी वास्तविक संख्याओं x के लिए $g(x) = \dfrac{3xe^x + 3x - \alpha e^x - \alpha x}{3(e^x + 1)}$ परिभाषित कीजिये।

तब निम्नलिखित कथनों में से कौन सा एक कथन सत्य है? **[JEE(Adv.)2025]**

(A) $\alpha = 2$ के लिए, $\lim\limits_{x \to x_0} \left| \dfrac{g(x) + e^{x_0}}{x - x_0} \right| = 0$ है

(B) $\alpha = 2$ के लिए, $\lim\limits_{x \to x_0} \left| \dfrac{g(x) + e^{x_0}}{x - x_0} \right| = 1$ है

(C) $\alpha = 3$ के लिए, $\lim\limits_{x \to x_0} \left| \dfrac{g(x) + e^{x_0}}{x - x_0} \right| = 0$ है

(D) $\alpha = 3$ के लिए, $\lim\limits_{x \to x_0} \left| \dfrac{g(x) + e^{x_0}}{x - x_0} \right| = \dfrac{2}{3}$ है

**Ans.** C

**Sol.** $g(x) = x - \dfrac{\alpha(e^x + x)}{3(e^x + 1)}$

$$g'(x) = 1 - \frac{\alpha}{3}\left\{\frac{(e^x + 1)^2 - (e^x + x)\cdot e^x}{(e^x + 1)^2}\right\}$$

$$g'(x_0) = 1 - \frac{\alpha}{3}\left\{\frac{(e^{x_0} + 1)^2 - 0}{(e^{x_0} + 1)^2}\right\}$$

$$g'(x_0) = 1 - \frac{\alpha}{3}$$

$$\lim_{x \to x_0} \left| \frac{g(x) + e^{x_0}}{x - x_0} \right| \left(\frac{0}{0}\right)$$

Using L'Hospital Rule

$$= \lim_{x \to x_0} \left| \frac{g'(x)}{1} \right| = \left| g'(x_0) \right|$$

$$= \left| 1 - \frac{\alpha}{3} \right|$$

for $\alpha = 3 \Rightarrow L = 0$

## Exercise C-1

1. Let $f(x)=\begin{cases} x\sin\left(\frac{1}{x}\right)+\sin\left(\frac{1}{x^2}\right) & x\neq 0 \\ 0 & x=0 \end{cases}$, then $\lim\limits_{x\to\infty} f(x)$ equals :

(A) 0 (B) – 1/2 (C) 1 (D) none of these

माना $f(x)=\begin{cases} x\sin\left(\frac{1}{x}\right)+\sin\left(\frac{1}{x^2}\right) & x\neq 0 \\ 0 & x=0 \end{cases}$, तो $\lim\limits_{x\to\infty} f(x)=$

(A) 0 (B) – 1/2 (C) 1 (D) इनमें से कोई नहीं

**Ans.** C

**Sol.** $\lim\limits_{x\to\infty} x\sin\left(\frac{1}{x}\right)+\sin\left(\frac{1}{x^2}\right)$

$$\lim_{x\to\infty}\frac{\sin\frac{1}{x}}{\frac{1}{x}}+\frac{1}{x^2}\frac{\sin\frac{1}{x^2}}{\sin\frac{1}{x^2}}$$

Let $\frac{1}{x}=t\Rightarrow x\to\infty, t\to 0$

$$\lim_{t\to 0}\frac{\sin t}{t}+t^2\frac{\sin t}{t}=1+0=1$$

2. Let $\alpha, \beta$ be the roots of $ax^2+bx+c=0$, where $1<\alpha<\beta$. Then $\lim\limits_{x\to x_0}\frac{|ax^2+bx+c|}{ax^2+bx+c}=1$ then which of the following statements is incorrect :

(A) $a>0$ and $x_0<1$ (B) $a>0$ and $x_0>\beta$

(C) $a<0$ and $\alpha<x_0<\beta$ (D) $a<0$ and $x_0<1$

माना समीकरण $ax^2+bx+c=0$ के मूल $\alpha, \beta$ हैं, जहाँ $1<\alpha<\beta$. यदि $\lim\limits_{x\to x_0}\frac{|ax^2+bx+c|}{ax^2+bx+c}=1$ हो, तो निम्न कथनों में से असत्य कथन है :

(A) $a>0$ और $x_0<1$ (B) $a>0$ और $x_0>\beta$

(C) $a<0$ और $\alpha<x_0<\beta$ (D) $a<0$ और $x_0<1$

**Ans.** D

**Sol.** $\alpha, \beta$ be the roots of the equation $ax^2+bx+c=0$

where $1<\alpha<\beta$.

(i) if $a>0$

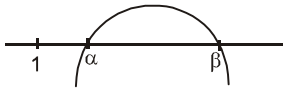



So $ax^2 + bx + c > 0$ when $x \in (-\infty, \alpha) \cup (\beta, \infty)$

So $\lim\limits_{x \to x_0} \dfrac{|ax^2 + bx + c|}{ax^2 + bx + c} = 1$

when $a > 0$ and $x \in (-\infty, \alpha) \cup (\beta, \infty)$

(ii) If $a < 0$




So $ax^2 + bx + c > 0$ when $x \in (\alpha, \beta)$

$\lim\limits_{x \to x_0} \dfrac{|ax^2 + bx + c|}{ax^2 + bx + c} = 1$

when $a < 0$ and $x \in (\alpha, \beta)$

So (A) $a > 0$ and $x_0 < 1$ right

(B) $a > 0$ and $x_0 > \beta$ right

(C) $a < 0$ and $\alpha < x_0 < \beta$ right

(D) $a < 0$ and $x_0 < 1$ wrong

3. If $\lim\limits_{x \to 0} \dfrac{x^3}{\sqrt{a+x}\,(bx - \sin x)} = 1$ then the constants 'a' and 'b' are : (where $a > 0$)

यदि $\lim\limits_{x \to 0} \dfrac{x^3}{\sqrt{a+x}\,(bx - \sin x)} = 1$ हो, तो a एवं b के मान हैं : (जहाँ $a > 0$)

(A) $b = 1, a = 36$ (B) $a = 1, b = 6$ (C) $a = 1, b = 36$ (D) $b = 1, a = 6$

**Ans.** A

**Sol.** $\lim\limits_{x \to 0} \dfrac{x^3}{\sqrt{a+x}\,(bx - \sin x)} = 1$

$\Rightarrow \lim\limits_{x \to 0} \dfrac{x^3}{\sqrt{a}\,(bx - \sin x)} = 1$

$\Rightarrow \lim\limits_{h \to 0} \dfrac{3x^2}{\sqrt{a}\,(b - \cos x)} = 1$

$\therefore b = 1$

$\therefore 1 = \lim\limits_{x \to 0} \dfrac{3x^2}{\sqrt{a}\,(1 - \cos x)}$

$$= \lim_{x\to 0} \frac{3x^2}{\sqrt{a}\,2\sin^2\frac{x}{2}} = \frac{6}{\sqrt{a}}$$

$\therefore$ a = 26

4. If $\ell = \lim\limits_{x\to\infty} (\sin\sqrt{x+1} - \sin\sqrt{x})$ and $m = \lim\limits_{x\to-\infty} [\sin\sqrt{x+1} - \sin\sqrt{x}]$ where [.] denotes the greatest integer function then :

(A) $\ell = m = 0$ (B) $\ell = 0$ ; m is undefined

(C) $\ell$, m both do not exist (D) $\ell = 0$, $m \neq 0$ (although m exist)

यदि $\ell = \lim\limits_{x\to\infty} (\sin\sqrt{x+1} - \sin\sqrt{x})$ और $m = \lim\limits_{x\to-\infty} [\sin\sqrt{x+1} - \sin\sqrt{x}]$ है, जहाँ [.] महत्तम पूर्णांक फलन को प्रदर्शित करता है, तो :

(A) $\ell = m = 0$ (B) $\ell = 0$ ; m अपरिभाषित है

(C) $\ell$, m दोनों विद्यमान नहीं है (D) $\ell = 0$, $m \neq 0$ (यद्यपि m विद्यमान है)

**Ans.** B

**Sol.** (i) $\ell = \lim\limits_{x\to\infty} \left(\sin\sqrt{x+1} - \sin\sqrt{x}\right)$

$$\Rightarrow \quad \ell = \lim_{x\to\infty} 2\cos\left(\frac{\sqrt{x+1}+\sqrt{x}}{2}\right).\sin\left(\frac{\sqrt{x+1}-\sqrt{x}}{2}\right)$$

$$\Rightarrow \quad \ell = \lim_{x\to\infty} 2\cos\left(\frac{\sqrt{x+1}+\sqrt{x}}{2}\right).\sin\left(\frac{x+1-x}{2\left(\sqrt{x+1}+\sqrt{x}\right)}\right)$$

$$\Rightarrow \quad \ell = \lim_{x\to\infty} 2\cos\left(\frac{\sqrt{x}+\sqrt{x+1}}{2}\right)\sin\left(\frac{1}{2\left(\sqrt{x}+\sqrt{x+1}\right)}\right)$$

$\Rightarrow \quad \ell = (\text{oscillating value} -1 \text{ to } 1) \times 0 = 0$

(ii) $m = \lim\limits_{x\to-\infty} \left[\sin\sqrt{x+1} - \sin\sqrt{x}\right]$

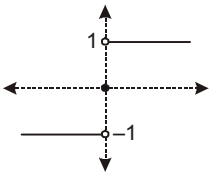
when $x \to -\infty$

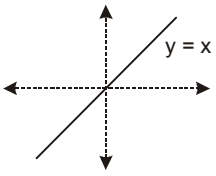
then $\sqrt{x}$ undefined

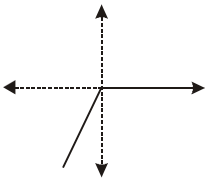
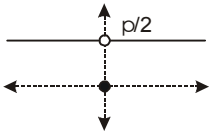
m is undefined

5. The graph of the function $f(x) = \underset{t\to 0}{\text{Lim}} \left(\frac{2x}{\pi}\cot^{-1}\frac{x}{t^2}\right)$, is :

फलन $f(x) = \underset{t\to 0}{\text{Lim}} \left(\frac{2x}{\pi}\cot^{-1}\frac{x}{t^2}\right)$ का आरेख है :

(A) 

(B) 

(C) 
(D) 


**Ans.** C

**Sol.** $f(x) = \lim\limits_{t\to 0}\left(\frac{2x}{\pi}\cot^{-1}\frac{x}{t^2}\right)$

Case-I : when $x = 0$

$f(x) = 0$

Case-II : when $x > 0$

$$f(x) = \lim_{t\to 0}\left(\frac{2x}{\pi}\cot^{-1}\frac{x}{t^2}\right) = \frac{2x}{\pi}\times\cot^{-1}(\infty) = \frac{2x}{\pi}.0 = 0$$

Case-III when $x < 0$

$$f(x) = \lim_{t\to 0}\left(\frac{2x}{\pi}\cot^{-1}\frac{x}{t^2}\right) = \frac{2x}{\pi}\times\cot^{-1}(-\infty) = \frac{2x}{\pi}.\pi = 2x \Rightarrow \quad f(x) = 2x$$

Hindi. $f(x) = \lim\limits_{t\to 0}\left(\frac{2x}{\pi}\cot^{-1}\frac{x}{t^2}\right)$

स्थिति-I : जब $x = 0$

$f(x) = 0$

स्थिति-II : जब $x > 0$

$$f(x) = \lim_{t\to 0}\left(\frac{2x}{\pi}\cot^{-1}\frac{x}{t^2}\right) = \frac{2x}{\pi}\times\cot^{-1}(\infty) = \frac{2x}{\pi}.0 = 0$$

**स्थिति-III** जब $x < 0$

$$f(x) = \lim_{t\to 0}\left(\frac{2x}{\pi}\cot^{-1}\frac{x}{t^2}\right) = \frac{2x}{\pi}\times\cot^{-1}(-\infty) = \frac{2x}{\pi}.\pi = 2x \Rightarrow \quad f(x) = 2x$$

6. If $\lim\limits_{x\to\infty} f(x)$ exist and is finite & nonzero and if $\lim\limits_{x\to\infty}\left(f(x)+\frac{3f(x)-1}{f^2(x)}\right) = 3$ then the value of $\lim\limits_{x\to\infty} f(x)$ is :

(A) 1 (B) $-1$ (C) 2 (D) none of these

यदि $\lim\limits_{x\to\infty} f(x)$ विद्यमान है तथा परिमित एवं अशून्य है और $\lim\limits_{x\to\infty}\left(f(x)+\frac{3f(x)-1}{f(x)^2}\right) = 3$ हो, तो $\lim\limits_{x\to\infty} f(x)$ का मान है :

(A) 1 (B) $-1$ (C) 2 (D) इनमें से कोई नहीं

**Ans.** A

**Sol.** $\lim\limits_{x\to\infty}\left(f(x)+\frac{3f(x)-1}{f^2(x)}\right) = 3$

Let $\lim_{x\to\infty} f(x) = y$

or $\left( \lim_{x\to\infty} f(x) + \frac{3\lim_{x\to\infty} f(x) - 1}{\left(\lim_{x\to\infty} f(x)\right)^2} \right) = 3$

or $\left( y + \frac{3y-1}{y^2} \right) = 3$

or $y^3 - 3y^2 + 3y - 1 = 0$

or $(y-1)^3 = 0$

or $y = 1$

7. $\underset{x\to\infty}{\text{Limit}} \frac{e^x\left(\left(2^{x^n}\right)^{\frac{1}{e^x}} - \left(3^{x^n}\right)^{\frac{1}{e^x}}\right)}{x^n}$, $n \in N$ is equal to :

(A) 0 (B) ln(2/3) (C) ln(3/2) (D) none

$\underset{x\to\infty}{\text{Limit}} \frac{e^x\left(\left(2^{x^n}\right)^{\frac{1}{e^x}} - \left(3^{x^n}\right)^{\frac{1}{e^x}}\right)}{x^n}$, $n \in N$ का मान है :

(A) 0 (B) $\ell$n(2/3) (C) $\ell$n(3/2) (D) इनमें से कोई नहीं

**Ans.** B

**Sol.** $\underset{x\to\infty}{\ell\text{im}} \frac{e^x\left(\left(2^{x^n}\right)^{\frac{1}{e^x}} - \left(3^{x^n}\right)^{\frac{1}{e^x}}\right)}{x^n}$, $n \in N$

$\underset{x\to\infty}{\ell\text{im}} \frac{(2)^{\frac{x^n}{e^x}} - (3)^{\frac{x^n}{e^x}}}{\frac{x^n}{e^x}}$

when $x \to \infty$, $\frac{x^n}{e^x} \to 0$

Put $\frac{x^n}{e^x} = t$

$\underset{t\to0}{\ell\text{im}} \left( \frac{2^t - 3^t}{t} \right) = \ell n2 - \ell n3 = \ell n\left(\frac{2}{3}\right)$

8. If $\underset{x\to 0}{\text{Limit}} \frac{\left((a-n)\,nx-\tan x\right)\sin nx}{x^2} = 0$, where n is non-zero real number, then a is equal to :

यदि $\underset{x\to 0}{\text{Limit}} \frac{\left((a-n)\,nx-\tan x\right)\sin nx}{x^2} = 0$, जहाँ n अशून्य वास्तविक संख्या है, तो a का मान है :

(A) 0 (B) $\frac{n+1}{n}$ (C) n (D) $n+\frac{1}{n}$

**Ans.** D

**Sol.** Given, $\lim\limits_{x\to 0}\frac{\left\{(a-n)nx-\tan x\right\}\sin nx}{x^2}=0$

$$\Rightarrow \lim_{x\to 0}\left\{(a-n)n-\frac{\tan x}{x}\right\}\frac{\sin nx}{nx}\times n=0$$

$$\Rightarrow \qquad \{(a-n)n-1\}n=0$$

$$\Rightarrow (a-n)n=1 \Rightarrow a=n+\frac{1}{n}$$

9. $\underset{x\to 0}{\text{Limit}} \frac{x\tan 2x-2x\tan x}{(1-\cos 2x)^2}$ is :

$\underset{x\to 0}{\text{Limit}} \frac{x\tan 2x-2x\tan x}{(1-\cos 2x)^2}$ का मान है :

(A) 2 (B) $-2$ (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

**Ans.** C

**Sol.** $\lim\limits_{x\to 0}\frac{x\tan 2x-2x\tan x}{\left(1-\cos 2x\right)^2}$ is

$$=\lim_{x\to 0}\frac{x\dfrac{2\tan x}{1-\tan^2 x}-2x\tan x}{\left(2\sin^2 x\right)^2}$$

$$=\lim_{x\to 0}\frac{2x\tan x\left[\dfrac{1}{1-\tan^2 x}-1\right]}{4\sin^4 x}$$

$$=\lim_{x\to 0}\frac{2x\tan x\left[\dfrac{1-1+\tan^2 x}{1-\tan^2 x}\right]}{4\sin^4 x}$$

$$=\lim_{x\to 0}\frac{x\cdot\tan^3 x}{2\sin^4 x\left(1-\tan^2 x\right)}$$

$$= \lim_{x\to 0} \frac{1}{2} \cdot \frac{x\left(\frac{\tan x}{x}\right)^3 x^3}{\sin^4 x\left(1-\tan^2 x\right)}$$

$$= \frac{1}{2} \lim_{x\to 0} \frac{\left(\frac{\tan x}{x}\right)^3}{\left(\frac{\sin x}{x}\right)^4 \left(1-\tan^2 x\right)}$$

$$= \frac{1(1)^3}{2(1)^4(1-0)} = \frac{1}{2}$$

10. The value of $\underset{x\to\infty}{\text{Limit}} \frac{\cot^{-1}\left(x^{-a}\log_a x\right)}{\sec^{-1}\left(a^x \log_x a\right)}$ $(a > 1)$ is equal to :

(A) 1 (B) 0 (C) $\pi/2$ (D) does not exist

$\underset{x\to\infty}{\text{Lim}} \frac{\cot^{-1}\left(x^{-a}\ \log_a x\right)}{\sec^{-1}\left(a^x\ \log_x a\right)}$ $(a > 1)$ का मान होगा :

(A) 1 (B) 0 (C) $\pi/2$ (D) अस्त्विहीन

**Ans.** A

**Sol.** $\underset{x\to\infty}{\text{Limit}} \frac{\cot^{-1}\left(\frac{\log_a x}{x^a}\right)}{\sec^{-1}\left(\frac{a^x}{\log_a x}\right)}$ ; $\underset{x\to\infty}{\text{as}} \left(\frac{\log_a x}{x^a}\right) \to 0$ and $\left(\frac{a^x}{\log_a x}\right) \to \infty$ (using L'opital rule)

$\therefore \quad l = \frac{\pi/2}{\pi/2} = 1$

11. $\underset{x\to 0}{\text{Lim}} \frac{\left(\sqrt{(1-\cos x)+\sqrt{(1-\cos x)+\sqrt{(1-\cos x)+.....\infty}}}\right)-1}{x^2}$ equals :

$\underset{x\to 0}{\text{Lim}} \frac{\left(\sqrt{(1-\cos x)+\sqrt{(1-\cos x)+\sqrt{(1-\cos x)+.....\infty}}}\right)-1}{x^2} =$

(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) 2

**Ans.** B

**Sol.** Let $\left(\sqrt{(1-\cos x)+\sqrt{(1-\cos x)+\sqrt{(1-\cos x)+.....\infty}}}\right)=y$

$y=\sqrt{(1-\cos x)+y}$

$\Rightarrow y^2-y+(\cos x-1)=0$

$y=\dfrac{1+\sqrt{5-4\cos x}}{2}$

Now $\lim\limits_{x\to 0}\dfrac{\sqrt{5-4\cos x}-1}{2x^2}$

$=\lim\limits_{x\to 0}\dfrac{4(1-\cos x)}{2x^2\left(\sqrt{5-4\cos x}+1\right)}=\dfrac{1}{2}$

12. $\lim\limits_{x\to\infty} x\left(\tan^{-1}\dfrac{x+1}{x+2}-\cot^{-1}\dfrac{x+2}{x}\right)$ is

(A) $-1$ (B) $\dfrac{1}{2}$ (C) $-\dfrac{1}{2}$ (D) non existent

$\lim\limits_{x\to\infty} x\left(\tan^{-1}\dfrac{x+1}{x+2}-\cot^{-1}\dfrac{x+2}{x}\right)$ का मान है –

(A) $-1$ (B) $\dfrac{1}{2}$ (C) $-\dfrac{1}{2}$ (D) विद्यमान नहीं है

**Ans.** B

**Sol.** For $x>0 \Rightarrow \cot^{-1}\dfrac{x+2}{2}=\tan^{-1}\left(\dfrac{x}{x+2}\right)$

$\lim\limits_{x\to\infty} x\left(\tan^{-1}\left(\dfrac{x+1}{x+2}\right)-\tan^{-1}\left(\dfrac{x}{x+2}\right)\right)$

$=\lim\limits_{x\to\infty}\left\{x\left(\tan^{-1}\left(\dfrac{\frac{x+2}{x+2}-\frac{x}{x+2}}{1+\left(\frac{x+1}{x+2}\times\frac{x}{x+2}\right)}\right)\right)\right\}$

$=\lim\limits_{x\to\infty}\left\{x\left(\tan^{-1}\left(\dfrac{(x+2)}{(x+2)^2+x(x+1)}\right)\right)\right\}$

$$= \lim_{x\to\infty}\left\{x\left(\tan^{-1}\left(\frac{x+2}{2x^2+5x+4}\right)\right)\right\}$$

$$= \lim_{x\to\infty}\left(\frac{\tan^{-1}\left(\frac{x+2}{2x^2+5x+4}\right)}{\left(\frac{x+2}{2x^2+5x+4}\right)}\right)\times\left(\frac{x(x+2)}{2x^2+5x+4}\right)$$

$$= 1\times\frac{1}{2} = \frac{1}{2}$$

13. Which one of the following statement is true?

(A) If $\lim_{x\to c} f(x)\cdot g(x)$ and $\lim_{x\to c} f(x)$ exist, then $\lim_{x\to c} g(x)$ exists.

(B) If $\lim_{x\to c} f(x)\cdot g(x)$ exists, then $\lim_{x\to c} f(x)$ and $\lim_{x\to c} g(x)$ exist.

(C) If $\lim_{x\to c}(f(x)+g(x))$ and $\lim_{x\to c} f(x)$ exist, then $\lim_{x\to c} g(x)$ exist.

(D) If $\lim_{x\to c}(f(x)+g(x))$ exists, then $\lim_{x\to c} f(x)$ and $\lim_{x\to c} g(x)$ exist

निम्न में से कौनसा कथन सत्य है ?

(A) यदि $\lim_{x\to c} f(x)\cdot g(x)$ तथा $\lim_{x\to c} f(x)$ विद्यमान है तो $\lim_{x\to c} g(x)$ विद्यमान होगी

(B) यदि $\lim_{x\to c} f(x)\cdot g(x)$ विद्यमान है तो $\lim_{x\to c} f(x)$ तथा $\lim_{x\to c} g(x)$ विद्यमान होगी

(C) यदि $\lim_{x\to c}(f(x)+g(x))$ तथा $\lim_{x\to c} f(x)$ विद्यमान है तो $\lim_{x\to c} g(x)$ विद्यमान होगी

(D) यदि $\lim_{x\to c}(f(x)+g(x))$ विद्यमान है तो $\lim_{x\to c} f(x)$ तथा $\lim_{x\to c} g(x)$ विद्यमान होगी

**Ans.** C

**Sol.** For option A and B, take $f(x) = x$ and $g(x) = \frac{1}{x}$.

Limit of g(x) doesn't exist at $x = 0$, but for $\lim_{x\to 0} f(x)\cdot g(x)$ and $f(x) = x$ limit exists

For option C,

Let assume $h(x) = f(x) + g(x)$,

Take limit on both sides as $x \to c$

$\lim_{x\to c} h(x) = \lim_{x\to c}(f(x) + g(x))$

And it is given that $\lim_{x\to c} f(x)$ exists.

Using sum law of limits :

If $\lim_{x\to c} F(x)$ and $\lim_{x\to c} G(x)$ exists, then $\lim_{x\to c}(F(x) \pm G(x))$ also exists.

$\therefore \lim_{x\to c}(h(x)-f(x))=\lim_{x\to c}(f(x)+g(x)-f(x))=\lim_{x\to c}g(x)$ exists.

For option D, take $f(x)=-\frac{\cos x}{x^2}$ and $g(x)=\frac{1}{x^2}$.

Limit of g(x) and $f(x)=-\frac{\cos x}{x^2}$ doesn't exist at $x=0$, but for $\lim_{x\to 0} f(x)+g(x)$ limit exists.

Hence option C.

14. If $f(x)=\cos x,\ x=n\pi,\ n=0,1,2,3,.....$
$=3$, otherwise and

$$\phi(x)=\begin{cases} x^2+1 & \text{when } x\neq 3,\ x\neq 0 \\ 3 & \text{when } x=0 \\ 5 & \text{when } x=3 \end{cases} \quad \text{then } \lim_{x\to 0} f(\phi(x))=$$

(A) 1 (B) 3 (C) 5 (D) none

यदि $f(x)=\cos x,\ x=n\pi,\ n=0,1,2,3,.....$
$=3$ अन्यथा तथा

$$\phi(x)=\begin{cases} x^2+1 & \text{जब } x\neq 3,\ x\neq 0 \\ 3 & \text{जब } x=0 \\ 5 & \text{जब } x=3 \end{cases} \quad \text{है तो } \lim_{x\to 0} f(\phi(x))=$$

(A) 1 (B) 3 (C) 5 (D) कोई नही

**Ans.** B

**Sol.** $\lim_{x\to 0}\phi(x)=\lim_{x\to 0}(x^2+1)$
$=1$

$\lim_{x\to 0} f(\phi(x))=f(1)=3$

15. $\lim_{n\to\infty}\left(\frac{\sqrt[n]{p}+\sqrt[n]{q}}{2}\right)^n$, $p, q>0$ equals :

$\lim_{n\to\infty}\left(\frac{\sqrt[n]{p}+\sqrt[n]{q}}{2}\right)^n$, $p, q>0$ का मान होगा :

(A) 1 (B) $\sqrt{pq}$ (C) pq (D) $\frac{pq}{2}$

**Ans.** B

**Sol.** $L=\lim_{n\to\infty}\left(\frac{p^{1/n}+q^{1/n}}{2}\right)^n=e^{\lim_{n\to\infty} n\left(\frac{p^{1/n}+q^{1/n}}{2}-1\right)}=e^{\lim_{n\to\infty} n\left(\frac{(p^{1/n}-1)+(q^{1/n}-1)}{2}\right)}=e^{\lim_{n\to\infty}\frac{1}{2}\left(\frac{(p^{1/n}-1)}{1/n}+\frac{(q^{1/n}-1)}{1/n}\right)}$

$= e^{\left(\frac{\ln p+\ln q}{2}\right)} = e^{\left(\ln\sqrt{pq}\right)} = \sqrt{pq}$

16. If $f(x) = \cot^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$ and $g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$, then $\lim\limits_{x\to a}\frac{f(x)-f(a)}{g(x)-g(a)}$, $0 < a < \frac{1}{2}$ is :

यदि $f(x) = \cot^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$ तथा $g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ है तो $\lim\limits_{x\to a}\frac{f(x)-f(a)}{g(x)-g(a)}$, $0 < a < \frac{1}{2}$ का मान होगा :

(A) $\frac{3}{2(1+a^2)}$ (B) $\frac{3}{2}$ (C) $\frac{-3}{2(1+a^2)}$ (D) $-\frac{3}{2}$

**Ans.** D

**Sol.** $f(x) = \cot^{-1}\frac{3x-x^3}{1-3x^2} = \cot^{-}(\tan 3\theta)$

$\theta = \tan^{-1}x = \frac{\pi}{2} - 3\tan^{-1}x$

$g(x) = \cos^{-1}\frac{1-x^2}{1+x^2} = 2\tan^{-1}x$

$\lim\limits_{x\to\infty}\frac{\frac{\pi}{2}-3\tan^{-1}x-\frac{\pi}{2}+3\tan^{-1}a}{2\tan^{-1}x-2\tan^{-1}a}$

$= \frac{-3}{2}\times\frac{\tan^{-1}x-\tan^{-1}a}{\tan^{-1}x-\tan^{-1}a}$

$= \frac{-3}{2}$

17. If $\lim\limits_{x\to c^-}\{\ln x\}$ and $\lim\limits_{x\to c^+}\{\ln x\}$ exists finitely but they are not equal (where {.} denotes fractional part function), then :

(A) 'c' can take only rational values

(B) 'c' can take only irrational values

(C) 'c' can take infinite values in which only one is irrational

(D) 'c' can take infinite values in which only one is rational

यदि $\lim\limits_{x\to c^-}\{\ln x\}$ तथा $\lim\limits_{x\to c^+}\{\ln x\}$ परिमित रूप से विद्यमान है लेकिन दोनों का मान बराबर नहीं है (जहां {.} भिन्नात्मक भाग फलन को दर्शाता है) तो :

(A) 'c' केवल परिमेय संख्यायें ले सकता है

(B) 'c' केवल अपरिमेय संख्याये ले सकता है

(C) 'c' अनन्त संख्यायें ले सकता है जिनमें ठीक एक अपरिमेय हो

(D) 'c' अनन्त संख्यायें ले सकता है जिनमें ठीक एक परिमेय हो

**Ans.** D

**Sol.** $\{\ln x\}$ is discontinuous at $\ln x \notin I$

$x \notin e^I$

$x \neq e^I$

18. Let $a = \lim\limits_{x\to 1}\left(\frac{x}{\ln x} - \frac{1}{x\ln x}\right)$; $b = \lim\limits_{x\to 0}\frac{x^3-16x}{4x+x^2}$; $c = \lim\limits_{x\to 0}\frac{\ln(1+\sin x)}{x}$ and $d = \lim\limits_{x\to -1}\frac{(x+1)^3}{3\left[\sin(x+1)-(x+1)\right]}$, then the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is :

(A) Idempotent (B) Involutary (C) Non-singular (D) Nilpotent

माना $a = \lim\limits_{x\to 1}\left(\frac{x}{\ln x} - \frac{1}{x\ln x}\right)$; $b = \lim\limits_{x\to 0}\frac{x^3-16x}{4x+x^2}$; $c = \lim\limits_{x\to 0}\frac{\ln(1+\sin x)}{x}$ तथा $d = \lim\limits_{x\to -1}\frac{(x+1)^3}{3\left[\sin(x+1)-(x+1)\right]}$ है तो आव्यूह $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ होगा :

(A) वर्गसम आव्यूह (B) अर्न्तवलनीय आव्यूह (C) व्युत्क्रमणीय आव्यूह (D) शून्यभावी आव्यूह

**Ans.** D

**Sol.** $a = \lim\limits_{x\to 1}\left(\frac{x^2-1}{x\ln x}\right) = \lim\limits_{x\to 1}\frac{x^2-1}{x(x-1)} = \lim\limits_{x\to 1}\frac{x+1}{x}$

$= 2$

$b = \lim\limits_{x\to 0}\frac{x(x-4)(x+4)}{x(x+4)} = -4$

$c = \lim\limits_{x\to 0}\frac{\ln(1+\sin x)}{x} = \lim\limits_{x\to 0}\frac{\sin x}{x} = 1$

$d = \lim\limits_{x\to -1}\frac{(x+1)^3}{3\left[(x+1) - \frac{(x+1)^3}{3!} + ..... - (x+1)\right]} = -2$

$$\begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix} = 0$$

19. Consider the function $f(x) = \begin{cases} \dfrac{\max\left(x, \dfrac{1}{x}\right)}{\min\left(x, \dfrac{1}{x}\right)}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$, then $\lim\limits_{x\to 0^-}\{f(x)\} + \lim\limits_{x\to 1^-}\{f(x)\} + \lim\limits_{x\to -1^-}[f(x)] =$

(where {.] denotes fraction part function [.] denotes greatest integer function)

माना फलन $f(x) = \begin{cases} \dfrac{\max\left(x, \dfrac{1}{x}\right)}{\min\left(x, \dfrac{1}{x}\right)}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$ है तो $\lim\limits_{x\to 0^-}\{f(x)\} + \lim\limits_{x\to 1^-}\{f(x)\} + \lim\limits_{x\to -1^-}[f(x)] =$

(जहां {.] भिन्नात्मक भाग फलन को दर्शाता है, [.] महत्तम पूर्णांक फलन को दर्शाता है)

(A) 0 (B) 1 (C) 2 (D) 3

**Ans.** A

**Sol.** $f(x) = \begin{cases} \dfrac{1}{x^2} & x < -1 \\ x^2 & -1 < x < 0 \\ 1 & x = 0 \\ \dfrac{1}{x^2} & 0 < x < 1 \\ x^2 & x > 1 \end{cases}$

$$\lim_{x\to 0^-}\{f(x)\} + \lim_{x\to 1^-}\{f(x)\} + \lim_{x\to 1^-}[f(x)]$$

$$\lim_{x\to 0^-}\{x^2\} + \lim_{x\to 1^-}\left\{\frac{1}{x^2}\right\} + \lim_{x\to 1^-}\left[\frac{1}{x^2}\right] = 0 + 0 + 0 = 0$$

20. Let $\alpha = \lim\limits_{n\to\infty} \dfrac{(1^3 - 1^2) + (2^3 - 2^2) + \ldots + (n^3 - n^2)}{n^4}$, then $\alpha$ is equal to :

(A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) non existent

माना $\alpha = \lim\limits_{n\to\infty}\dfrac{\left(1^3-1^2\right)+\left(2^3-2^2\right)+.....+\left(n^3-n^2\right)}{n^4}$ है तो $\alpha$ का मान होगा :

(A) $\dfrac{1}{3}$ (B) $\dfrac{1}{4}$ (C) $\dfrac{1}{2}$ (D) विद्यमान नहीं है

**Ans.** B

**Sol.** $\alpha = \lim\limits_{n\to\infty}\dfrac{\sum n^3-\sum n^2}{n^4}$

$$\alpha = \lim_{n\to\infty}\frac{\left(\frac{n[n+1]}{2}\right)^2-\frac{n(n+1)(2n+1)}{6}}{n^4}=\frac{1}{4}$$

21. If $\lim\limits_{x\to0}\left[1+x\ln(1+b^2)\right]^{\frac{1}{x}} = 2b\sin^2\theta,\ b>0$ and $\theta\in(-\pi,\pi]$, then the value of $\theta$ is : **[JEE-2011]**

यदि $\lim\limits_{x\to0}\left[1+x\ln(1+b^2)\right]^{\frac{1}{x}} = 2b\sin^2\theta,\ b>0$ तथा $\theta\in(-\pi,\pi]$, तो $\theta$ का मान होगा : **[JEE-2011]**

(A) $\pm\dfrac{\pi}{4}$ (B) $\pm\dfrac{\pi}{3}$ (C) $\pm\dfrac{\pi}{6}$ (D) $\pm\dfrac{\pi}{2}$

**Ans.** D

**Sol.** $\lim\limits_{x\to\infty}\left(\dfrac{x^2+x+1}{x+1}-ax-b\right)=4$

$$\Rightarrow \lim_{x\to\infty}\left(\frac{(1-a)x^2+x(1-a-b)+1-b}{x+1}\right)=4$$

$$\Rightarrow \lim_{x\to\infty}\frac{\left((1-a).x+1-a-b+\left(\frac{1-b}{x}\right)\right)}{1+\frac{1}{x}}=4$$

for limit to exist finitely

$1 - a = 0$ and $1 - a - b = 4$

$\Rightarrow a = 1$ and $b = -4$.

## ONE OR MORE THAN ONE CORRECT TYPE

22. Let $f(x)=\begin{cases}|x|, & -2\le x\le 0\\ e^x-1, & 0<x\le 5\end{cases}$, $g(x)=\begin{cases}-x-1, & -2\le x\le 0\\ \dfrac{4x-3}{3}, & 0<x\le 3\end{cases}$ then :

माना $f(x)=\begin{cases}|x|, & -2\le x\le 0\\ e^x-1, & 0<x\le 5\end{cases}$, $g(x)=\begin{cases}-x-1, & -2\le x\le 0\\ \dfrac{4x-3}{3}, & 0<x\le 3\end{cases}$ तो :

(A) $\lim\limits_{x\to 0} f(g(x))=1$

(B) $\lim\limits_{x\to 0^-} f(g(2^x))=1$

(C) $\lim\limits_{x\to 0^+} f(g(2^x))=e^{1/3}-1$

(D) $\lim\limits_{x\to 0^+} f(g(2^x-1))=\lim\limits_{x\to 0^-} f(g(2^x-1))$

**Ans.** ACD

**Sol.** (A) L.H.L. $=\lim\limits_{x\to 0^-} f\left(g(x)\right)=\lim\limits_{x\to 0^-} f(-x-1)$

$=f(-1)=1$

R.H.L. $=\lim\limits_{x\to 0^+} f\left(g(x)\right)=\lim\limits_{x\to 0^+} f\left(\dfrac{4x-3}{3}\right)$

$=f(-1)=1$

(B) & (C) $\lim\limits_{x\to 0} f\left(g\left(2^x\right)\right)=\lim\limits_{x\to 0} f\left(\dfrac{4\cdot 2^x-3}{3}\right)$

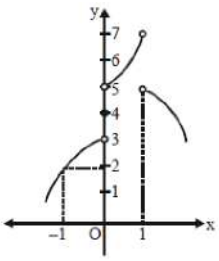
$=f\left(\dfrac{1}{3}\right)=e^{\frac{1}{3}}-1$

(D) $\lim\limits_{x\to 0^+} fog\left(2^x-1\right)=fog\left(0^+\right)=f(-1)=1$

$\lim\limits_{x\to 0^-} fog\left(2^x-1\right)=fog\left(0^-\right)=f(-1)=1$

23. The graph of $y=f(x)$ is as shown. The which of the following hold(s) good? [**Note** : [k] and {k} denote largest integer less than or equal k and fractional part of k respectively]

$y=f(x)$ का आरेख दिया गया है, तो निम्न में से कौनसे कथन सत्य है : (जहाँ [.] तथा {.} महत्तम पूर्णांक तथा भिन्नात्मक भाग फलन दर्शाता है)




(A) $\lim\limits_{x\to 0^+} f([x-\sin x]) = 4$

(B) $\lim\limits_{x\to 0^-} f([x-\tan x]) = 2$

(C) $\lim\limits_{x\to 0^+} f\left(\left\{\dfrac{x}{\tan x}\right\}\right) = 7$

(D) $\lim\limits_{x\to \frac{\pi}{2}^+} f(\{\cos x\}) = 5$

**Ans.** AC

**Sol.** (A) $x > \sin x$

$$\lim_{x\to 0^+} f([x-\sin x]) = \lim_{x\to 0^+} f(0) = 4$$

(B) $x < 0 \Rightarrow x > \tan x$

$\Rightarrow x - \tan x > 0$

$$\lim_{x\to 0} f([x-\tan x]) = \lim_{x\to 0^+} f(0) = 4$$

(C) $\tan x > x$

$$\Rightarrow \frac{x}{\tan x} < 1$$

$$\lim_{x\to 0^+} f\left\{\frac{x}{\tan x}\right\} = \lim_{x\to 0^+} f(1^-) = 7$$

(D) $$\lim_{x\to \frac{\pi}{2}^+} f\{\cos x\} = \lim_{x\to \frac{\pi}{2}^+} f\{-1\} = \lim_{x\to \frac{\pi}{2}^+} f(-1) = 7$$

24. Let $f(x) = \dfrac{\cos^{-1}(\cos x)}{x}$, then which of the following is (are) correct : (where [.] denotes greatest integer function)

माना $f(x) = \dfrac{\cos^{-1}(\cos x)}{x}$, तो निम्न में से कौनसे कथन सत्य है : (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है ) ?

(A) $\lim\limits_{x\to 0^+} [f(x)] = 1$

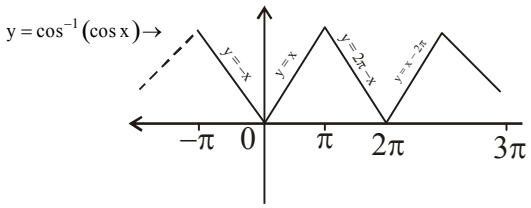
(B) $\lim\limits_{x\to 0^-} [f(x)] = 1$

(C) $\lim\limits_{x\to 0^+} [f(x)] = 0$

(D) $\lim\limits_{x\to 0^-} [f(x)] = -1$

**Ans.** AD

**Sol.** $f(x)=\dfrac{\cos^{-1}(\cos x)}{x}$




So, clearly when $x \to 0^+$,

$$\to \lim_{x\to 0^+}\left[\frac{\cos^{-1}(\cos x)}{x}\right]$$

$$=\lim_{h\to 0}\left[\frac{\cos^{-1}(\cos(0+h))}{0+h}\right]$$

$$=\lim_{h\to 0}\left[\frac{h}{h}\right]=1$$

$$=\lim_{x\to 0^-}\left[\frac{\cos^{-1}(\cos x)}{x}\right]$$

$$=\lim_{h\to 0}\left[\frac{\cos^{-1}(\cos(0-h))}{0-h}\right]=\lim_{h\to 0}\left[\frac{h}{-h}\right]$$

$=-1$

$\therefore$ Hence, Option (A), (D) are correct.

25. Which of the following limits does not exist? (where [.] represents greatest integer function)
निम्न में से कौनसी सीमा विद्यमान नहीं होगी : (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

(A) $\lim\limits_{x\to 1^+}\left([x]\right)^{\frac{1}{x-1}}$

(B) $\lim\limits_{x\to 3}\dfrac{(x^2-9-\sqrt{x^2-6x+9})}{|x-1|-2}$

(C) $\lim\limits_{x\to 0^+}(x)^{\ln x}$

(D) $\lim\limits_{x\to 0^+}\left(\dfrac{1-\cos(\sin^2 x)}{x^2}\right)^{\frac{\ln(1-2x^2)}{\sin^2 x}}$

**Ans.** BCD

**Sol.** (A) $\lim\limits_{x\to 1^+}[x]^{\frac{1}{x-1}}=\lim\limits_{x\to 1^+}1^{\frac{1}{x-1}}=1$

(B) $\lim\limits_{x\to3}\dfrac{x^2-9\sqrt{(x-3)^2}}{|x-1|-2}=\lim\limits_{x\to3}\dfrac{(x-3)(x+3)-|x-3|}{(x-3)}$

L.H.L. $=\lim\limits_{x\to3^-}\dfrac{(x-3)(x+3)+(x-3)}{(x-3)}=\lim\limits_{x\to3^-}x+4=7$

R.H.L. $=\lim\limits_{x\to3^+}\dfrac{(x-3)(x+3)-(x-3)}{(x-3)}=\lim\limits_{x\to3^+}x+2=5$

LHL $\neq$ RHL limit DNE

(C) $\lim\limits_{x\to0^+}(x)^{\ln x}=\lim\limits_{x\to0^+}e^{(\ln x)^2}=e^{\infty}\neq$ finite

Limit DNE

(D) $\lim\limits_{x\to0^+}\left(\dfrac{1-\cos\sin^2 x}{x^2}\right)^{\frac{\ln(1-2x^2)}{\sin^2 x}}=\lim\limits_{x\to0^+}(f(x))^{g(x)}$

$\lim\limits_{x\to0^+}f(x)=\lim\limits_{x\to0^+}\dfrac{1-\cos\sin^2 x}{x^2}=\lim\limits_{x\to0^+}\dfrac{1-\cos\sin^2 x}{\sin^4 x}\times\dfrac{\sin^2 x}{x^2}\times\sin^2 x=\dfrac{1}{2}\times1\times0=0$

$\lim\limits_{x\to0^+}g(x)=\lim\limits_{x\to0^+}\dfrac{\ln(1-2x^2)}{\sin^2 x}=\lim\limits_{x\to0^+}\dfrac{-2x^2}{\sin^2 x}=-2$

$\lim\limits_{x\to0^+}(f(x))^{g(x)}=0^{-2}\neq$ Neither finite nor indeterminate form.

26. Which of the following limits are non existent ?

(A) $\lim\limits_{x\to0}\dfrac{1}{x}\sqrt{\dfrac{1-\cos x}{1+\cos x}}$ (B) $\lim\limits_{x\to0}\dfrac{|x|}{x}$ (C) $\lim\limits_{x\to0}\dfrac{\cot^{-1}\frac{1}{x}}{x}$

(D) $\lim\limits_{x\to\pi/2}\cos^{-1}[\cot x]$ where [ ] is greatest integer function

निम्न में से कौनसी सीमा विद्यमान नही होगी :

(A) $\lim\limits_{x\to0}\dfrac{1}{x}\sqrt{\dfrac{1-\cos x}{1+\cos x}}$ (B) $\lim\limits_{x\to0}\dfrac{|x|}{x}$ (C) $\lim\limits_{x\to0}\dfrac{\cot^{-1}\frac{1}{x}}{x}$

(D) $\lim\limits_{x\to\pi/2}\cos^{-1}[\cot x]$ (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

**Ans.** ABCD

**Sol.** (A) $\lim\limits_{x\to0}\dfrac{|\tan x|}{x}$

$\text{LHL} = -1, \text{RHL} = 1$

(B) $\lim\limits_{x\to 0} \dfrac{|x|}{x} \Rightarrow \text{LHL} = -1, \text{RHL} = 1$

(C) $\lim\limits_{x\to 0} \dfrac{\cot^{-1}\dfrac{1}{x}}{x}$

$\text{LHL} = \lim\limits_{x\to 0^-} \dfrac{\pi + \tan^{-1} x}{x} \neq \text{finite}(\text{DNE})$

$\text{RHL} = \lim\limits_{x\to 0^-} \dfrac{\tan^{-1} x}{x} = 1$

(D) $\lim\limits_{x\to 0} \cos^{-1}[\cot x]$

$\text{LHL} = \lim\limits_{x\to 0^-} \cos^{-1}(-1) = \pi$

$\text{RHL} = \lim\limits_{x\to 0^+} \cos^{-1}(0) = 0$

27. Let $f(x) = \begin{cases} \dfrac{x}{\tan x} & , \quad -\infty < x < 0 \\ 2 & , \quad x = 0 \\ \dfrac{\ln(1+2x)}{x} & , \quad 0 < x < \infty \end{cases}$ and $g(x) = \begin{cases} x+4 & , \quad -\infty < x < 1 \\ x^2 - 5x + 11 & , \quad 1 \le x < 2 \\ x - 3 & , \quad 2 \le x < \infty \end{cases}$, then which of the following is(are) correct?

माना $f(x) = \begin{cases} \dfrac{x}{\tan x} & , \quad -\infty < x < 0 \\ 2 & , \quad x = 0 \\ \dfrac{\ln(1+2x)}{x} & , \quad 0 < x < \infty \end{cases}$ तथा $g(x) = \begin{cases} x+4 & , \quad -\infty < x < 1 \\ x^2 - 5x + 11 & , \quad 1 \le x < 2 \\ x - 3 & , \quad 2 \le x < \infty \end{cases}$, तो निम्न में से कौनसे कथन सत्य है :

(A) $\lim\limits_{x\to 0^-} g(f(x)) = 5$

B) $\lim\limits_{x\to 0^-} g(f(x)) = 7$

(C) $\lim\limits_{x\to 0^+} g(f(x)) = 5$

(D) $\lim\limits_{x\to 0^+} g(f(x)) = -1$

**Ans.** AC

**Sol.** $\dfrac{x}{\tan x} < 1$

(A) $\lim\limits_{x\to 0^-} g(f(x)) = \lim\limits_{x\to 0^-} g\left(\dfrac{x}{\tan x}\right) = \lim\limits_{x\to 0^-} g\left(1^-\right) = 5$

(C) $\lim\limits_{x\to 0^+} g(f(x)) = \lim\limits_{x\to 0^+} g\left(\dfrac{x}{\tan x}\right) = \lim\limits_{x\to 0^+} g\left(1^-\right) = 5$

28. Let $f(x) = \begin{cases} |x-2| + a^2 - 6a + 9 , & x < 2 \\ 5 - 2x , & x \geq 2 \end{cases}$

If $\lim\limits_{x\to 2}$ [f(x)] exists, the possible values a can take is/are (where [.] represents the greatest integer function)

माना $f(x) = \begin{cases} |x-2| + a^2 - 6a + 9 , & x < 2 \\ 5 - 2x , & x \geq 2 \end{cases}$

यदि $\lim\limits_{x\to 2}$ [f(x)] विद्यमान है, तो a के सम्भावित मान जो a ले सकता है (जहां [.] महत्तम पूर्णांक फलन को दर्शाता है)

(A) 2 (B) $\dfrac{5}{2}$ (C) 3 (D) $\dfrac{7}{2}$

**Ans.** BCD

**Sol.** $LHL = \lim\limits_{x\to 2^-} \left[|x-2| + (a-3)^2\right] = \left[(a-3)^2\right]$

$RHL = \lim\limits_{x\to 2^+} [5-2x] = 0$

$(a-3)^2 \in [0,1) \Rightarrow a \in (2,4)$

29. Let $L = \underset{x\to 0}{Lim} \dfrac{a - \sqrt{a^2 - x^2} - \dfrac{x^2}{4}}{x^4}$, $a > 0$. If L is finite, then : **[JEE-2009]**

माना कि $L = \underset{x\to 0}{Lim} \dfrac{a - \sqrt{a^2 - x^2} - \dfrac{x^2}{4}}{x^4}$, $a > 0$ यदि L एक परिमित संख्या हो तो सही विकल्प चुनो : **[JEE-2009]**

(A) $a = 2$ (B) $a = 1$ (C) $L = \dfrac{1}{64}$ (D) $L = \dfrac{1}{32}$

**Ans.** AC

**Sol.** $\dfrac{a - a\left(1 - \dfrac{x^2}{a^2}\right)^{\frac{1}{2}} - \dfrac{x^2}{4}}{x^4} = \dfrac{a - a\left(1 - \dfrac{x^2}{2a^2} - \dfrac{1}{8}\dfrac{x^4}{a^4}\right) - \dfrac{x^2}{4}}{x^4}$

$a = 2$, (coefficient of $x^2 = 0$)

$\therefore\ L = \dfrac{1}{16}$

1. Let $x_1 = -\sqrt{3}$, $x_2 = -\sqrt{3+\sqrt{3}}$, $x_3 = -\sqrt{3+\sqrt{3+\sqrt{3}}}$, ..... . Evaluate $\lim_{n\to\infty} x_n$ :

यदि $x_1 = -\sqrt{3}$, $x_2 = -\sqrt{3+\sqrt{3}}$, $x_3 = -\sqrt{3+\sqrt{3+\sqrt{3}}}$, ..... हो, तो $\lim_{n\to\infty} x_n$ का मान ज्ञात कीजिए :

**Ans.** $\frac{-1-\sqrt{13}}{2}$

**Sol.** $x_1 = -\sqrt{3}$ $\quad x_2 = -\sqrt{3+\sqrt{3}}$

$x_3 = -\sqrt{3+\sqrt{3+\sqrt{3}}}$

$\lim_{n\to\infty} x_n = x_\infty = x$ $\quad$ Let $= -\sqrt{3+\sqrt{3+\sqrt{3+.....}}}$

$\Rightarrow x = -\sqrt{3-x}$

$\Rightarrow x^2 + x - 3 = 0$ $\quad x = \frac{-1\pm\sqrt{13}}{2}$

Also $x_i < 0$ $\quad \Rightarrow x_\infty = x < 0$

hence $\quad x = \frac{-1-\sqrt{13}}{2}$

2. Let $L = \lim_{x\to0} \frac{e^{x\sin(2007)x} - 1}{x\,\ell n(1+x)}$ and $M = \lim_{x\to\infty} x\left(2x - \left(\sqrt[3]{x^3+x^2+1} + \sqrt[3]{x^3-x^2+1}\right)\right)$ then find the value of LM.

यदि $L = \lim_{x\to0} \frac{e^{x\sin(2007)x} - 1}{x\,\ell n(1+x)}$ और $M = \lim_{x\to\infty} x\left(2x - \left(\sqrt[3]{x^3+x^2+1} + \sqrt[3]{x^3-x^2+1}\right)\right)$ तो LM का मान बताओं।

**Ans.** 446

**Sol.** $L = \lim_{x\to0} \frac{e^{x\sin(2007)x} - 1}{x\cdot\ell n(1+x)}$

$L = \lim_{x\to0} \frac{e^{x\cdot\sin(2007)x} - 1}{x\cdot\sin(2007)x} \cdot \frac{\sin(2007)x}{2007x} \frac{(2007)x}{\ell n(1+x)}$

$L = .1.1..2007 = 2007$

$M = \lim_{x\to\infty} x\left[2x - \left(x^3+x^2+1\right)^{\frac{1}{3}} - \left(x^3-x^2+1\right)^{\frac{1}{3}}\right]$

$$= \lim_{x\to\infty} x^2\left[2-\left(1+\frac{1+x^2}{x^3}\right)^{\frac{1}{3}}-\left(1+\frac{1-x^2}{x^3}\right)^{\frac{1}{3}}\right]$$

$$= \lim_{x\to\infty} x^2\left[2-1-\frac{1}{3}a+\frac{2}{9}\frac{1}{2!}a^2-1-\frac{1}{3}b+\frac{2}{9}\frac{1}{2!}b^2\right]$$

$$\begin{cases} a=\dfrac{1+x^2}{x^3} \\ b=\dfrac{1-x^2}{x^3} \end{cases}$$

$$= \lim_{x\to\infty} x^2\left[-\frac{1}{3}\frac{1+x^2}{x^3}-\frac{1}{3}\frac{1-x^2}{x^2}+\frac{2}{9}a^2+\frac{2}{9}b^2\right]$$

$$= \lim_{x\to\infty} -\frac{2}{3}\frac{x^2}{x^3}+\frac{1}{9}\cdot x^2\left\{\frac{\left(1+x^2\right)^2}{x^6}+\frac{\left(1-x^2\right)^2}{x^6}\right\}$$

$$= 0+\frac{2}{9}\frac{1+x^4}{x^4}+0=\frac{2}{9}$$

$$LM = (2007)\ \frac{2}{9} = 446$$

3. Let $L = \lim_{n\to\infty} \dfrac{1}{\frac{1}{\sqrt{2}}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}}\ldots\ldots\ldots \text{n terms}}$. Find the value of $\left(2\sqrt{3}\cot L\right)$.

माना $L = \lim_{n\to\infty} \dfrac{1}{\frac{1}{\sqrt{2}}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}}\ldots\ldots\ldots \text{n पद}}$. $\left(2\sqrt{3}\cot L\right)$ का मान होगा।

**Ans.** 0

**Sol.** $T_1 = \dfrac{1}{\sqrt{2}} = \cos\theta \qquad \theta = \dfrac{\pi}{4}$

$$T_2 = \sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}} = \sqrt{\frac{1+\cos\theta}{2}} = \cos\frac{\theta}{2}$$

$$T_3 = \sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}} = \sqrt{\frac{1+\cos\frac{\theta}{2}}{2}} = \cos\left(\frac{\theta}{2^2}\right)$$

$$T_n = \cos\left(\frac{\theta}{2^{n-1}}\right)$$

$$L = \lim_{n\to\infty} \frac{1}{T_1, T_2, .....T_n}$$

$$L = \lim_{n\to\infty} \frac{1}{\cos\theta \cdot \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2^5}\right).....\cos\left(\frac{\theta}{2^{n-1}}\right)}$$

$$\lim_{n\to\infty} \frac{1}{\frac{\sin\left(2^n \cdot \frac{\theta}{2^{n-1}}\right)}{2^n \cdot \sin\left(\frac{\theta}{2^{n-1}}\right)}}$$

$$= \lim_{n\to\infty} \frac{2^n \cdot \sin\left(\frac{\theta}{2^{n-1}}\right)}{\sin 2\theta}$$

$$= \lim_{n\to\infty} \frac{2\theta}{\sin 2\theta} \cdot \frac{\sin\left(\frac{\theta}{2^{n-1}}\right)}{\left(\frac{\theta}{2^{n-1}}\right)}$$

$$L = \frac{2\theta}{\sin 2\theta}; \qquad \theta = \frac{\pi}{4}$$

$$L = \frac{\pi}{2} \qquad \cot L = 0$$

$$2\sqrt{3}\cot L = 0$$

4. $\lim\limits_{x\to\infty} x^2 \sin\left(\ell n \sqrt{\cos\frac{\pi}{x}}\right).$

**Ans.** $-\frac{\pi^2}{4}$

**Sol.** $t = \lim\limits_{x\to\infty} x^2 \sin\left(\ell n\sqrt{\cos\left(\frac{\pi}{x}\right)}\right)$

put $x = \frac{1}{y}$ as $x \to \infty,$ $y \to O^+$

$$\Rightarrow t=\lim_{y\to 0}\frac{\sin\left\{\frac{1}{2}\ell n\left(\cos \pi y\right)\right\}}{\frac{1}{2}\ell n\left(\cos \pi y\right)}\cdot\frac{1}{2}\frac{\ell n\cos \pi y}{y^2}$$

$$\Rightarrow t=\lim_{y\to 0}\frac{1}{2}\cdot\frac{\ell n\left[1-\{1-\cos \pi y\}\right]}{1-\cos \pi y}\cdot\frac{1-\cos \pi y}{\left(\pi y\right)^2}\pi^2$$

$$\Rightarrow t=\frac{1}{2}(-1)\frac{1}{2}\cdot\pi^2$$

$$\Rightarrow t=\frac{-\pi^2}{4}$$

5. $\lim\limits_{x\to\infty}\left[\cos\left(2\pi\left(\frac{x}{1+x}\right)^a\right)\right]^{x^2}$ $a\in Q$

**Ans.** $e^{-2\pi^2a^2}$

**Sol.** $L=\lim\limits_{x\to\infty}\left[\cos\left\{2\pi\left(\frac{x}{1+x}\right)^a\right\}\right]^{x^2}$ $a\in Q$

$$L=e^{\lim\limits_{x\to\infty}\left[\cos 2\pi\left(\frac{x}{1+x}\right)^a-1\right]\cdot x^2}$$

$$L=e^{\lim\limits_{x\to\infty}\frac{\cos\left\{2\pi-2\pi\left(\frac{\pi}{1+x}\right)^a\right\}-1}{4\pi^2\left[1-\left(\frac{x}{1+x}\right)^a\right]^2}\cdot 4\pi^2\cdot x^2\left[1-\left(\frac{x}{1+x}\right)^a\right]^2}$$

Put $x=\frac{1}{y}$, $x\to\infty$, $y\to 0^+$

$$L=e^{-\frac{4}{2}\pi^2\lim\limits_{y\to 0}\frac{1}{y^2}\left[1-(1+y)^{-a}\right]^2}$$

$$=e^{-2\pi^2\lim\limits_{y\to 0}\frac{\left[1-ay-\frac{(-a)(-a-1)}{21}y^2.....\right]^2}{y^2}}$$

$$=e^{-2\pi^2a^2}$$

6. Let $f(x)=\frac{\sin^{-1}(1-\{x\}).\cos^{-1}(1-\{x\})}{\sqrt{2\{x\}}\ .\ (1-\{x\})}$ then find $\lim\limits_{x\to 0^+}$ f(x) and $\lim\limits_{x\to 0^-}$ f(x), where $\{x\}$ denotes the fractional part function.

माना $f(x)=\frac{\sin^{-1}(1-\{x\}).\cos^{-1}(1-\{x\})}{\sqrt{2\{x\}}\,.\,(1-\{x\})}$ तो $\underset{x\to0^+}{\text{Lim}}\ f(x)$ तथा $\underset{x\to0^-}{\text{Lim}}\ f(x)$ का मान ज्ञात करो जहाँ पर $\{x\}$ भिन्नांश फलन है।

**Ans.** $\frac{\pi}{2},\frac{\pi}{2\sqrt{2}}$

**Sol.** $\text{R.H.L.}=\lim\limits_{x\to0^+}\frac{\sin^{-1}(1-x)\cos^{-1}(1-x)}{\sqrt{2x}\,(1-x)}$

$=\lim\limits_{x\to0^+}\frac{\sin^{-1}(1-x)}{(1-x)}\frac{\sin^{-1}\left[\sqrt{1-(1-x)^2}\right]}{\sqrt{2x-x^2}}\frac{\sqrt{2x-x^2}}{\sqrt{2x}}$

$=\frac{\pi}{2}\cdot(1)\cdot(1)$

$=\frac{\pi}{2}$

$\text{L.H.S.}=\lim\limits_{x\to0^-}\frac{\sin^{-1}(1-(x+1))}{(1-(x+1))}\frac{\cos^{-1}(1-(1+x))}{\sqrt{2(1+x)}}$

$\lim\limits_{x\to0^-}\frac{\sin^{-1}(-x)}{(-x)}\cdot\frac{\cos^{-1}(-x)}{\sqrt{2(1+x)}}$

$-x=h,\ x\to0^-,\ h\to0^+$

$=\lim\limits_{h\to0}\frac{\cos^{-1}(h)}{\sqrt{2}\sqrt{1-h}}$

$=\frac{1}{\sqrt{2}}\lim\limits_{h\to0}\frac{\cos^{-1}(h)}{\sqrt{1-h}}=\frac{1}{\sqrt{2}}\frac{\pi}{2}$

7. $\underset{n\to\infty}{\text{Lim}}\left(\frac{\sqrt{n^2+n}-1}{n}\right)^{2\sqrt{n^2+n}-1}$

**Ans.** $e^{-1}$

**Sol.** $\lim\limits_{n\to\infty}\left(\frac{\sqrt{n^2+n}-1}{n}\right)^{2\sqrt{n^2+n}-1}\qquad(1^\infty)$

$=e^{\lim\limits_{n\to\infty}\left[\frac{\sqrt{n^2+n}-(n+1)}{n}\right]\left[2\sqrt{n^2+n}-1\right]}$

$$= e^{\lim_{n\to\infty} \frac{-n-1}{n\left[\sqrt{n^2+n}+n+1\right]} \frac{4\left(n^2+n\right)-1}{2\sqrt{n^2+n}+1}}$$

$$= e^{\lim_{n\to\infty} \frac{-(n+1)}{n} \frac{4\left(n^2+n\right)-1}{\left[\sqrt{n^2+n}+n+1\right]\left[2\sqrt{n^2+n}+1\right]}}$$

$$= e$$

$$= e^{-1}$$

8. $\lim\limits_{x\to a} \dfrac{1}{(a^2-x^2)^2}\left(\dfrac{a^2+x^2}{ax} - 2\sin\left(\dfrac{a\pi}{2}\right)\sin\left(\dfrac{\pi x}{2}\right)\right)$ where a is an odd integer.

$\lim\limits_{x\to a} \dfrac{1}{(a^2-x^2)^2}\left(\dfrac{a^2+x^2}{ax} - 2\sin\left(\dfrac{a\pi}{2}\right)\sin\left(\dfrac{\pi x}{2}\right)\right)$ जहाँ a एक विषम पूर्णांक है।

**Ans.** $\dfrac{\pi^2a^2+4}{16a^4}$

**Sol.** $\lim\limits_{x\to a} \dfrac{1}{(a+x)^2\,ax} \qquad \dfrac{a^2+x^2-2ax.\sin\left(\dfrac{a\pi}{2}\right)\sin\left(\dfrac{\pi x}{2}\right)}{(a-x)^2}$

$$= \frac{1}{4a^4}\lim_{x\to a}\frac{a^2+x^2-2a\cdot\sin\left(\dfrac{a\pi}{2}\right)x\cdot\sin\dfrac{\pi}{2}x}{(a-x)^2}$$

By L – Hospital rule

$$= \frac{1}{4a^4}\lim_{x\to a}\frac{x-a\sin\left(\dfrac{a\pi}{2}\right)\left\{\sin\dfrac{\pi}{2}x+\dfrac{\pi}{2}x\cdot\cos\dfrac{\pi}{2}x^4\right\}}{x-a}$$

Again L – Hospital rule

$$= \frac{1}{4a^4}\lim_{x\to a}\left[1-a\sin\left(\frac{a\pi}{2}\right)\left\{\frac{\pi}{2}\cos\frac{\pi}{2}x+\frac{\pi}{2}\cos\frac{\pi}{2}x-\frac{\pi^2}{4}x\sin\frac{\pi}{2}x\right\}\right]$$

$$= \frac{1}{4a^4}\left[1-a\sin\frac{a\pi}{2}\left(\pi.\cos\frac{\pi}{2}a-\frac{\pi^2}{4}a\sin\frac{\pi}{2}a\right)\right]$$

$$= \frac{1}{4a^4}\left[1+\frac{a^2\pi^2}{4}\sin^2\left(\frac{\pi}{2}a\right)\right]$$

$$= \frac{4+(a\pi)^2}{16a^4}$$

9. If $L = \lim_{x\to 1} \frac{(1-x)(1-x^2)(1-x^3).....(1-x^{2n})}{[(1-x)(1-x^2)(1-x^3).....(1-x^n)]^2}$ then show that L can be equal to :

(a) $\prod_{r=1}^{n} \frac{n+r}{r}$ (b) $\frac{1}{n!}\prod_{r=1}^{n}(4r-2)$

(c) The sum of the coefficients of two middle terms in the expansion of $(1+x)^{2n-1}$.

(d) The coefficient of $x^n$ in the expansion of $(1+x)^{2n}$.

यदि $L = \lim_{x\to 1} \frac{(1-x)(1-x^2)(1-x^3).....(1-x^{2n})}{[(1-x)(1-x^2)(1-x^3).....(1-x^n)]^2}$ तो सिद्ध करो कि L निम्नलिखित के बराबर हो सकता है :

(a) $\prod_{r=1}^{n} \frac{n+r}{r}$ (b) $\frac{1}{n!}\prod_{r=1}^{n}(4r-2)$

(c) $(1+x)^{2n-1}$ के विस्तार में दो मध्य पदों के गुणांकों का योग

(d) $(1+x)^{2n}$ के विस्तार में $x^n$ का गुणांक

**Ans.**

**Sol.** $L = \lim_{x\to 1} \frac{(1-x^{n+1})(1-x^{n+2}).....(1-x^{2n})}{(1-x)(1-x^2).....(1-x^n)}$

$L = \lim_{x\to 1} \frac{1-x^{n+1}}{1-x}\cdot\frac{1-x^{n+2}}{1-x^2}\cdot\frac{1-x^{2n}}{1-x^n}$

$= \frac{(n+1)}{1}\frac{(n+2)}{2}.....\frac{(2n)}{n} = \prod_{r=1}^{n}\frac{n+r}{r}$ (a)

$= \frac{1\cdot 2\cdot.....\cdot n}{1\cdot 2\cdot 3\cdot.....n}\cdot\frac{(n+1).....(2n)}{1\cdot 2\cdot.....n} = \frac{(2n)!}{(n!)^2} = {}^{2n}C_n$ _____(1)

(c) $(1+x)^{2n-1} \to$ cofficients of middle term $= {}^{2n-1}C_{n-1}$ and ${}^{2n-1}C_n$ (c)

sum $= {}^{2n-1}C_{n-1} + {}^{2n-1}C_n = {}^{2n}C_n$

from (1)

$\Rightarrow \frac{[1\cdot 3\cdot 5\cdot.....(2n-1)]}{n!n!}(2\cdot 4\cdot.....2n)$

$\Rightarrow \frac{2\cdot 6\cdot 10.....(4n-2)}{1\cdot 2\cdot 3\cdot\quad n}\cdot\frac{(1\cdot 2\cdot....n)}{n!}$

$\prod_{r=1}^{n}\frac{4r-2}{n!} = \frac{1}{n!}\prod_{r=1}^{n}(4r-2)$ (b)

10. Let $x_0 = 2\cos\frac{\pi}{6}$ and $x_n = \sqrt{2+x_{n-1}}$, $n = 1, 2, 3, .....$, find $\lim\limits_{n\to\infty} 2^{(n+1)}\cdot\sqrt{2-x_n}$.

माना कि $x_0 = 2\cos\frac{\pi}{6}$ तथा $x_n = \sqrt{2+x_{n-1}}$, $n = 1, 2, 3, .....$, तो $\lim\limits_{n\to\infty} 2^{(n+1)}\cdot\sqrt{2-x_n}$ का मान बताओं।

**Ans.** $\frac{\pi}{3}$

**Sol.** $x_0 = 2\cos\frac{\pi}{6}$

$$x_1 = \sqrt{2+2\cos\frac{\pi}{6}} = 2\cos\left(\frac{\pi}{6}\cdot\frac{1}{2}\right)$$

$$x_2 = \sqrt{2+2\cos\left(\frac{\pi}{6}\cdot\frac{1}{2}\right)} = 2\cos\left(\frac{\pi}{6}\cdot\frac{1}{2^2}\right)$$

llly $x_n = 2\cos\left(\frac{\pi}{6}\frac{1}{2^n}\right)$

Now $\lim\limits_{n\to\infty} 2^{n+1}\sqrt{2-2\cos\frac{\theta}{2^n}}$ $\qquad\qquad \theta = \frac{\pi}{6}$

$$= \lim_{n\to\infty} 2^{n+1}2\cdot\sin\left(\frac{\theta}{2^{n+1}}\right)$$

$$= \lim_{n\to\infty} 2\theta\cdot\frac{\sin\left(\frac{\theta}{2^{n+1}}\right)}{\left(\frac{\theta}{2^{n+1}}\right)} = 2\theta = 2\left(\frac{\pi}{6}\right) = \frac{\pi}{3}$$

11. $\lim\limits_{x\to0}\left[\frac{\ell n\,(1+x)^{1+x}}{x^2} - \frac{1}{x}\right]$.

**Ans.** 1/2

**Sol.** $\lim\limits_{x\to0}\frac{(1+x)\ell n(1+x)-x}{x^2}$

$$= \lim_{x\to0}\frac{(1+x)\left[x-\frac{x^2}{2}+\frac{x^3}{3}.....\right]-x}{x^2} = \frac{1}{2}$$

12. Let $L = \prod_{n=3}^{\infty}\left(1-\frac{4}{n^2}\right)$; $M = \prod_{n=2}^{\infty}\left(\frac{n^3-1}{n^3+1}\right)$ and $N = \prod_{n=1}^{\infty}\frac{(1+n^{-1})^2}{1+2n^{-1}}$, then find the value of $L^{-1} + M^{-1} + N^{-1}$.

माना कि $L = \prod_{n=3}^{\infty}\left(1-\frac{4}{n^2}\right)$ ; $M = \prod_{n=2}^{\infty}\left(\frac{n^3-1}{n^3+1}\right)$ तथा $N = \prod_{n=1}^{\infty}\frac{(1+n^{-1})^2}{1+2n^{-1}}$ तो $L^{-1} + M^{-1} + N^{-1}$ का मान ज्ञात कीजिए।

**Ans.** 8

**Sol.** $L = \prod_{n=3}^{\infty}\frac{(n-2)(n+2)}{n\quad n} = \lim_{k\to\infty}\prod_{n=3}^{k}\frac{n-2}{n}\cdot\frac{n+2}{n}$

$$L = \lim_{k\to\infty}\left(\frac{1}{3}\cdot\frac{5}{3}\right)\left(\frac{2}{4}\cdot\frac{6}{4}\right)\left(\frac{3}{5}\cdot\frac{7}{5}\right)\cdot\left(\frac{4}{6}\cdot\frac{8}{6}\right).....\left(\frac{k-2}{k^3}\cdot\frac{k+2}{k}\right)$$

$$= \lim_{k\to\infty}\left(\frac{1}{3}\right)\left(\frac{2}{4}\right).....\left(\frac{k-3}{k-1}\right)\left(\frac{k+2}{k}\right)$$

$$L = \frac{1}{6}$$

$$M = \lim_{k\to\infty}\prod_{n=2}^{k}\frac{(n-1)}{(n+1)}\frac{\left(n^2+n+1\right)}{\left(n^2-n+1\right)}$$

$$= \lim_{k\to\infty}\left(\frac{1}{3}\cdot\frac{7}{3}\right)\left(\frac{2}{4}\cdot\frac{13}{7}\right)\cdot\left(\frac{3}{5}\cdot\frac{21}{13}\right).....\left(\frac{k-1}{k+1}\cdot\frac{k^2+k+1}{k^2-k+1}\right)$$

$$= \lim_{k\to\infty}\frac{1\cdot 2}{3}\cdot\frac{k^2+k+1}{k\cdot(k+1)} = \frac{2}{3}$$

$$N = \lim_{k\to\infty}\prod_{n=1}^{k}\frac{(n+1)^2}{n(n+2)} = \lim_{k\to\infty}\left(\frac{2}{1}\cdot\frac{2}{3}\right)\left(\frac{3}{2}\cdot\frac{3}{4}\right)\cdot\left(\frac{4}{3}\cdot\frac{4}{5}\right).....\left(\frac{k}{k-1}\cdot\frac{k}{k+1}\right)\left(\frac{k+1}{k}\cdot\frac{k+1}{k+2}\right)$$

$$\lim_{k\to\infty}\frac{2}{1}\cdot\frac{k+1}{k+2} = 2$$

$$L^{-1} + M^{-1} + N^{-1} = 6 + \frac{3}{2} + \frac{1}{2} = 8$$

13. Let $f(x) = \lim_{n\to\infty}\sum_{n=1}^{n}3^{n-1}\sin^3\frac{x}{3^n}$ and $g(x) = x - 4f(x)$. Evaluate $\lim_{x\to 0}(1+g(x))^{\cot x}$.

माना कि $f(x) = \lim_{n\to\infty}\sum_{n=1}^{n}3^{n-1}\sin^3\frac{x}{3^n}$ तथा $g(x) = x - 4f(x)$ तो $\lim_{x\to 0}(1+g(x))^{\cot x}$ का मान ज्ञात कीजिए।

**Ans.** g (x) = sin x and $l = e$

**Sol.**

Now

14. If $f(n, \theta)=$ , then compute

यदि $f(n, \theta)=$ , तो का मान ज्ञात कीजिए।

**Ans.**

**Sol.**

Now

15. If  = 6, then find the value of a.

यदि  = 6 तो a का मान ज्ञात कीजिये।

**Ans.** $e^3$

**Sol.** $x \to 0^+$

Now

16. Evaluate .

का मान ज्ञात कीजिये।

**Ans.**

**Sol.**

neglate $y^2$ and higher power

17. Through a point A on a circle, a chord AP is drawn & on the tangent at A a point T is taken such that AT = AP. If TP produced meet the diameter through A at Q, prove that the limiting value of AQ when P moves upto A is double the diameter of the circle.

वृत्त पर स्थित एक बिन्दु A से एक जीवा AP खींची गई है तथा A पर स्पर्श रेखा पर एक बिन्दु T इस प्रकार से लिया गया है कि AT = AP. यदि TP रेखा को बढ़ाने पर यह A से खीचीं जाने वाली व्यास को यदि Q पर मिलती हो तो सिद्ध करों कि जब $P \to A$ तो AQ का मान वृत्त की व्यास का दुगना होगा।

**Ans.**

**Sol.** $\Delta ACM$



$AP = AT = 2r \sin\theta$



$\Delta AT\ Q$







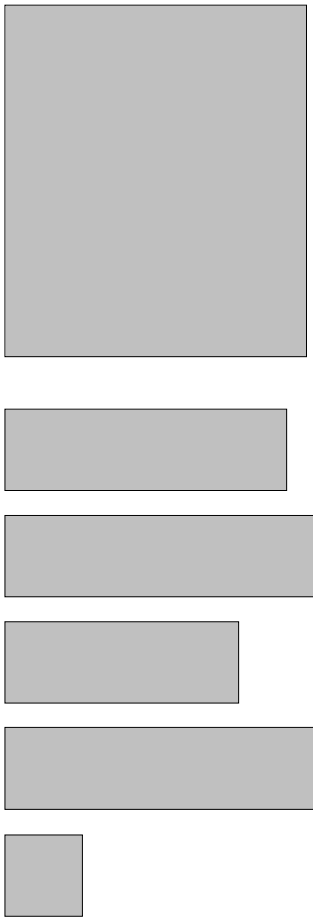
as $P \to A$ $Q \to 0$ $AQ = 4r = 2(2r) = 2$ (diameter)

18. At the end points A, B of the fixed segment of length L, lines are drawn meeting in C and making angles $\theta$ and $2\theta$ respectively with the given segment. Let D be the foot of the altitude CD and let $x$ represents the length of AD. Find the value of $x$ as $\theta$ tends to zero i.e.  .

L लम्बाई वाले रेखा खण्ड के सिरे A व B से, रेखाखण्ड से क्रमशः $\theta$ तथा $2\theta$ कोण बनाती हुयी, रेखा खीची जाती है जो कि C पर मिलती है। यदि रेखा खण्ड पर C से डाले गये लम्ब CD हो तथा AD की लम्बाई x है तो  का मान ज्ञात करो।

**Ans.** 

**Sol.**



19. At the end-points and the midpoint of a circular arc AB tangent lines are drawn, and the points A and B are joined with a chord. Prove that the ratio of the areas of the two triangles thus formed tends to 4 as the arc AB decreases indefinitely.

किसी वृत्तीय चाप AB के अन्त बिन्दुओं (End points) तथा मध्य बिन्दु (mid point) पर स्पर्श रेखायें खीची जाती है तथा बिन्दु A तथा Bको एक जीवा के द्वारा मिलाया जाता है। सिद्ध करों कि जब चाप AB की लम्बाई घटते–घटते शून्य पर आयेगी तो इस प्रकार से बने दोनों त्रिभुजो के क्षेत्रफल का अनुपात का मान 4 हो जायेगा।

**Ans.**

**Sol.**



RB = $l$ – m





Now

ΔPOB





_____(1)

r = $l$ cot θ _____(2)

ΔROB



_____(3)



from (2) and (3)









_____(4)

from (1) and (4)







20. If L =  then find the value of  .

यदि L =  तो  का मान ज्ञात कीजिए।

**Ans.** 307

**Sol.** 

















21. Let $f(x) =$  then find.

यदि $f(x) =$  तो निम्नलिखित का मान ज्ञात कीजिए।

**(a)**  , **(b)**  , **(c)**  , **(d)** 

**Ans.** (a) 2, (b) D.N.E., (c) 0, (d) 0

**Sol.** 

(a) 

(b)  



(c) 

(d) 

22. If  = 8,  = λ and  , then λ =

यदि  = 8,  = λ तथा  , है तो λ =

**Ans.** 8

**Sol.**  if 

then L.H.S. = $\to \infty$

it means 





it is possible when 

Now















## COMPREHENSION TYPE QUESTIONS

### Paragraph for Question Nos. 23 to 25

Given $g(x) = \sin$  , $h(x) = x^2 + 2x + (\lambda + 1)$ and $u(x) =$ 

Let 

On the basis of above information, answer the following questions :

दिया गया है $g(x) = \sin$  , $h(x) = x^2 + 2x + (\lambda + 1)$ तथा $u(x) =$ 

माना 

उपरोक्त जानकारी के आधार पर निम्न प्रश्नों के उतर दीजिए :

23. If  $f(x) = 2$, then the value of $\lambda$ is :

(A) 10 (B) 5 (C) 2 (D) Does not exist

यदि  $f(x) = 2$, तो $\lambda$ होगा :

(A) 10 (B) 5 (C) 2 (D) विद्यमान नहीं है

**Ans.** B

**Sol.** 





$\lambda = 5$

24. $\lim_{x \to 0}$ f(x) =

(A) 0 (B) 2 (C) 5 (D) 1

**Ans.** D

**Sol.** 

25. If $\lim_{x \to 0}$ f(x) = L, then [L] (where [.] denotes greatest integer function) is equal to :

यदि $\lim_{x \to 0}$ f(x) = L, तो [L] होगा : (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

(A) 0 (B) 1 (C) –1 (D) 2

**Ans.** A

**Sol.** 

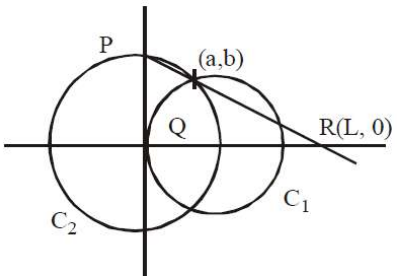
[L] = 0

**Paragraph for Question Nos. 26 to 28**

The figure shows a fixed circle $C_1$ and shrinking circle $C_2$. The equation of the fixed circle is $(x-1)^2 + y^2 = 1$. The centre of shrinking circle $C_2$ is at origin and it cuts positive y-axis at P. The points of intersection of $C_1$ and $C_2$ is Q (a, b) where b > 0, PQ cuts x-axis at R(L, 0). Let r be the radius of circle $C_2$ and let L = f(r). On the basis of above information, answer the following questions :

चित्र में एक अचर वृत $C_1$ तथा चर वृत $C_2$ दिया गया है। वृत $C_1$ का समीकरण $(x-1)^2 + y^2 = 1$ है तथा वृत $C_2$ का केन्द्र मूल बिन्दु है तथा यह धनात्मक y-अक्ष को P पर काटता है। $C_1$ तथा $C_2$ का प्रतिछेदन बिन्दु Q (a, b) है जहाँ b > 0, PQ x-अक्ष को R(L, 0) पर काटता है। माना वृत $C_2$ की त्रिज्या r है तथा L = f(r) ।

उपरोक्त जानकारी के आधार पर निम्न प्रश्नों के उतर दीजिए।




26. [redacted] f(r) =

(A) 0 (B) 2 (C) 4 (D) 8

**Ans.** C

**Sol.** [redacted]

P(0, r)

[redacted] _____(1)

[redacted] _____(2)

(1) and (2) $2a - 1 = r^2 - 1$ $\Rightarrow$ [redacted]

[redacted]

[redacted] equation of PR [redacted]

(a, b) lies on PR [redacted]

[redacted]

[redacted]

[redacted]

Now [redacted]

27. If $f(r).g(r) = r^2$, then [redacted] is :

यदि $f(r).g(r) = r^2$, तो [redacted] होगा :

(A) 0 (B) 1 (C) 1/2 (D) 2

**Ans.** B

**Sol.** [redacted]

28. Range of f is :

f का परिसर होगा :

(A) [2, 4] (B) (2, 4] (C) [2, 4) (D) (2, 4)

**Ans.** D

**Sol.** [redacted]

Range $\in (2, 4)$

**Paragraph for Question Nos. 29 to 31**

Let f(x) is a function continuous for all $x \in R$ except at $x = 0$. Such that $f'(x) < 0$ [redacted] $x \in (-\infty, 0)$ and $f'(x) > 0$ [redacted] $x \in (0, \infty)$. Let [redacted] $f(x) = 2$, [redacted] $f(x) = 3$ and $f(0) = 4$.

माना फलन f(x), $x = 0$ को छोड़कर सभी $x \in R$ के लिए इस प्रकार सतत है, कि $f'(x) < 0$ [redacted] $x \in (-\infty, 0)$ तथा $f'(x) > 0$ [redacted] $x \in (0, \infty)$. माना [redacted] $f(x) = 2$, [redacted] $f(x) = 3$ तथा $f(0) = 4$.

29. The value of $\lambda$ for which $2$ [redacted] $= \lambda$ [redacted] is :

λ का मान होगा जिसके लिए 2 = λ :

(A) (B) 2 (C) 3 (D) 5

**Ans.** C

**Sol.**

$x \to 0^-$ $x^3 = -ve$ $x^2 = +ve$ $x^3 - x^2 < 0$

$\Rightarrow f(X^3 - X^2) > f(0)$

$f(x^3 - x^2) > 4$

$x \to 0^+$ $2x^4 - x^5 > 0$ $x \to 0^+$ $f(2x^4 - x^5) = 2^+$

$f(2x^4 - x^5) > f(0)$

$f(2x^4 - x^5) > 4$

$\Rightarrow 2(3) = \lambda(2) \Rightarrow \lambda = 3$

30. The values of where [ · ] denote greatest integer function and { · } denote fraction part function:

का मान होगा जहां [ · ] महत्तम पूर्णांक फलन तथा { · } भिन्नात्मक भाग फलन को दर्शाता है :

(A) 6 (B) 12 (C) 18 (D) 24

**Ans.** B

**Sol.** $x \to 0^+ \Rightarrow -x \to 0^-$ 

 $[f(x)] = 2$

$x \to 0^+$ 



31.  where [ · ] denote greatest integer function :

 है जहां [ · ] महत्तम पूर्णांक फलन को दर्शाता है :

(A) 3 (B) 5 (C) 7 (D) 9

**Ans.** B

**Sol.** $x \to 0^-$ $x^3 < \sin^3 x$ 





and

$x \to 0^-$ $\sin^3 x > x$  

Now



$= 9 - 4 = 5$

## MATCH THE COLUMN

32. Match the following column :

| **Column-I** | **Column-II** |
|---|---|
| (A)  is equal to | (P) 0 |
| (B)  | (Q) 1 |
| (C)  is equal to | (R) 2 |
| (D) The absolute value of  is | (S) 3 |

(where [.] denotes greatest integer function)

निम्न कॉलम को सुमेलित कीजिए :

| **Column-I** | **Column-II** |
|---|---|
| (A)  | (P) 0 |
| (B)  | (Q) 1 |
| (C)  | (R) 2 |
| (D)  का निरपेक्ष मान होगा | (S) 3 |

(जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

**Ans.** A → P, B → Q, C → S, D → R

**Sol.** (A) 



Hence A → P

(B)  $[1^{\infty})$



$B \rightarrow Q$

(C) 





$= [\pi] = 3$ $C \rightarrow S$

(D) 





$\Rightarrow$ absolute value = 2

33. Match the following column :

| **Column-I** | | **Column-II** |
|---|---|---|
| (A)  | , where *n* is an integer, equals | (P)  |
| (B)  | $(n \in N)$ equals | (Q)  |
| (C)  | is (where $n \in N$) | (R) $\pi$ |

(D) If [redacted] where 'a' is some real constant then the value of 'a' is equal to

(S) non existent

निम्न कॉलम को सुमेलित कीजिए :

| **Column-I** | **Column-II** |
|---|---|
| (A) [redacted], जहाँ $n$ पूर्णांक है, का मान होगा | (P) [redacted] |
| (B) [redacted] ($n \in N$) का मान होगा | (Q) [redacted] |
| (C) [redacted] (जहाँ $n \in N$) का मान होगा | (R) $\pi$ |
| (D) यदि [redacted] जहाँ 'a' कोई वास्तविक नियतांक है तब 'a' का मान होगा। | (S) अस्तित्व विहीन |

**Ans.** $A \to Q; B \to R; C \to P; D \to P$

**Sol.** [redacted]

[redacted]

[redacted]

[redacted]

Now

[redacted]

[redacted]









$A \rightarrow Q$

(B) 





= 

Now







$= 1 \cdot (\pi - 0) = \pi$

$B \rightarrow R$

(C) 







C $\rightarrow$ P

(D) 





D $\rightarrow$ P

34. Match the following column :

| **Column-I** | **Column-II** |
|---|---|
| (A)  = | (P) 2 |
| (B) Let f(x) =  , then  f(x) = | (Q) 0 |
| (C)  | (R) 1 |
| (D) If  , then L + 2 = | (S) 3 |
| | (T) Non existent |

निम्न कॉलम को सुमेलित कीजिए :

| **Column-I** | **Column-II** |
|---|---|
| (A)  = | (P) 2 |
| (B) माना f(x) =  है, तो  f(x) = | (Q) 0 |
| (C)  | (R) 1 |
| (D) यदि  है, तो L + 2 = | (S) 3 |
| | (T) विद्यमान नहीं |

**Ans.** A → P; B → Q; C → R; D → S

**Sol.** (A)  $1^{\infty}$

 where 



$A \to P$

(B)  



$f(x) = x$ $B \to Q$

$x \to 0^+$ $f(x) \to 0$

(C)  



$= 1$ $C \to R$

(D) 



By L-Hospital Rule



$L + 2 = 3$